

Selection, Adoption, and Efficiency in Flexible Healthcare Supply: Evidence from Hospital-at-Home

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ABSTRACT: Hospital capacity is costly to expand, and the value of an available inpatient bed varies with congestion. Hospital-at-Home (HaH) provides inpatient-level care in a patient's residence and can release a brick-and-mortar bed without requiring permanent capacity expansion. However, this release is only realized when the hospital offers HaH and the patient accepts. I study how these decisions determine the use and value of flexible hospital capacity. Using clinical, accounting, census, and staffing data from four hospitals, supplemented by a novel patient survey, I jointly estimate patient acceptance and hospital offers, accounting for persistent patient preferences and endogenous offer selection. Patients differ systematically in their willingness to receive home care, and hospitals target likely adopters while balancing financial returns, clinical needs, and capacity relief. At median occupancy levels, the estimated gross capacity-relief component is equivalent to between \$830 and \$1,200 per discounted bed-day at the two largest hospitals. A counterfactual 10 percent increase in HaH reimbursement generates approximately 11,300 additional bed-days freed. Induced transfers increase predicted treatment costs, making the hospital's net financial gain dependent upon the realized use of released capacity. These findings show how consumer adoption shapes the deployment of flexible supply and why capacity relief can motivate its use even without treatment-cost savings.

KEYWORDS: Hospital-at-home, healthcare innovation, capacity management, demand-mediated capacity-constrained production

JEL CLASSIFICATION: D12, D24, I11, I18, L21, L23

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1 Introduction

The United States healthcare system faces an overcrowding problem. This challenge is particularly acute in inpatient hospital care, with many facilities operating at or near capacity. Available bed capacity has declined since the Covid-19 public health emergency, even as patient demand has remained strong. Population aging is expected to increase annual hospitalizations by approximately 11 percent between 2025 and 2035, pushing national average hospital occupancy to 85 percent as early as 2032 if bed supply does not expand (Leuchter et al., 2025).¹ This presents both a clinical and economic problem. When capacity binds, admitting one patient can delay care for others, raise treatment costs, and worsen health outcomes. A ten-minute increase in emergency department wait time raises hospital costs for seriously ill patients by approximately 6 percent, emergency department crowding increases mortality, and ICU congestion during the Covid-19 pandemic increased inpatient mortality (Woodworth and Holmes, 2020; Woodworth, 2020; Gowrisankaran et al., 2026). Hospitals cannot costlessly eliminate these losses by simply building more beds. Physical expansion requires large, long-lived investments and complementary clinical resources. Stochastic demand, capital constraints, and regulation make capacity adjustment slow and difficult to reverse (Pindyck, 1988; Adelino et al., 2015; Janke and Venkatesh, 2025; Courtemanche and Garuccio, 2025; Chiu, 2021). Against this backdrop, the recent expansion of Hospital-at-Home (HaH), an innovative care delivery model that provides inpatient-level care in patients' homes, offers a flexible way to relieve capacity pressure without requiring permanent capacity expansion.

HaH turns this capacity problem into a two-sided allocation problem. The hospital determines which clinically appropriate patients receive an offer, while patients determine whether that offer translates into treatment at home and released B&M capacity. These interacting decisions motivate the paper's central question: how do firm selection and consumer preferences shape the use and value of flexible capacity when conventional production is capacity constrained? I combine evidence on treatment-location effects with structural models of patient acceptance and hospital offers to characterize these decisions and recover the private value of released capacity. The framework then evaluates how reimbursement, eligibility rules, information, and hospital capacity shocks affect HaH use, capacity relief, and associated patient and hospital outcomes.

The empirical analysis links four complementary sources of information from four hospitals in a large southeastern U.S. health system between September 2021 and February 2026. Detailed electronic health records (EHR) identify clinical eligibility, health outcomes, and each stage of the

¹An occupancy rate of 85 percent is a commonly used healthcare standard for capacity pressure because hospitals require slack to accommodate stochastic demand and patient flow. Occupancy at this level is associated with regular bed shortages and periodic capacity crises (Bagust et al., 1999; National Guideline Centre, 2018). However, recent evidence suggests that 85 percent is a conservative upper bound for the onset of capacity strain, which may occur at lower occupancy rates depending on the ward type and care setting (Pratt and Wood, 2021).

HaH screening, offer, and acceptance process. Granular census, capacity, and staffing records measure the hospital state in which these decisions are made and trace how admissions, HaH transfers, and discharges affect subsequent congestion. Payor negotiated rates and patient-level revenues and costs reveal the pecuniary incentives associated with treating a patient at home rather than in the hospital. Finally, a novel survey linked to the EHR provides supplementary evidence on patients' expectations about care quality, comfort, safety, and convenience, together with household non-medical spending and caregiver circumstances that administrative records do not capture.

I jointly estimate a panel mixed-logit model of patient acceptance and a structural model of hospital offers using 31,944 inpatient stays from 21,354 unique patients. Repeated encounters identify persistent differences in willingness to receive care at home, while the model accounts for unobserved factors that jointly influence hospital offers and patient acceptance. Hospitals anticipate patient acceptance when weighing financial returns, clinical needs, HaH program strain, and the B&M bed-capacity released by a transfer. Expected remaining treatment duration determines the extent of the release, and current occupancy and staffed capacity determine the measured pressure it relieves. I estimate both static and dynamic versions of the hospital offer model, examining how offers reflect the value of released capacity under current operating conditions and the additional consequences for future capacity allocation. Complementary fuzzy spatial regression discontinuity evidence which exploits the program's geographic eligibility boundary provides additional context on health and utilization effects.

The estimates show that effective capacity depends on selection by both hospitals and patients. Willingness to receive home care varies persistently across patients, and first HaH treatment remains positively associated with subsequent acceptance. Survey evidence connects these preferences to expectations about care quality and experience. Hospitals target patients likely to accept while balancing financial returns, clinical needs, and capacity relief. The estimated capacity component rises with occupancy and, at 90% utilization, is equivalent to approximately \$1,200 and \$830 per discounted bed-day at the two largest hospitals. Evidence from the RD design indicates mixed health effects, including lower mortality and resource utilization during the acute-care episode but more subsequent inpatient readmissions.

The reimbursement policy counterfactual illustrates how capacity relief can motivate expansion even without treatment-cost savings. A 10% reimbursement increase generates approximately 11,300 additional potential B&M bed-days across the observed assessment cohort. Predicted treatment costs rise by more than additional reimbursement, but potential backfill produces a combined hospital cash gain of \$4.0–\$16.9 million under the stated accounting and utilization scenarios. These results highlight the importance of distinguishing the costs of treating patients at home from the value of the hospital capacity their transfers release.

This paper builds on a clinical health literature showing that HaH can safely substitute for conventional B&M hospitalization for selected patients and may improve satisfaction, reduce resource utilization, or lower treatment costs in some settings (Leff et al., 2005; Levine et al., 2020; Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021). More recent evidence under the federal Acute Hospital Care at Home initiative documents the feasibility of operating HaH at broader scale but continues to emphasize the importance of patient and program selection (Adams et al., 2023; Centers for Medicare & Medicaid Services, 2024). Existing studies primarily evaluate clinical outcomes among patients who enter HaH programs. This paper instead treats entry itself as an economic outcome, modeling both the hospital’s decision to make HaH available and the patient’s decision to accept it.

The economic value of HaH begins with the limits of conventional B&M expansion. Hospitals balance the costs of reserve capacity against its benefits for managing patient flow and stochastic demand (Joskow, 1980; Gaynor and Anderson, 1995; Carey, 1998; Bagust et al., 1999; Pratt and Wood, 2021). Permanent expansion is long-lived, partly irreversible, and depends on reimbursement, internal cash flow, and financing conditions (Pindyck, 1988; Wedig et al., 1989; Adelino et al., 2015; Gao et al., 2022), while certificate-of-need regulation can further constrain adjustment (Chiu, 2021; Courtemanche and Garuccio, 2025). HaH introduces a temporary capacity margin whose deployment can respond to current operating conditions, provided that patients accept treatment at home.

The paper’s first central contribution is to recover the model-implied shadow value of released-bed capacity, which I refer to as simply the private capacity value. Existing work shows that empty-bed availability affects hospital assignment and admission decisions (Kleiner, 2019; Freedman, 2016), while congestion changes care and generates health losses across patients (Woodworth, 2020; Francetic et al., 2024). Most closely related, Gowrisankaran et al. (2026) model how patient selection and congestion interact across hospitals during the Covid-19 pandemic. I study a within-hospital margin, in which transferring a patient to HaH makes B&M capacity available for operating slack or another admission. This connects the paper to research on production under capacity constraints and the opportunity cost of reserving capacity for future demand (Jofre-Bonet and Pesendorfer, 2003; Ryan, 2012; Kalouptsi, 2014; Jeziorski and Krasnokutskaya, 2016). Patient-level offer decisions reveal how the hospital values measured capacity relief relative to accounting returns, conditional on expected patient acceptance.

The paper’s second central contribution is to model patient acceptance as an input into the firm production process. The hospital determines which patients receive access to HaH, but a B&M bed is released only when the patient accepts. Hospital selection and patient acceptance therefore jointly determine whether additional capacity is produced. Gandhi (2023), which studies selective admissions at skilled nursing facilities, is closest in spirit because firm admission decisions

vary with census and patient type, but the firm in that setting controls the relevant capacity margin directly. On the supply side, I allow offers to reflect financial returns, clinical needs, offering frictions, and congestion relief, consistent with research showing that provider objectives need not reduce to contemporaneous profit maximization (Ellis and McGuire, 1986; Jack, 2005; Brekke et al., 2011). On the demand side, existing work studies hospital choice, the value of access, and intermediary incentives (Capps et al., 2003; Ho and Pakes, 2014). In this setting, the hospital first determines whether HaH enters the patient’s choice set, making patient demand a problem of endogenous consideration (Goeree, 2008). I address correlated hospital selection and patient acceptance using a control function (Petrin and Train, 2010) and repeated encounters, while supplementary survey evidence connects choice heterogeneity to beliefs about care and household circumstances (Ben-Akiva et al., 1994; Allcott and Wozny, 2014; Fuster and Zafar, 2016; Stantcheva, 2023; Ferrario and Stantcheva, 2025; Haaland et al., 2025). More broadly, the paper provides a framework for flexible supply technologies in which firms control access but consumers determine whether additional capacity is produced.

These findings make the allocation of offers central to evaluating HaH policy. Changes in reimbursement, eligibility, and information can expand opportunities for home treatment, but realized capacity relief depends on which patients receive offers, whether they accept, and when and how released beds are used. Hospital financial gains need not coincide with social resource savings, because reimbursement changes provider incentives while treatment costs and benefits extend across hospitals, patients, and households. More broadly, the paper provides a framework for understanding flexible supply technologies whose deployment requires consumer adoption. Their value depends on the joint distribution of gains from reallocating production, consumers’ willingness to adopt, and the shadow value of production capacity.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 develops a framework which distinguishes the private and social value of HaH. Section 4 describes the novel data and reduced-form evidence that supports the model. Section 5 develops a joint model of patient acceptance and hospital offers, and Section 6 reports and discusses the estimates. Section 7 evaluates the counterfactual analyses, and Section 8 concludes.

2 Institutional Background: Hospital-at-Home

Hospital-at-Home (HaH) provides acute, hospital-level care in a patient’s residence as a substitute for treatment that would otherwise require an inpatient admission. It is distinct from conventional home health, which primarily provides intermittent post-acute or chronic care, and from telemedicine alone. HaH episodes may include physician oversight, in-person nursing or paramedic visits, remote monitoring, medication and intravenous therapy, laboratory and diagnos-

tic services, oxygen delivery, and transfer back to a physical hospital when necessary. Programs are commonly classified as either admission-avoidance models, in which the patient enters HaH directly, or early-transfer models, in which part of an inpatient episode is completed at home (Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021; Leong et al., 2021). The program studied in this paper contains elements of both: patients generally enter through a hospital emergency department and may transfer to HaH immediately or after receiving inpatient care in the B&M hospital.

Clinical experimentation with home-based acute care dates at least to randomized trials in the United Kingdom in the 1970s (Hill et al., 1978). In the United States, Johns Hopkins developed an early physician-led HaH program in the 1990s. Its pilot and subsequent multi-site evaluation demonstrated the feasibility of treating selected older patients with conditions such as pneumonia, heart failure, chronic obstructive pulmonary disease, and cellulitis outside a conventional hospital (Leff et al., 1999, 2005). Nevertheless, U.S. adoption remained limited for the next two decades. Operating a program required hospitals to develop patient-selection protocols, mobile clinical teams, pharmacy and diagnostic delivery, remote monitoring, and emergency escalation systems, while reimbursement for home-based inpatient care remained uncertain (Applebaum and Rosemary, 2023; Brody et al., 2019). HaH consequently remained concentrated in demonstration programs, integrated delivery systems, and alternative-payment arrangements.

The Covid-19 pandemic produced the regulatory change underlying the present U.S. HaH market. CMS introduced its Hospitals Without Walls flexibilities in March 2020 and launched the Acute Hospital Care at Home (AHCAH) initiative in November 2020 (Centers for Medicare & Medicaid Services, 2020). The initiative permits approved Medicare-certified hospitals to treat qualifying patients as hospital inpatients in their homes through waivers of selected hospital Conditions of Participation. Participating hospitals must perform an in-person evaluation before home treatment, apply medical and non-medical screening protocols, provide daily evaluation by a physician or advanced-practice provider and registered nurse, arrange at least two in-person visits per day, maintain immediate remote communication, and provide a response to clinical deterioration within 30 minutes. The initiative has been extended beyond the public-health emergency and is currently authorized through September 30, 2030 (Centers for Medicare & Medicaid Services, 2026). Regulatory flexibility, inpatient reimbursement, and pandemic-era capacity pressure generated rapid adoption, with several hundred hospitals entering the initiative during its first years (Applebaum and Rosemary, 2023).

The waiver changes both the production technology and the financial incentives for acute care. It creates a capacity margin that does not require new B&M construction, but capacity is released only when the hospital offers HaH and the patient accepts it. HaH patients retain inpatient status, so hospital reimbursement is generally determined by the inpatient episode and diagnosis-related

group rather than by the physical location of each service. A hospital may therefore receive similar revenue while substituting mobile labor, monitoring, transportation, and household inputs for hospital rooms and facility-based services. The resulting accounting return need not equal the change in social resource use. HaH may be privately profitable without generating equivalent resource savings, or may conserve resources while remaining unattractive under a particular payer contract. Hospital accounting data also omit inputs supplied by patients and households, including time, utilities, housing space, and caregiving. These distinctions make reimbursement, hospital costs, and social costs separate empirical objects.

Access to HaH is determined through a sequential and highly selective process. Patients must first satisfy condition, payer, and geographic eligibility rules. In the program studied here, geographic eligibility is based on residence within an operational service area constructed around CMS's 30-minute emergency response requirement. Hospital staff then assess clinical stability, anticipated procedures and monitoring needs, and whether the patient's condition can be managed safely outside the hospital. A subsequent social-stability assessment may consider the safety of the residence, access to utilities and communication, caregiver or household support, and the patient's ability to participate in home treatment. Operational constraints, including mobile staffing and current program capacity, can further affect whether an otherwise suitable patient receives an offer. Only after the hospital makes HaH available does the patient or caregiver decide whether to accept. Acceptance may depend on perceived safety, trust in home-based care, prior experience, caregiver burden, housing conditions, and the convenience or comfort of remaining at home (Saenger et al., 2020; Centers for Medicare & Medicaid Services, 2024). A review of patient and caregiver perceptions likewise highlights the importance of the experience of receiving acute care at home, motivating attention to preferences beyond clinical outcomes (Wang et al., 2024). The hospital therefore determines whether HaH enters the patient's feasible choice set, while the patient determines whether the alternative is used.

Clinical evidence generally finds that HaH can provide safe care for appropriately selected patients. Trials and systematic reviews report outcomes that are often comparable to, and in some settings better than, conventional hospitalization, including possible reductions in mortality, readmissions, institutional care, and measured treatment expenditures (Leff et al., 2005; Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021). A prominent randomized trial found comparable or improved clinical outcomes and substantially lower measured expenditures under HaH (Levine et al., 2020).² The findings are not uniform across admission-avoidance and early-transfer programs, clinical conditions, outcome windows, or cost perspectives. Many trials are small and use tightly controlled selection protocols, while observational comparisons face

²Approximately 63% of eligible patients in the study declined participation in HaH, illustrating why clinical eligibility does not translate automatically into adoption or effective capacity expansion.

selection by both providers and patients. Evidence from the AHCAH period similarly suggests favorable or comparable short-run outcomes among treated patients but documents substantial differences between HaH and conventional inpatient populations (Adams et al., 2023; Centers for Medicare & Medicaid Services, 2024; Vakkalanka et al., 2026). Existing evidence therefore establishes the clinical feasibility of HaH more clearly than its causal effects in routine implementation or its value as a system-wide capacity instrument.

These institutions make HaH a sequential allocation problem rather than a simple comparison between two treatment locations. The hospital determines which patients receive access to the alternative, the patient determines whether access translates into treatment at home, and an accepted offer changes health outcomes, resource use, hospital returns, and the capacity available for other patients. Section 3 formalizes these channels and characterizes the resulting wedge between the private and social value of an HaH offer.

3 Economic Framework and Welfare

The central economic object is the shadow value of B&M capacity released by an accepted HaH offer. The hospital determines whether home treatment is available, and the patient determines whether the release occurs. This section describes how patient demand, treatment returns, and the value of freed capacity shape hospital and social planner incentives. Three wedges govern their potential disagreement: patient and health benefits, resource costs and profitability, and the value assigned to released capacity.

Patient choice and the consequences of HaH. Consider an HaH-eligible patient i admitted when hospital congestion is c . If offered HaH, the patient chooses between home treatment, h , and B&M care, b . Let

$$\Delta u_i = u_{ih} - u_{ib}$$

denote the patient's money-metric value of HaH relative to B&M, including an idiosyncratic component observed by the patient. Acceptance occurs when $\Delta u_i \geq 0$. Given the hospital's pre-offer information, the acceptance probability and expected patient surplus from an offer are

$$\rho_i = \mathbb{P}(\Delta u_i \geq 0), \quad \mathcal{Q}_i = \mathbb{E}[\max\{\Delta u_i, 0\}].$$

The probability ρ_i determines whether an offer reallocates treatment, while $\mathcal{Q}_i$ measures the value of the additional treatment option.

Let $\Delta \mathcal{H}_i$ denote the monetized social value of health changes under HaH beyond those incorporated in the patient's own valuation, and let $\Delta \mathcal{C}_i$ denote the reduction in real treatment resources

relative to B&M. Positive values favor home treatment. Acceptance-contingent consequences are understood as conditional expectations among accepting patients given the pre-offer information.

An accepted transfer also makes inpatient bed space available for other uses. Let $r_i^S(c)$ denote the social shadow value of this release: the value of additional capacity over the period in which the patient would otherwise occupy a B&M bed. It includes shorter waits, fewer delayed or forgone admissions, and lower crowding exposure for current and future patients. This value generally rises with utilization and expected demand (Joskow, 1980; Gaynor and Anderson, 1995). The finding that ICU strain increased mortality among hospitalized Covid-19 patients illustrates the consequences of scarce capacity for other patients (Gowrisankaran et al., 2026).

The shadow value reflects both the duration of the release and the scarcity of capacity during that period. A release during peak demand can be valuable even if brief, while a longer release preserves flexibility across subsequent arrivals. Thus, $r_i^S(c)$ values a temporary release conditional on acceptance; its expected contribution at the offer stage is $\rho_i r_i^S(c)$.

Social value of an HaH offer. A social planner chooses whether to offer HaH while respecting the patient's treatment choice. Let k_i denote the real cost of screening, coordinating, and extending an offer, borne by the hospital regardless of acceptance. Transfer and treatment costs incurred upon acceptance enter $\Delta\mathcal{C}_i$. The expected social gain from offering HaH is

$$G_i^S(c) = \mathcal{Q}_i + \rho_i \left[\Delta\mathcal{H}_i + \Delta\mathcal{C}_i + r_i^S(c) \right] - k_i. \quad (1)$$

The planner offers when $G_i^S(c) > 0$. Patient surplus already incorporates the patient's choice, while health, resource, and capacity gains are weighted by the probability of acceptance. Reimbursement transfers resources between payers and providers; its welfare consequences operate through behavioral incentives and any financing costs included in $\Delta\mathcal{C}_i$.

Hospital incentives. Let

$$\Delta\Pi_i = \Pi_{ih} - \Pi_{ib}$$

denote incremental hospital accounting profit from an accepted HaH placement. It reflects reimbursement and costs borne by the hospital, while $\Delta\mathcal{C}_i$ encompasses real resources used across providers and households.

Let $r_i^F(c)$ denote the hospital's private shadow value of the released B&M capacity. Freed bed space gives the hospital the option to accommodate another admission or retain operating slack. Its value combines the returns from these uses over the remaining period of the release. Benefits can extend across departments because inpatient patient-flow protocols influence emergency-

department bottlenecks and delays (Armony et al., 2015). The private and social shadow values differ to the extent that the hospital internalizes these consequences for patients, staff, and the broader healthcare system.

Hospitals may also value patient welfare, health, or access to home care. Let m_i denote the expected nonpecuniary value of extending the offer. The hospital's expected gain is

$$G_i^F(c) = m_i + \rho_i [\Delta\Pi_i + r_i^F(c)] - k_i. \quad (2)$$

The hospital offers when $G_i^F(c) > 0$. For $\rho_i > 0$, this condition can be written as $r_i^F(c) > (k_i - m_i)/\rho_i - \Delta\Pi_i$: capacity value can justify an offer even when home treatment lowers accounting profit. Greater congestion raises the incentive to release a bed when it increases this shadow value, while higher acceptance makes the release more likely relative to the cost of extending the offer.

Private and social incentives. The difference between social and private offer values is

$$\begin{aligned} \Delta G_i(c) &= G_i^S(c) - G_i^F(c) \\ &= \underbrace{Q_i + \rho_i \Delta\mathcal{H}_i - m_i}_{\text{demand/health wedge}} + \underbrace{\rho_i (\Delta\mathcal{C}_i - \Delta\Pi_i)}_{\text{resource/profitability wedge}} + \underbrace{\rho_i [r_i^S(c) - r_i^F(c)]}_{\text{capacity wedge}}. \end{aligned} \quad (3)$$

The first wedge compares patient and health benefits with the hospital's nonpecuniary valuation. The second compares social resource savings with accounting returns. Payment rules and the allocation of costs between the hospital and households can therefore make a transfer privately attractive even when its resource savings are small, or financially unattractive despite substantial social savings.

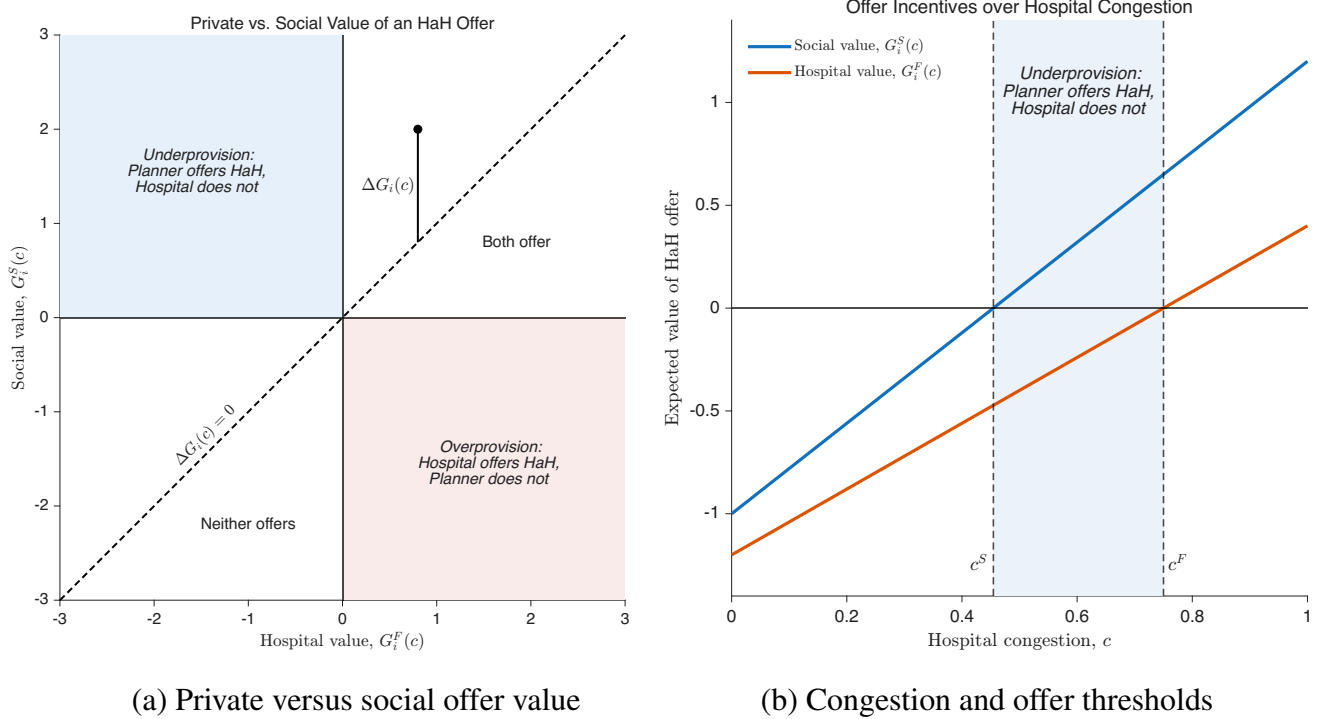
The capacity wedge compares the two shadow values of freed bed space. When the hospital captures only part of the benefit to other patients, its offer incentive falls short of the social value of the release. This gap can widen during congestion peaks as the consequences of delayed care increase. The combined wedge varies across patients and operating states with demand, health gains, costs, and the timing of the capacity release.

HaH is underprovided for patient i when $G_i^S(c) > 0$ but $G_i^F(c) < 0$, and overprovided when $G_i^F(c) > 0$ but $G_i^S(c) < 0$. The hospital and planner can therefore disagree about both program scale and which patients should receive offers. Patient acceptance governs the capacity, health, and resource gains realized under either policy. Valuable candidates combine willingness to receive home care with substantial gains from the resulting transfer.

Figure 1 illustrates these allocation differences. Panel (a) compares private and social offer values: the upper-left region denotes underprovision and the lower-right region denotes overprovi-

sion. The 45-degree line represents agreement in valuations. Panel (b) illustrates a case in which social offer value crosses zero at a lower congestion level than private offer value. The interval between these thresholds identifies states in which the planner offers HaH but the hospital does not.

Figure 1: Private and Social Incentives for Hospital-at-Home Offers



Empirical scope. The empirical analysis uses this framework to connect patient acceptance, hospital offer incentives, and the shadow value of freed capacity. The joint model uses variation in accounting returns and operating conditions, conditional on patient characteristics and acceptance beliefs, to recover the hospital's valuation of capacity relief in profit-equivalent units. A separate acceptance-belief term captures additional incentives to target likely adopters. Accounting data measure hospital costs, and the survey provides evidence on household preferences and non-medical spending. The health evidence in Section 4 motivates the baseline focus on treatment allocation, with clinical needs and escalation risk entering hospital offer incentives. Counterfactuals report hospital returns, potential bed-days released, and patient option value in normalized utility units. These outcomes quantify the principal allocation mechanisms, with the three wedges organizing their interpretation from private and social perspectives.

4 Data and Reduced-Form Evidence

4.1 Data, Assessment Sample, and Offer Documentation

The introduction of hospital-at-home care at a large southeastern health system provides a unique opportunity to study patient adoption and hospital capacity allocation from the program’s inception to current maturation. I construct a novel proprietary dataset combining information from four sources. From the health system, I obtain (i) rich electronic health records (EHR), (ii) administrative data on hourly bed census, and (iii) accounting data on costs and revenues, spanning September 2021 through February 2026. The data cover two large urban medical centers and two satellite hospitals in less densely populated cities. I supplement these data with (iv) a novel survey I administer to patients who are eligible for HaH. This survey includes additional information on unobserved household expenses from receiving HaH, patient beliefs and preferences, and caregiver income and employment. Appendix D describes the survey and its supplementary role in characterizing patient preferences.

In addition to data on patient diagnoses and procedures, the EHR data provides information on patient demographics, primary and secondary payors, post-stay outcomes, and patient acquisition through the HaH screening process (see Figure A.1 for a detailed overview of the stages of screening). My analysis also employs patient-level revenues and costs. Additional proprietary data on negotiated payor contracts help characterize provider returns and patient cost-sharing. Baseline price difference across treatment locations has little variation, thus highlighting the importance of the survey to provide supplementary information on additional hospitalization-related household expenses. These comprehensive data allow me to model the complex process of patient admission to a HaH program, while accounting for the various trade-offs hospitals and patients face in choosing in-hospital or at-home care.

The selection stages are summarized in Table 1. Roughly 80% of condition-location-payor eligible encounters are screened, about 18% of screened encounters have a recorded offer, and approximately half of these observed offers are accepted. These recorded frequencies describe how patients progress from eligibility to treatment, although incomplete documentation understates offers in earlier periods. The program expanded substantially during the study period. Figure 2 plots monthly observed offers and accepted placements. Accepted placements are observed throughout and therefore provide a consistent measure of treatment activity. Recorded offers also reflect changes in EHR procedures that improved documentation of patient declinations.³

To account for undocumented declined offers, I use overlapping reporting indicators to con-

³The October 2025 federal government shutdown interrupted HaH activity, followed by a shorter partial shutdown in January 2026. These operational interruptions are distinguished from the documentation changes marked in Figure 2.

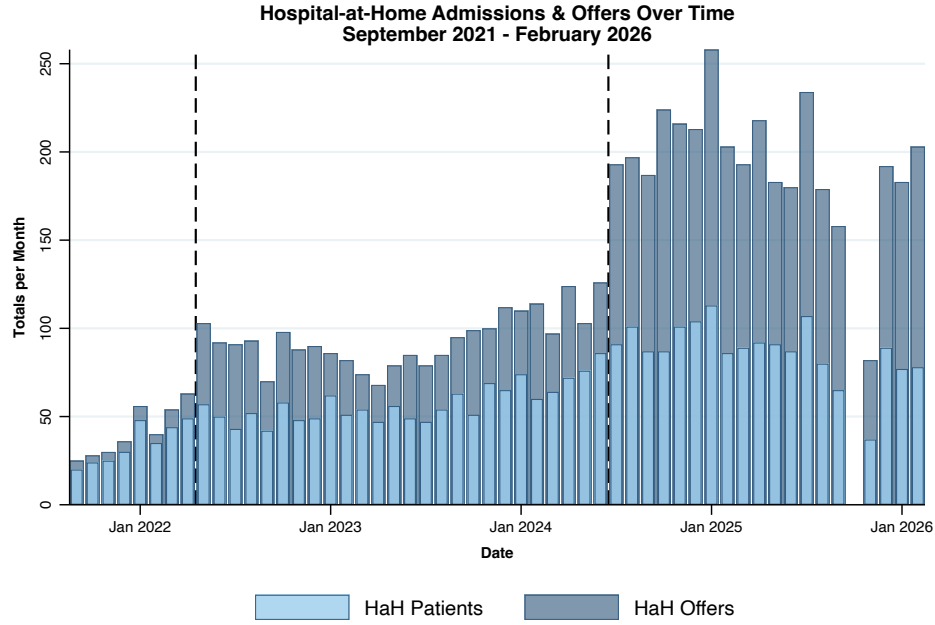
struct hospital-regime documentation probabilities, q_i , treating the most recent “modern regime” documentation as complete. These probabilities and the number of assessments inform a separate reconstruction of offer totals for the reduced-form evidence. Appendix C.1 describes this adjustment and its assumptions. The subsequent analysis examines how hospitals choose whom to offer HaH and how patients respond. It therefore focuses on encounters with an observed HaH assessment, taking the preceding screening decision as given. Section 6 reports the final estimation sample and the restrictions needed to construct its inputs.

TABLE 1: Hospital admissions and HaH selection, September 2021–February 2026

	All	Location & Payor eligible	Screened	Screened, not offered	Offered	Declined	Accepted
Encounters	76,481	41,595	34,563	28,234	6,329	2,938	3,391
Unique patients	49,830	26,613	22,881	19,535	5,219	2,597	2,899
Age (SD)	66.2 (17.1)	70.0 (16.3)	71.1 (15.8)	71.0 (16.0)	71.4 (14.8)	71.8 (14.3)	71.1 (15.1)
Male	0.468	0.461	0.464	0.468	0.447	0.432	0.461
White	0.653	0.682	0.681	0.672	0.724	0.708	0.737
Black	0.257	0.233	0.233	0.248	0.167	0.199	0.139
Latino/Hispanic	0.042	0.041	0.042	0.038	0.059	0.048	0.069
Commercial	0.135	0.072	0.054	0.052	0.063	0.055	0.069
Medicare	0.680	0.797	0.815	0.815	0.815	0.834	0.798
Medicaid	0.108	0.069	0.080	0.083	0.066	0.071	0.062
Dual Medicaid coverage	0.148	0.169	0.167	0.177	0.123	0.143	0.106
Multiple payors	0.364	0.484	0.482	0.487	0.460	0.452	0.468
Total LOS, days (SD)	6.4 (5.7)	6.2 (5.3)	6.4 (5.4)	6.5 (5.6)	6.0 (4.2)	5.4 (4.2)	6.5 (4.1)
DRG weight (SD)	1.5 (0.9)	1.4 (0.8)	1.4 (0.8)	1.4 (0.8)	1.2 (0.6)	1.3 (0.6)	1.2 (0.6)
ED entry	0.826	0.915	0.945	0.938	0.975	0.972	0.978
Nearest hospital, mins. (SD)	38.5 (35.4)	23.5 (14.0)	22.7 (12.6)	22.8 (12.8)	22.4 (11.8)	22.7 (11.5)	22.2 (12.0)
Net revenue, \$000 (SD)	18.8 (26.0)	15.7 (18.3)	15.2 (16.8)	15.8 (17.8)	12.3 (10.9)	12.6 (10.7)	12.0 (11.0)
<i>Neighborhood traits</i>							
Household-size (SD)	2.6 (4.4)	2.5 (1.0)	2.5 (1.0)	2.5 (1.0)	2.5 (0.6)	2.5 (0.5)	2.5 (0.6)
College/graduate share	0.426	0.496	0.500	0.499	0.506	0.491	0.518
Log per-capita income (SD)	10.6 (0.5)	10.7 (0.5)	10.7 (0.5)	10.7 (0.5)	10.8 (0.4)	10.7 (0.4)	10.8 (0.4)
Share below 150% poverty line	0.197	0.169	0.168	0.170	0.159	0.165	0.154

Notes: Subset to NC residents and inpatient conditions with 3+ HaH discharges. Numbers reflect shares unless SD is indicated. Screened means evidence of human-review for HaH suitability or evidence of HaH treatment. Offered includes recorded offers or accepted HaH; declined means recorded offer without acceptance. ACS education status is bachelor plus graduate share, income is $\log(1 + \text{per-capita income})$, and poverty is the source share below 150% of the poverty line. All monetary values described in this manuscript are expressed in February 2026 U.S. dollars (\$).

Figure 2: Hospital-at-Home Offers and Admissions Over Time



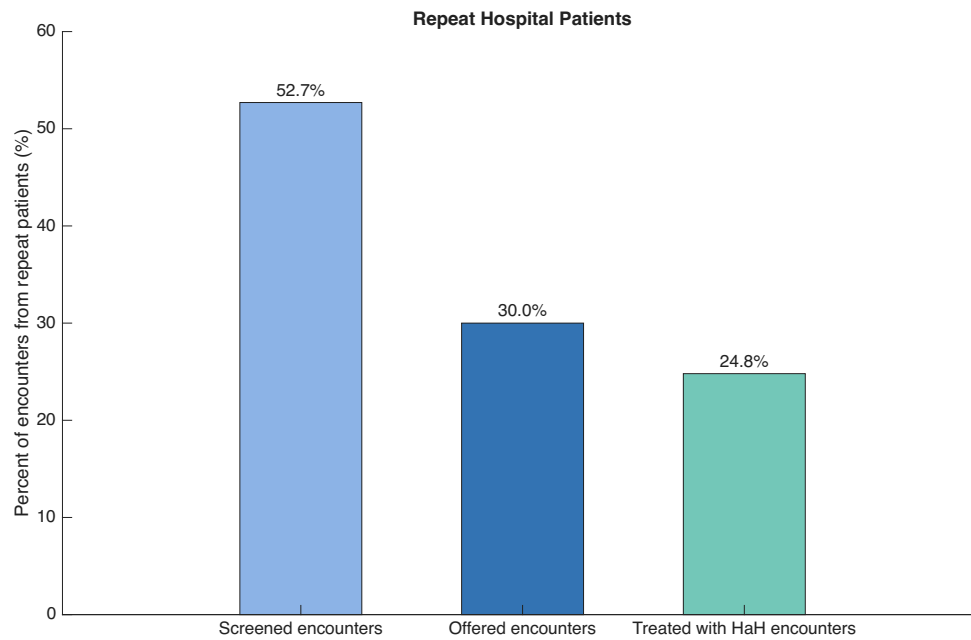
4.2 Selection and Repeated Patient Decisions

Patients who receive HaH offers differ from other screened patients, and those who accept differ further from those who decline. Relative to screened patients without an observed offer, offered patients have lower clinical severity, are more likely to be White and less likely to be Black or Medicaid insured, and have lower reimbursement. Among patients with an observed offer, those who accept are somewhat younger and more likely to be male, White, and commercially insured than those who decline. These differences motivate examining hospital offers and patient acceptance as separate decision margins. The hospital may also use information about patient suitability or preferences that is unobserved by the econometrician, thus accounting for observed patient characteristics alone may not fully explain which patients receive offers.

The repeated nature of inpatient encounters provides information on persistent willingness to receive HaH. Figure 3 shows that repeat patients account for approximately 53% of screened encounters, 30% of offered encounters, and 25% of accepted placements. Repeated choices by the same patient can therefore help distinguish persistent differences in willingness to accept HaH from variation across encounters.

Figure 4 shows that acceptance conditional on an observed offer rises most between zero and one prior acceptance, in this first increment by 14.8 percentage points. It rises more modestly

Figure 3: Share of Encounters Accounted for by Repeat Patients



at two prior acceptances and then levels off, whereas offer rates continue to increase with prior HaH offers. Repeated declinations, especially without any prior acceptance, predict much lower subsequent acceptance. These patterns motivate an initial treatment-associated shift alongside persistent patient preferences and hospital targeting based on prior interactions. They do not by themselves distinguish stable tastes from the effects of earlier treatment or outreach, since selection into treatment and subsequent offers also shapes the observed rates (Heckman, 1981; Dube et al., 2010). The estimated model of HaH offers and patient acceptances that follows therefore combines a persistent patient effect with a rich measures of patient history and interaction with the HaH program.

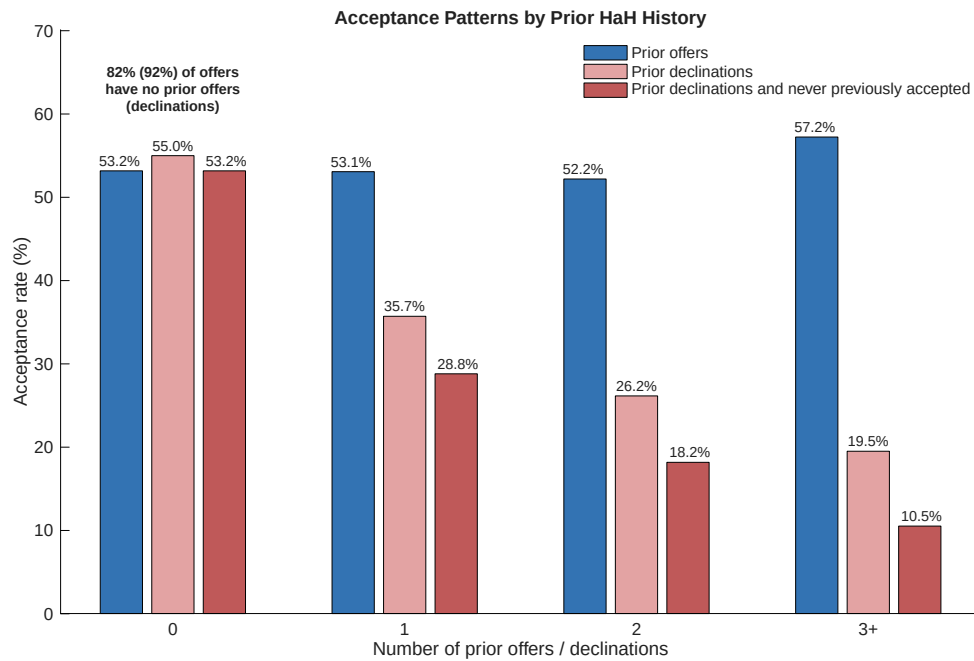
4.3 Capacity Pressure and Hospital Offers

Administrative census data capture the persistently high inpatient utilization in this market. Figure 5 plots the distribution of hourly capacity utilization separately by hospital. The two large medical centers, Hospitals A and B, operate at high utilization for a large share of hours with mass concentrated around or above 90% utilization, while the two satellite facilities exhibit lower and more dispersed utilization. These levels are operationally meaningful because hospitals generally require slack bed capacity to move patients and resources throughout the system; utilization above roughly 85% is often associated with regular bed shortages and periodic capacity crises (Bagust et al., 1999; National Guideline Centre, 2018; Pratt and Wood, 2021). Thus, the same HaH tech-

Figure 4: Probability of Outcome Across Offer and Acceptance History



(a) Offer and Acceptance Rates by Prior Interactions



(b) Acceptance Rates by Prior Offers and Declinations

nology is embedded in hospitals facing different degrees of capacity pressure, with the strongest congestion motive arising at the two urban facilities.⁴

Figure 5: Average Brick-and-Mortar Capacity Utilization

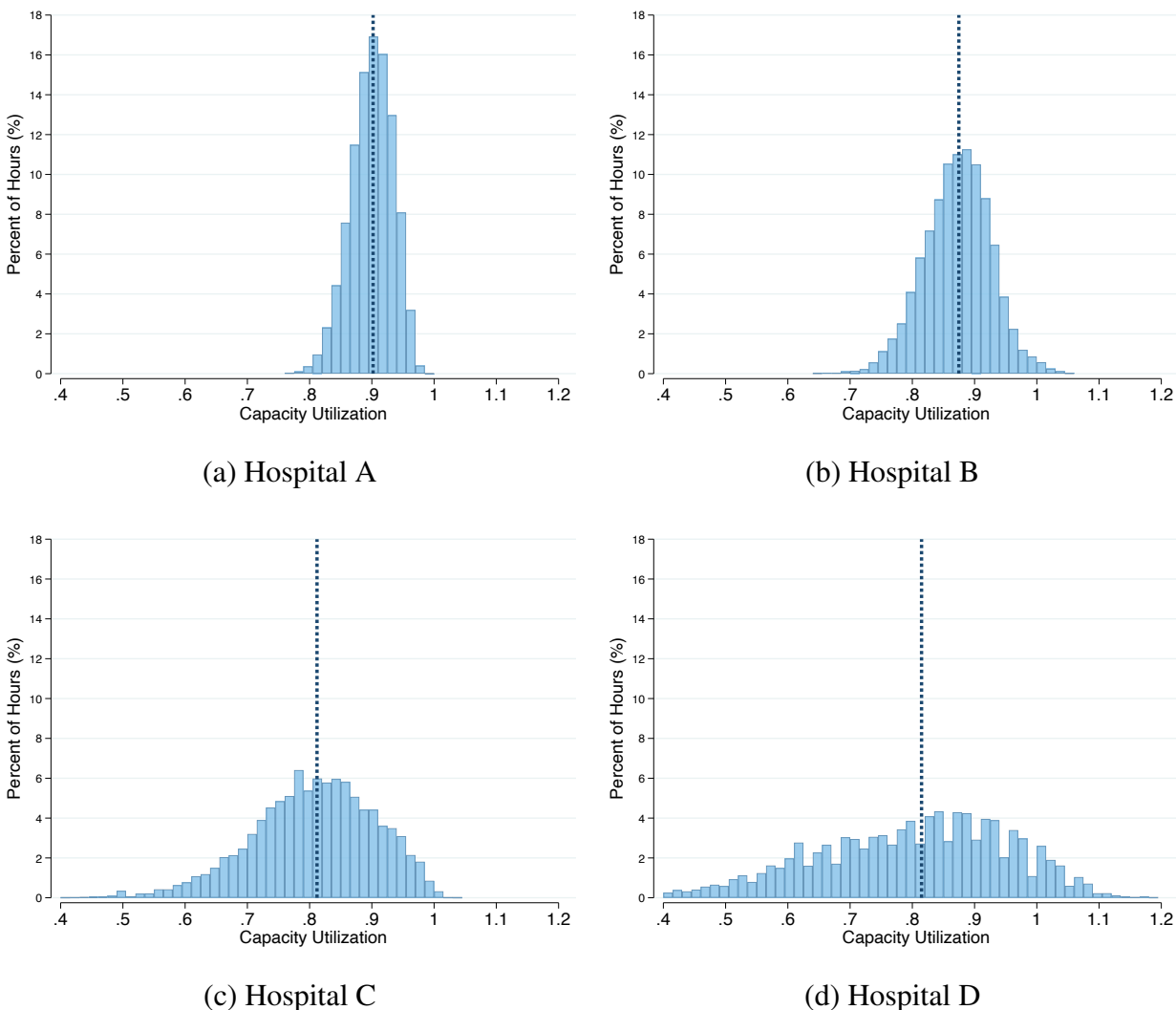
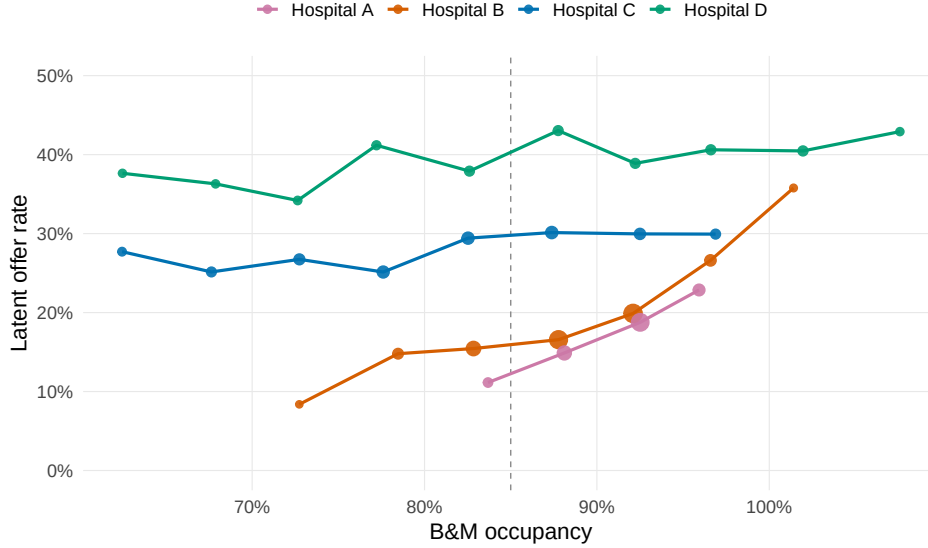


Figure 6 relates documentation-adjusted empirical offer rates to B&M occupancy. Offer rates at the two large hospitals exhibit the most visible increase in occupancy, across the full distribution of occupancy measures and the upper-tail of their support. The satellite facilities have higher offer rates over much of their support but flatter occupancy profiles. These level differences need not represent differences in capacity value alone: patient composition, operational differences, and persistent hospital policy also vary across facilities. The relevant comparison I shall examine is offer behavior changes with capacity pressure within a given hospital.

⁴The time-series analogues are reported in Figure A.2. These plots show substantial within-hospital variation in utilization over time, which provides within-hospital variation for estimating the conditional congestion response in

Figure 6: Documentation-Adjusted Offer Rates by B&M Occupancy



Notes: Bubble sizes correspond to the share of total hospital-hours spent in the corresponding B&M occupancy range.

Figure 7 provides evidence that HaH activity responds to capacity pressure. I estimate linear probability models for two sequential margins. The first outcome is whether an HaH-eligible patient is screened for home-care. The second is whether a reviewed patient receives an offer. In each specification, the slope on B&M occupancy varies by hospital. The figure reports the hospital-specific marginal effect of a 10 percentage-point increase in B&M occupancy. I estimate

$$Screened_{ih} = \alpha^S + \beta_h^S Occ_{h,t_i^A}^{BM} + \theta_h^S Stock_{h,t_i^A} + X_i' \gamma^S + \delta_h + \tau_{ym(t_i^A)}^S + \eta_{dow(t_i^A)}^S + \varepsilon_{ih}^S,$$

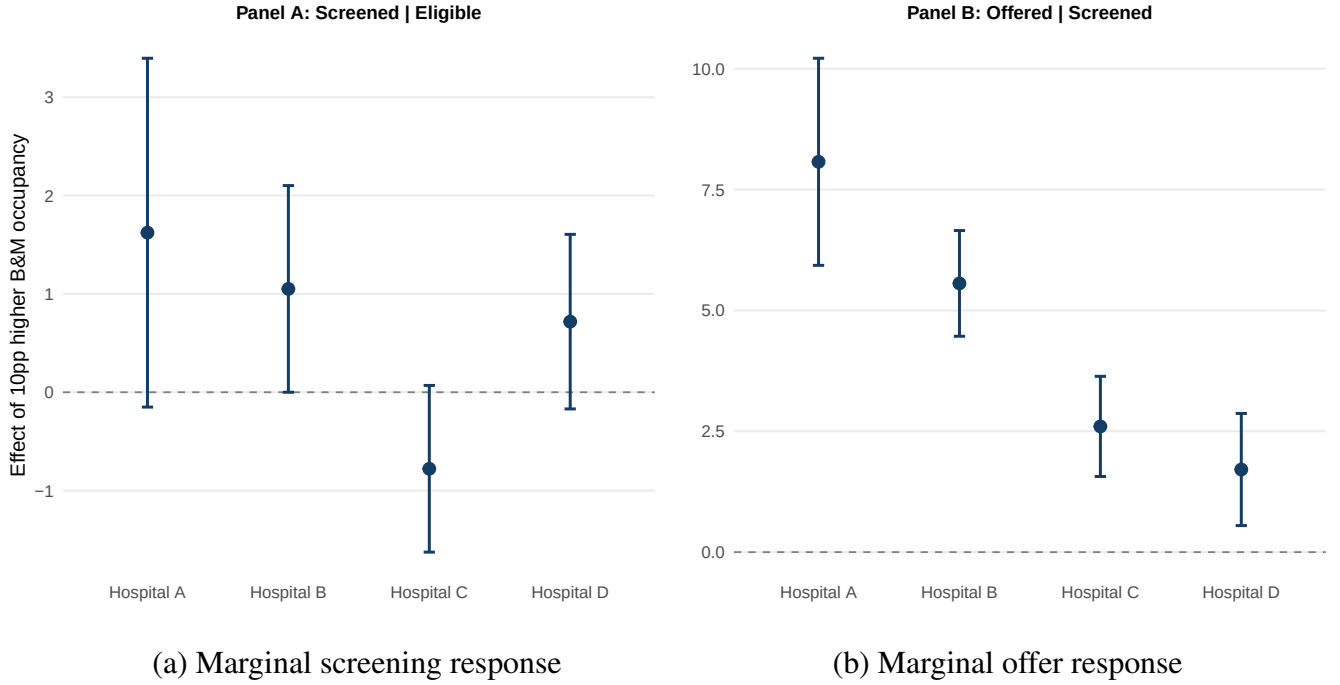
$$Offer_{ih} = \alpha^O + \beta_h^O Occ_{h,t_i^O}^{BM} + \theta_h^O Stock_{h,t_i^O} + X_i' \gamma^O + \delta_h + \tau_{ym(t_i^O)}^O + \eta_{dow(t_i^O)}^O + \varepsilon_{ih}^O.$$

Here, t_i^A is the admission time and t_i^O is the offer-decision time. $Occ_{h,t}^{BM}$ is B&M occupancy. $Stock_{h,t}$ contains the HaH census and the B&M emergency-department stock. The vector X_i contains patient demographics, payor, condition, and illness severity. The regressions include hospital, year-month, and day-of-week fixed effects. Standard errors are clustered at the hospital-day level.

Screening responses are weak, while offer responses are positive at all four hospitals. The two large urban hospitals have the largest offer responses. These estimates do not directly identify the value of released B&M bed capacity, but they show that observed offer behavior changes with temporal capacity pressure. This is directly relevant to patient welfare, as substantial evidence supports the claim that meaningful capacity pressure also affects clinical quality and patient outcomes

screening and offer behavior in the analyses to come.

Figure 7: Marginal Effects of Capacity Utilization by Decision Margin



(Woodworth, 2020; Francetic et al., 2024; Gowrisankaran et al., 2026). The linear regressions summarize average conditional gradients.

Health outcomes and modeling scope. A fuzzy spatial regression discontinuity design exploits the program’s geographic eligibility boundary to study health and utilization effects; Appendix B presents the design and summarizes the evidence developed in Capron et al. (2026). The estimates point to mixed effects, including lower mortality and fewer falls during the acute-care episode but more subsequent inpatient readmissions, with limited evidence of differences in several other health- and resource-related outcomes. These findings motivate a baseline model that abstracts from health outcomes directly while retaining escalation risk and acute-care needs as determinants of HaH offers.

5 Empirical Model of Hospital-at-Home Offers and Patient Acceptance

In this section, I outline a tractable and estimable joint model of demand for and supply of hospital-at-home care. Jointly modeling these decisions characterizes heterogeneity in patient willingness to receive inpatient-level care at home while recovering hospital offer incentives in a framework that accounts for endogenous offer selection and the dependence of realized capacity relief on

patient acceptance. Because access to HaH occurs only after a patient presents to a hospital or its ED, I condition on the admitting hospital and abstract from hospital choice and competition across hospitals. The model instead focuses on the subsequent treatment-location margin: whether the hospital offers HaH and, conditional on an offer, whether the patient accepts care at home rather than remaining in B&M care.

The model conditions on an observed HaH assessment and does not separately estimate the preceding screening decision. An assessment produces two sequential decisions. The hospital first decides whether to offer at-home care to an eligible patient. If an offer is made, the patient decides whether to accept HaH or remain in B&M. Given most consequences of an offer are realized only after an accepted transfer, the hospital anticipates and internalizes the probability that the patient will accept. The model has three observed outcomes: an accepted HaH placement, a documented rejection, and no observed offer. The panel structure of the model allows willingness to accept to vary persistently across patients, with repeat encounters exploited to distinguish this heterogeneity from encounter-specific choice variation. The hospital values the capacity released by an accepted transfer while balancing this benefit against the corresponding exposure to HaH staffing demands, location profitability, and patient care needs.

5.1 Decisions and Offer Documentation

Let i index HaH assessments and let $j(i)$ denote the patient associated with assessment i admitted to hospital $b(i)$. Let $O_i \in \{0, 1\}$ indicate that the hospital makes an actual HaH offer and let $A_i \in \{0, 1\}$ indicate that the patient accepts. Define

$$p_i = \mathbb{P}(O_i = 1 \mid \mathcal{J}_i^O), \quad \rho_i = \mathbb{P}(A_i = 1 \mid O_i = 1, \mathcal{J}_i^O),$$

where $\mathcal{J}_i^O$ is the hospital's information at the assessment. Section 4.1 describes the hospital-regime rejection-capture probabilities q_i , which are constructed from reporting patterns and fixed before estimation. Accepted placements are always observed; a rejected offer is recorded with probability q_i . These inputs link actual offers to the three observed EHR outcomes.

Let $Y_i \in \{a, d, n\}$ denote an accepted placement, a documented rejection, or no observed offer. Conditional on p_i and ρ_i , the observed outcome probabilities are

$$\mathbb{P}(Y_i = a) = p_i \rho_i, \tag{4}$$

$$\mathbb{P}(Y_i = d) = p_i (1 - \rho_i) q_i, \tag{5}$$

$$\mathbb{P}(Y_i = n) = 1 - p_i + p_i (1 - \rho_i) (1 - q_i). \tag{6}$$

The last outcome combines actual non-offers with latent rejected offers that were not documented.

The probabilities in equations (4)–(6) sum to one for every assessment.

The maintained measurement assumption is that, conditional on hospital and documentation regime, q_i describes capture among rejected offers and does not independently shift offer behavior or acceptance utility after conditioning on the included controls. Appendix C.1 describes this adjustment. The empirical offer frequencies in Section 4.1 do not use fitted p_i or ρ_i ; the posterior probabilities below are separate model objects used to update patient histories.

Let $\hat{O}_i \equiv \mathbb{P}(O_i = 1 \mid Y_i)$ denote the posterior probability that a true offer occurred. Accepted placements and documented rejections reveal an offer, so $\hat{O}_i = 1$ for these outcomes. When no offer is recorded, either no offer occurred or an offer was declined and not documented. Therefore, the posterior probability of a true offer when no offer is documented is

$$\hat{O}_i = \mathbb{P}(O_i = 1 \mid Y_i = n) = \frac{p_i(1 - \rho_i)(1 - q_i)}{1 - p_i + p_i(1 - \rho_i)(1 - q_i)}. \quad (7)$$

These posterior probabilities update the patient’s prior HaH interaction histories, which enter the demand and supply specifications below. The update occurs only after evaluating the current outcome and therefore affects future assessments.

5.2 Patient Acceptance

Figure 3 documents the importance of repeated patients in the assessment, offer, and treatment samples. Figure 4 show that prior offers, acceptances, and rejections predict subsequent HaH decisions. These patterns motivate a model that distinguishes persistent heterogeneity in patient preferences from state dependence associated with prior outcomes.

Conditional on an offer, the patient chooses between HaH (h) and B&M ($b(i)$) care. Let ΔU_{ij} denote the utility of h relative to $b(i)$ at assessment i for patient $j(i)$ with persistent HaH preference ξ_j . I parameterize this indirect utility difference as

$$\Delta U_{ij} = X_i' \beta_U - \alpha P_i + \sigma_\xi \xi_j + \varepsilon_i, \quad (8)$$

where ε_i is a logistic utility-difference shock, as implied by independent Type-I extreme-value shocks to the two treatment alternatives. The vector X_i contains race, payor, age, sex, distance to the admitting hospital, condition severity, neighborhood characteristics, condition fixed effects, hospital-by-quarter fixed effects, and prior HaH interaction history, while P_i contains out-of-pocket price differences borne by the patient.⁵ An object of central interest is the term $\xi_j \sim N(0, 1)$,

⁵ Age, DRG weight, home-to-hospital distance, household size, neighborhood college share, log per-capita income, and the share below 150% of the poverty line are standardized using fixed reference means and standard deviations. Prior interaction histories are raw cumulative counts. Indicators and the generalized residual introduced below are not

which represents persistent unobserved willingness to receive acute care at home. Conditional on the persistent effect, the acceptance probability follows the standard logit formulation (Train, 2009).

5.3 Hospital Offer Decision

While patient acceptance determines whether a release actually occurs, hospital offer selection first determines which transfers are attempted. The hospital may value consequences beyond contemporaneous accounting profit. The model therefore uses observed and latent offers to recover the private capacity value of scarce bed stock together with the hospital's valuations on accounting returns, HaH program strain, and clinical risk.⁶

The hospital observes the assessed patient, their clinical condition, and the current operating state before choosing whether to offer HaH. Let ρ_i denote the probability that the patient will accept based on the information available to the hospital at the assessment. The deterministic difference in the hospital's value from offering rather than not offering is

$$\Gamma_i(\rho_i; \Theta_S) = Z_i' \kappa + \rho_i \Delta W_i, \quad (9)$$

where Z_i contains determinants of the offer decision whose effects do not require acceptance and ΔW_i is the incremental value of an accepted HaH placement. The hospital offers HaH when $\Gamma_i + v_i \geq 0$, where v_i is a logistic offer-payoff difference, obtained from independent Type-I extreme-value shocks to offering and not offering. The resulting offer probability is thus

$$p_i = \Lambda(\Gamma_i), \quad (10)$$

where $\Lambda(x) = \exp(x) / [1 + \exp(x)]$. The implemented empirical specification decomposes the value of an accepted placement as

$$\Delta W_i = \beta_{\Pi} \Delta \Pi_i + \beta_H H_i + \beta_E E_i^B + \lambda_B' M_i^B + \lambda_H M_i^H. \quad (11)$$

These terms measure expected accounting profits, escalatory care risks, hospitalization duration, released B&M capacity, and HaH program strain, respectively. The vector

$$\Theta_S = (\kappa, \beta_{\Pi}, \beta_H, \beta_E, \lambda_B, \lambda_H)$$

standardized. Appendix C.2 records the transformations.

⁶Consistent with health-economics models in which providers place weight on mission-driven payoffs as well as financial incentives (Ellis and McGuire, 1986; Jack, 2005; Brekke et al., 2011).

collects the supply-side parameters, with κ containing all direct offer coefficients, including the nuisance coefficients θ below.

The vector Z_i contains the offer-stage determinants that do not require patient acceptance. To distinguish the components of structural interest from nuisance controls, let

$$Z_i' \kappa = \kappa_O + \kappa_{\text{hist}} \text{Hist}_i^O + \gamma_S S_i + \gamma_P \rho_i + \zeta_i' \theta. \quad (12)$$

The parameter κ_O is the common net offer term. A negative value is consistent with outreach, vetting, and administrative effort incurred before acceptance, but its level also absorbs baseline benefits and depends on the reference categories and regressor centering. It is therefore interpreted as a baseline offer friction rather than a separately measured financial cost. The variable Hist_i^O is the strictly prior expected number of true offers made to the patient. It includes recorded offers and the posterior probabilities of undocumented declined offers. Its coefficient κ_{hist} allows prior interactions to affect current offer value through repeated outreach, familiarity with the patient and caregivers, or persistent targeting.

S_i measures acute severity and clinical care burden at the assessment based on rich EHR flow-sheet data. Care burden combines nursing-assessment (NWASS) and fall-risk (Morse) scores, while acute severity uses organ-failure-risk (SOFA) scores. These measures enter outside the acceptance multiplier because they affect the decision to attempt a transfer regardless of whether the patient ultimately accepts. The term $\gamma_P \rho_i$ allows the hospital to place greater value on offering HaH to patients with a higher predicted probability of acceptance, separately from the measured consequences of an accepted placement. This is consistent with a mission-driven preference for providing specific services to patients who are most likely to want them. The remaining vector ζ_i contains condition type indicators, hospital and department fixed effects, and indicators for missing clinical EHR inputs and the partial federal government shutdown described in Section 4.⁷

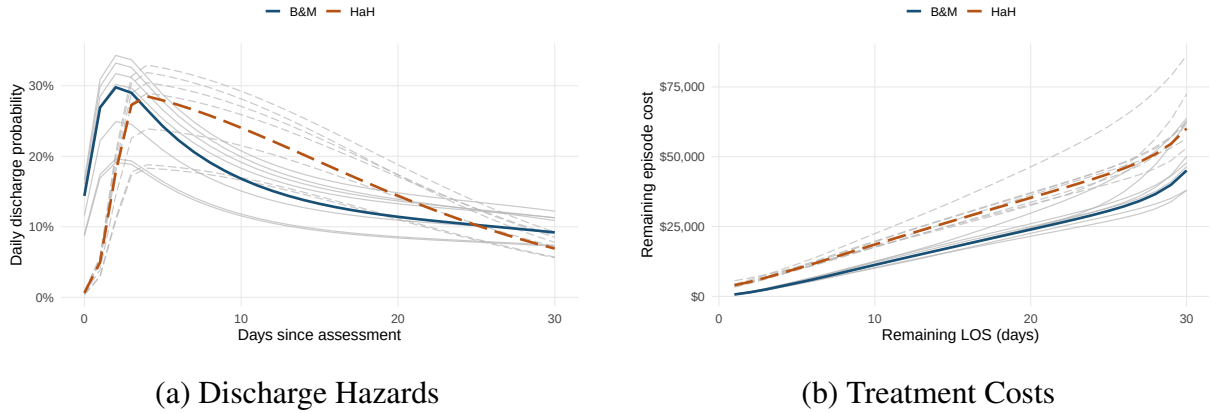
The components of ΔW_i summarize the predicted consequences of an accepted transfer. Let E_i^B and E_i^H denote expected discounted remaining days in B&M and HaH care, respectively. These measures are constructed from assessment-origin discharge models that condition on hospital, condition, treatment location, current length of stay, and day of week, using a daily discount factor of $\delta = 0.99$. Appendix C.3 details their construction. Expected remaining B&M duration enters separately and affects the firm's payoff through a conditional duration effect, meaning hospitals'

⁷For readability, equations (9)–(12) suppress input-transformation notation. The profit input subtracts a fixed hospital-quarter-condition reference, with the existing sparse-cell fallback, and is divided by \$1,000. B&M relief, HaH strain, and escalation risk subtract fixed global reference means without rescaling. Only the standalone acceptance-belief term uses $\rho_i - 0.4106788110$; the multipliers on placement consequences use the uncentered probability ρ_i . Clinical burden and severity indices are standardized, with separate missingness flags. Prior expected offers remain raw counts.

willingness to offer a patient can vary with the anticipated remaining stay conditional on accounting returns, clinical risk, and capacity consequences. This allows the capacity terms to fully capture how the value of releasing those bed-days varies with hospital utilization.

Let $\Delta\Pi_i^{\$} = \pi_i^H - \pi_i^B$ denote the conditional HaH-minus-B&M dollar profit difference, where $\pi_i^o = R_i^o - C_i^o$ is expected remaining revenue minus direct treatment costs under option $o \in \{H, B\}$. Reimbursement parity across treatment locations implies $\Delta\Pi_i^{\$} = C_i^B - C_i^H$. The offer index uses the centered difference in \$1,000 units, denoted by $\Delta\Pi_i$ and defined in equation (C.13). Figure 8 reports the remaining cost profiles and fitted discharge hazards by condition and treatment location. Appendix C.4 explains how episode-level cost estimates yield these conditional predictions of remaining direct treatment costs.

Figure 8: Predicted Discharge Hazards and Remaining Treatment Costs



Let H_i denote the probability that the patient would require return transfer from HaH back to B&M care due to escalation in acute care needs. Because this risk arises only under HaH treatment, it enters ΔW_i and is therefore scaled by the probability of acceptance.

To construct HaH program strain, let N_i^H denote HaH census and L_i denote HaH staffing at the time of the assessment. Define HaH utilization as

$$u^H(N^H, L) = \frac{N^H}{L}. \quad (13)$$

The program-strain function is

$$m^H(N^H, L) = L [u^H(N^H, L)]^2 = \frac{(N^H)^2}{L}. \quad (14)$$

The expected strain generated by adding the assessed patient is therefore

$$\begin{aligned} M_i^H &= E_i^H [m^H(N_i^H + 1, L_i) - m^H(N_i^H, L_i)] \\ &= E_i^H \frac{2N_i^H + 1}{L_i}. \end{aligned} \quad (15)$$

This quadratic specification implies that the contemporaneous strain from adding a patient is greater when the HaH program is more highly utilized. Multiplication by L_i allows the level of the strain function to scale with staffing levels.

The B&M term M_i^B is of central interest to the paper's primary research question. It measures occupancy-weighted bed-days released by an accepted HaH transfer. Let N_i^B denote B&M census and K_i denote staffed B&M capacity. Similarly, define B&M utilization as

$$u^B(N^B, K) = \frac{N^B}{K}. \quad (16)$$

The capacity-pressure function is

$$m^B(N^B, K) = K [u^B(N^B, K)]^2 = \frac{(N^B)^2}{K}. \quad (17)$$

The corresponding hospital-specific released-capacity component is

$$\begin{aligned} M_{ib}^B &= \mathbb{1}\{b(i) = b\} E_i^B [m^B(N_i^B, K_i) - m^B(N_i^B - 1, K_i)] \\ &= \mathbb{1}\{b(i) = b\} E_i^B \frac{2N_i^B - 1}{K_i}. \end{aligned} \quad (18)$$

This component increases with current utilization and expected remaining B&M duration. With E_i^B entering the payoff separately, the capacity coefficients capture how the value of releasing those bed-days varies with occupancy, conditional on other payoff inputs.

Let M_i^B collect the hospital-specific components $\{M_{ib}^B\}_{b \in \mathcal{B}}$, with corresponding coefficient vector λ_B . Consequently,

$$\lambda_B' M_i^B = \sum_{b \in \mathcal{B}} \lambda_{B,b} M_{ib}^B \quad (19)$$

is the measured B&M capacity contribution to the hospital's offer index, conditional on acceptance. Each hospital has its own coefficient on the smooth released-capacity component. Released capacity may be retained as operating slack or used to admit another patient. The model therefore identifies the composite private value of making additional capacity available. It does not separately identify the portion attributable to reduced operating pressure, improved patient throughput,

or additional pecuniary profits from backfill (i.e., admitting patients waiting in the queue). The capacity terms hold the hospital's current operating state fixed over the patient's expected remaining stay. The offer model is therefore a static representation that accounts for expected remaining treatment duration of the hospital's decision at the time of the assessment rather than a model of future hospital states.

Appendix F formalizes how expected acceptance strengthens the capacity incentive in the offer equation and characterizes the conditions under which greater congestion lowers the acceptance threshold needed to justify an offer. It also distinguishes these direct incentives from the response of jointly determined offer and acceptance probabilities.

5.4 Selection on Unobservables

Offers need not be conditionally random with respect to unobserved patient preferences. The hospital may observe information about home support, patient or caregiver reluctance, or clinical suitability that is unavailable to the econometrician yet is related to the patient's willingness to accept HaH. Estimating acceptance only among observed offers would then confound patient preferences with selection into the offer set. I account for this dependence by augmenting the acceptance utility in equation (8) with a control function derived from the offer probability:

$$\Delta U_{ij} = X_i' \beta_U - \alpha P_i + \sigma_\xi \xi_j + \varphi(1 - p_i) + \varepsilon_i, \quad (20)$$

Let the vector

$$\Theta_D = (\beta_U, \alpha, \sigma_\xi, \varphi)$$

collect the demand-side parameters of interest, and denote the collection of all demand and supply parameters as $\Theta = (\Theta_D, \Theta_S)$. For the offer equation, $1 - p_i$ is the generalized residual from the binary logit model conditional on an offer being made (Gourieroux et al., 1987). The coefficient φ allows acceptance utility to vary with offer selection after conditioning on the observed demand covariates and persistent patient heterogeneity. This specification links the two decisions directly: ρ_i depends on p_i through the generalized residual, while p_i depends on ρ_i through the hospital's beliefs over patient acceptance.

5.5 Joint Likelihood and Estimation

Offer–acceptance fixed point. Let ξ_g index the support points for persistent patient preferences, and let w_{ig}^- denote the probability, immediately before assessment i , that patient $j(i)$ is represented

by point g . The node-specific acceptance probability and the hospital's acceptance belief are

$$\rho_{ig}(p_i) = \Lambda(X_i' \beta_U - \alpha P_i + \sigma_\xi \xi_g + \varphi(1 - p_i)), \quad (21)$$

$$\rho_i(p_i) = \sum_g w_{ig}^- \rho_{ig}(p_i). \quad (22)$$

The weights w_{ig}^- depend only on the patient's strictly prior observed history and are updated as described below.

Within each assessment, the offer and acceptance probabilities must be mutually consistent. The offer probability depends on the hospital's acceptance belief through equation (9), while the acceptance probability depends on the offer probability through the control function. For each trial parameter vector, p_i therefore solves

$$F_i(p_i; \Theta) = p_i - \Lambda[\Gamma_i(\rho_i(p_i); \Theta_S)] = 0. \quad (23)$$

The solution jointly determines the offer probability p_i and the corresponding acceptance belief $\rho_i(p_i)$. This fixed point imposes consistency between the hospital's offer policy and its expectation of patient acceptance without altering the sequential timing in which the hospital first makes an offer and the patient then decides whether to accept.

Observed-outcome likelihood. Recall $Y_i \in \{a, d, n\}$ denotes an accepted offer, a documented rejection, or no observed offer, respectively. Conditional on preference node g , the probability of the observed outcome at assessment i is

$$\begin{aligned} \ell_{ig}(Y_i; \Theta) &= [p_i \rho_{ig}(p_i)]^{\mathbb{1}\{Y_i=a\}} [p_i \{1 - \rho_{ig}(p_i)\} q_i]^{\mathbb{1}\{Y_i=d\}} \\ &\quad \times [1 - p_i + p_i \{1 - \rho_{ig}(p_i)\} (1 - q_i)]^{\mathbb{1}\{Y_i=n\}}. \end{aligned} \quad (24)$$

Let $\mathcal{T}_j$ denote the chronologically ordered assessments for patient j that enter the likelihood, and let F_j^0 denote the distribution of the persistent patient taste immediately before the first assessment in $\mathcal{T}_j$. Conditional on the persistent patient effect, the patient-level likelihood contribution is the product of equation (24) across assessments. Integrating over the patient effect gives

$$\begin{aligned} L_j(\Theta) &= \int \prod_{i \in \mathcal{T}_j} \ell_i(Y_i; \Theta, \xi) dF_j^0(\xi) \\ &\approx \sum_{g=1}^G w_{jg}^0 \prod_{i \in \mathcal{T}_j} \ell_{ig}(Y_i; \Theta), \end{aligned} \quad (25)$$

where w_{jg}^0 denotes the node weight immediately before the first likelihood assessment for patient j . The joint log likelihood that is maximized is

$$\mathcal{L}(\Theta) = \sum_j \log L_j(\Theta). \quad (26)$$

Sequential integration over persistent heterogeneity. I approximate the distribution of persistent patient preferences using Gauss–Hermite quadrature, with transformed standard-normal nodes ξ_g and prior weights w_g . The estimator evaluates equation (25) recursively. Immediately before assessment i , the predictive likelihood contribution and subsequent posterior update are

$$L_i(\Theta) = \sum_{g=1}^G w_{ig}^- \ell_{ig}(Y_i; \Theta), \quad (27)$$

$$w_{ig}^+ = \frac{w_{ig}^- \ell_{ig}(Y_i; \Theta)}{\sum_{k=1}^G w_{ik}^- \ell_{ik}(Y_i; \Theta)}. \quad (28)$$

The updated weights become the prior weights for the patient’s next assessment. At the beginning of the full observed patient history, the weights equal the Gaussian quadrature weights w_g . Earlier EHR assessments outside the likelihood sample may update these weights before the first assessment in $\mathcal{T}_j$, thereby initializing w_{jg}^0 , but contribute zero directly to the objective. Because the offer probability and predictive likelihood are evaluated using w_{ig}^- before the current outcome updates the weights, the current outcome does not enter its own acceptance utility or offer index. Appendix C provides additional details on computation and estimation.

5.6 Identification

The fixed documentation adjustment and complete reporting in the modern regime distinguish the probabilities of actual non-offers and incompletely recorded rejections. Conditional on the fixed documentation probabilities, variation in acceptances, documented rejections, and no observed offers identifies the latent offer and acceptance probabilities. Repeated observations for the same patient identify persistent preference heterogeneity through serial dependence in acceptance and rejection outcomes.

Observed patient characteristics, prior interaction histories, and repeated decisions identify systematic and persistent heterogeneity in acceptance utility. The patient out-of-pocket price coefficient additionally requires variation in patient prices conditional on these characteristics. Because such variation is limited in the empirical application due to parity reimbursement and cost-sharing, patient utility is reported in its normalized choice scale rather than converted to dollars. The household beliefs and burden survey provides variation in the expected difference in non-medical

spending between HaH and B&M care, but coverage over the full sample is scarce. Appendix D provides an in-depth discussion of the survey-augmented demand model and its results.

Supply-side variables excluded from patient utility shift the offer probability and help identify selection into offers. These variables include current B&M capacity pressure, HaH staffing exposure, direct-variable savings, assessment department, and an indicator for the partial federal-shutdown. Identification requires these variables to be conditionally independent of unobserved acceptance utility after controlling for observed patient characteristics, clinical condition, hospital-by-quarter demand effects, and prior patient history. Under this exclusion restriction and the maintained logit control-function specification, variation in these offer shifters identifies φ . The offer intercept is a net baseline term under the stated normalizations; interpreting it as a literal marginal offering cost would additionally require ruling out omitted constant benefits.

The hospital payoff coefficients are identified in offer-index units because the scale of the logistic offer shock is normalized. The direct-variable savings coefficient provides a dollar normalization. Ratios to β_{Π} express other offer-index components in profit-equivalent units when combined with a \$1,000 scaling conversion. The duration coefficient is identified from variation in expected remaining B&M days conditional on the other payoff inputs. The hospital-specific capacity coefficients use within-hospital variation in occupancy and staffed capacity interacting with that duration, separately from the duration effect itself. The interpretation is conditional on the cost and duration predictions, the functional form of the capacity terms, and the included clinical and operational controls. The resulting capacity contribution is composite, it does not separately identify the financial return from admitting another patient. Normalization and centering affect component levels, while changes in capacity value between two states are invariant to additive centering constants.

5.7 Dynamic Model Counterpart

Appendix E develops and estimates the dynamic counterpart to the static specification. Both use the same payoff specification and distinguish offer-stage consequences from those contingent on acceptance. The static model already values expected discounted profits and capacity exposure over the remaining stay, evaluating capacity at the current operating state; the dynamic version additionally accounts for how today's transfer changes future operating states and subsequent offer decisions. The static specification permits joint maximum likelihood estimation, which is asymptotically efficient under correct specification and standard regularity conditions (Newey and McFadden, 1994). The dynamic implementation uses two-step GMM, whose efficiency depends on the moments and their weighting (Hansen, 1982); appropriately constructed two-step estimators can attain the efficiency of maximum likelihood (Aguirregabiria et al., 2021). The practical ad-

vantage of the static baseline is that it avoids the additional transition estimation and continuation-value approximations required by the dynamic implementation. The dynamic estimates indicate that continuation values contribute little to the hospital's offer incentives, providing further support for the static approximation. Unless otherwise explicitly stated, the main-text estimates and counterfactuals therefore refer to the static joint model.

6 Estimation and Results

6.1 Patient Acceptance

The panel nature of the model allows repeated choices to reveal both persistent differences in willingness to accept HaH and dependence on prior interactions. The estimated preference standard deviation indicates unobserved heterogeneity across patients, while completing a first HaH treatment is positively associated with subsequent acceptance even after accounting for these persistent differences. There is no clear evidence of further increases with additional acceptances, suggesting an initial treatment-associated shift without cumulative reinforcement. Prior expected rejections remain negatively associated with acceptance beyond the information they provide through posterior updating about patient preferences.

Appendix Tables A.1 and A.2 compare persistent heterogeneity alone with specifications adding previously completed HaH treatment, prior acceptance and rejection counts, or both. The preferred specification in Table 2 includes all three components. The estimates indicate both persistent differences in willingness to accept and a positive association with first treatment, without clear evidence of further increases from additional acceptances. The main supply estimates remain broadly stable across these specifications.

This distinction matters because expanding HaH requires hospitals to secure patient acceptance as well as identify clinically suitable transfers. Persistent preference differences suggest a role for targeting offers toward patients already willing to receive care at home. Prior decisions can help hospitals identify these patients and direct outreach more effectively. The first-treatment association raises a different possibility: willingness to accept may also change after experiencing HaH. If that association reflects an effect of treatment, facilitating a suitable patient's first use could increase acceptance at subsequent encounters, extending the benefit of outreach beyond the initial placement. Information about what treatment involves or support with an initial transition could therefore complement targeting based on existing preferences. Taken together, the estimates suggest that repeat acceptance reflects both persistent preferences and an additional shift associated with initial treatment. This provides a basis for targeting patients already inclined toward HaH, alongside suggestive evidence that facilitating suitable patients' first use could strengthen subse-

quent willingness to accept. The absence of clear gains from additional acceptances points toward an initial-exposure effect rather than cumulative reinforcement.

Conditional acceptance utility is lower for Black patients than for White patients and lower for Medicaid and Medicare patients than for commercially insured patients. Acceptance utility also declines with the neighborhood share of residents below 150 percent of the federal poverty line. Distance to the hospital, sex, household size, and neighborhood educational attainment are not statistically distinguishable from zero in the joint model. The demographic associations may reflect differences in household support, housing conditions, or preferences for home care, but the base-line coefficients do not separately identify these channels. Appendix D provides supplementary evidence from patient and caregiver responses.

The control-function coefficient is negative and precisely estimated. Since $1 - p_i$ is larger when the latent offer propensity is lower, the estimate implies that relatively unlikely offers are associated with lower patient acceptance utility after conditioning on observed covariates and persistent preference type. The coefficient is evidence against setting the control-function term to zero within the maintained specification. It should not be read as an independently estimated correlation between primitive patient and hospital shocks.

6.2 Hospital Offers

Table 3 reports selected estimates from the hospital offer equation. The coefficients describe the relative importance of financial returns, acceptance, capacity relief, and clinical concerns in the offer index. Their economic magnitudes depend on the units of the regressors and, for acceptance-contingent inputs, the probability of transfer.

Financial returns and expected acceptance. The direct-variable profit coefficient is 0.453 per \$1,000. At the fixed reference acceptance belief of approximately 0.411, an additional \$1,000 of profit conditional on an accepted transfer raises the offer index by about 0.186, equivalent to approximately 20% higher offer odds when the other components are held fixed. This however is a local financial incentive, so it does not imply that HaH has positive absolute accounting returns or lower total episode costs.

The standalone acceptance-belief coefficient is 2.898. A ten-percentage-point increase in expected acceptance raises this direct component of the offer index by 0.290, or approximately 34% higher offer odds. At the reference belief, that direct contribution is equivalent to about \$1,560 in additional profit conditional on an accepted transfer. The full effect of a belief change also scales the financial, capacity, and clinical consequences in $p_i \Delta W_i$. The estimates therefore indicate substantial targeting toward patients expected to accept, beyond the mechanical weighting of mea-

TABLE 2: Patient Acceptance Estimates

	Estimate	Standard error
Asian	0.303***	(0.087)
Black/African American	−0.088**	(0.038)
Latino/Hispanic	0.301***	(0.058)
Medicaid	−0.117*	(0.067)
Medicare	−0.184***	(0.053)
Home-to-hospital distance	0.012	(0.013)
Age	0.101***	(0.019)
Male	0.021	(0.024)
Neighborhood average household size	0.004	(0.014)
Neighborhood college education share	0.032	(0.019)
Neighborhood log(per-capita income)	−0.031	(0.023)
Neighborhood share below 150% FPL	−0.043**	(0.017)
Prior expected HaH declinations	−0.294***	(0.066)
Ever previously treated with HaH	0.263***	(0.080)
Count of prior HaH acceptances	−0.037	(0.071)
Condition severity weight	−0.097***	(0.028)
Generalized offer residual, $1 - p_i$	−3.272***	(0.204)
Persistent HaH preference SD	0.466***	(0.165)
Condition fixed effects	Yes	
Hospital-by-quarter fixed effects	Yes	
Assessments	31,944	
Patients	21,354	

Notes: Age, condition weight, home-to-hospital distance, household size, neighborhood household size, college share, log per-capita income, and neighborhood poverty share use fixed reference means and standard deviations. Prior acceptances and expected declinations are raw cumulative counts. Previously treated with HaH indicates a distinct prior HaH-treated encounter from this specific health system. Indicators and $1 - p_i$ are unstandardized. Omitted categories are White for race and Commercial for payor. The specification also includes condition and hospital-by-quarter fixed effects, and “other” race and payor categories. Standard errors are conditional on the generated inputs and use the observed information matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

sured placement consequences. This is consistent with a mission-related benefit from successful service provision, but should not be interpreted as a direct weight on patient surplus.

Clinical concerns and offering frictions. Care burden, acute severity, and HaH escalation risk reduce offer value. Because escalation risk enters conditional on acceptance, a ten-percentage-point increase in risk changes the index by approximately $0.411(-1.2915)(0.10) \approx -0.053$ at the

reference belief, corresponding to about 5% lower offer odds. Clinical burden and severity enter directly and are measured in standardized units.

The HaH-strain coefficient is positive and statistically significant ($p = 0.017$), but economically small over the observed range. An increase in the uncentered strain measure from approximately 2 to 6 raises the index by only $0.411(0.0032)(6 - 2) \approx .004$, or about 0.4% higher odds. This would constitute a tripling in the ratio of HaH patients to active staff, yet would lead to negligible change in HaH offer behavior. This therefore provides little evidence that the measured strain term materially discourages offers within the operating range represented in the data. However, it does not establish that staffing constraints would remain negligible under substantial program expansion.

Strictly prior expected offers predict additional offers, consistent with repeated outreach, familiarity, and persistent targeting. The negative common offer term captures a baseline friction in initiating HaH offers, consistent with the staff time and coordination required to assess suitability, contact patients and caregivers, and arrange a potential transfer. These efforts begin before acceptance is known, so the hospital weighs them against the expected benefits of an accepted placement when deciding whom to offer HaH.

Capacity pressure. The hospital-specific linear B&M relief coefficients are positive at all four hospitals and statistically significant at Hospitals A, B, and C; the estimate for Hospital D is not statistically distinguishable from zero. The separate B&M exposure coefficient is negative, distinguishing the baseline association with remaining treatment duration from the value of capacity relief as occupancy rises. Holding other inputs fixed, higher occupancy increases the estimated capacity contribution at each hospital. The central finding in profit-equivalent terms is described below.

6.3 Profit-Equivalent Capacity Value

The profit coefficient permits the capacity-relief component to be expressed in profit-equivalent units. For assessment i , the model-implied gross private value of B&M capacity released by an accepted HaH placement is

$$\widehat{CV}_i^B = \frac{1,000}{\widehat{\beta}_\Pi} \widehat{\lambda}_{B,b(i)} E_i^B \frac{2N_i^B - 1}{K_i}. \quad (29)$$

This measure values the focal patient's expected discounted B&M bed-days at the hospital's current occupancy and staffed capacity. Because both capacity relief and accounting profits enter conditional on acceptance, their common acceptance multiplier cancels in the coefficient ratio. Multiplying $\widehat{CV}_i^B$ by ρ_i gives the expected capacity contribution per offer.

TABLE 3: Hospital Offer Estimates

	Estimate	Standard error
Accounting profits (per \$1,000)	0.453***	(0.093)
HaH program strain	0.003**	(0.001)
HaH escalation risk	−1.292*	(0.707)
Care-burden	−0.182***	(0.013)
Acute-severity	−0.080***	(0.012)
B&M stay-duration exposure	−0.775***	(0.103)
Hospital A released-capacity	0.301***	(0.046)
Hospital B released-capacity	0.209***	(0.047)
Hospital C released-capacity	0.191***	(0.046)
Hospital D released-capacity	0.065	(0.048)
Strictly prior expected offers	0.298***	(0.027)
Acceptance belief	2.898***	(0.189)
Baseline net offer term	−0.717***	(0.079)
Supply-condition fixed effects	Yes	
Hospital fixed effects	Yes	
Department fixed effects	Yes	
Assessments	31,944	

Notes: All reported coefficients enter the offer equation additively. Acceptance-contingent variables are multiplied by ρ_i . Direct-variable costs are centered within hospital-quarter-condition cells and expressed in \$1,000 units. B&M relief, B&M stay-duration exposure, HaH strain, and escalation risk subtract fixed global reference means without rescaling. The stand-alone acceptance belief term uses centered $\rho_i - \bar{\rho}$; acceptance-contingent multipliers use uncentered ρ_i . Care burden and acute severity are standardized indices with zero-imputed underlying standardized scores and separate missingness flags. Prior expected offers are raw cumulative counts. The specification includes supply-condition, hospital, and department fixed effects as well as indicators for missing clinical measures. Standard errors are conditional on the generated inputs and use the observed information matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The point estimates imply positive capacity value at all four hospitals over observed operating conditions, although Hospital D's relief coefficient is not statistically distinguishable from zero. Dividing equation (29) by E_i^B gives the implied value per discounted bed-day released. At 90 percent occupancy, these values are approximately \$1,200, \$830, \$760, and \$260 at Hospitals A through D, respectively. These are illustrative valuations at a common occupancy level, rather than sample averages. Holding staffing and other inputs fixed, a ten-percentage-point increase in occupancy raises this value per discounted bed-day by approximately \$133, \$92, \$85, and \$29, respectively.

TABLE 4: Observed and Predicted Outcome Rates

Outcome	Empirical	Predicted	Difference
Accepted HaH placement	0.1028	0.1037	0.0009
Documented rejection	0.0903	0.0887	−0.0016
No observed offer	0.8069	0.8076	0.0007
Latent offer, fitted capture-adjusted	0.2213	0.2187	−0.0026

Notes: The sample contains 31,944 assessments from 21,354 patients. Empirical rates are observed frequencies for the first three rows and a separately reconstructed latent-offer rate for the fourth (Appendix C.1). Predicted rates average $p_i \rho_i$, $p_i(1 - \rho_i)q_i$, $1 - p_i + p_i(1 - \rho_i)(1 - q_i)$, and p_i , respectively. Differences are predicted minus empirical rates, calculated before rounding.

6.4 Model Fit

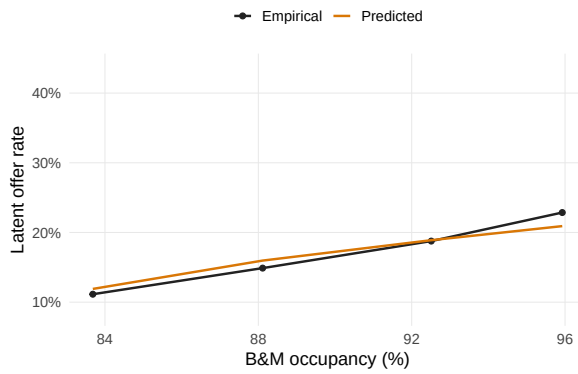
Table 4 shows that the model closely reproduces the aggregate frequencies of accepted placements, documented rejections, and assessments with no observed offer. Predicted latent offers also closely match the separate documentation-adjusted empirical estimate. The figures assess whether this aggregate fit extends across hospital occupancy, calendar time, and predicted probabilities. All comparisons are in sample, with patient histories evaluated using strictly prior outcomes.

Figure 9 compares empirical and predicted offer rates across occupancy bins. The model captures the increasing offer profiles at the large hospitals, especially Hospital B, and the flatter profiles at the satellite facilities. It nevertheless overpredicts offers in the lowest occupancy bins and tends to underpredict at high occupancy. These discrepancies remain relevant when interpreting the strength of the estimated capacity response.

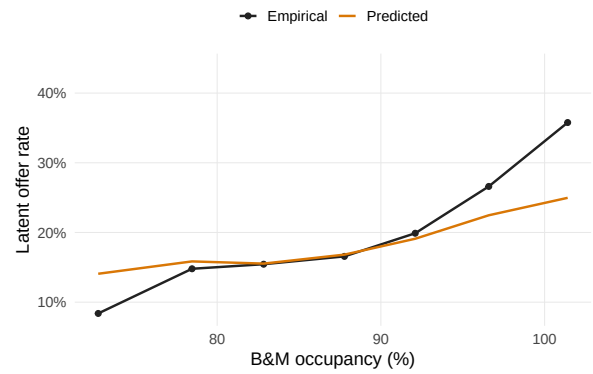
Figure 10 compares quarterly rates. Predicted offers track much of the broad variation over time, although some quarter-specific movements remain unexplained. Hospital D’s unusually high initial empirical rate should be interpreted in light of the number of assessments underlying that estimate.

Figure 11 shows close agreement between empirical frequencies and predicted probabilities across prediction bins. The accepted-placement panel evaluates the joint probability of an offer and acceptance, $p_i \rho_i$. The remaining panels assess documented rejections, no observed offers, and latent offers. Overall, the model captures aggregate outcomes and broad patterns of selection, while the occupancy comparisons identify where its representation of hospital responses is less precise.

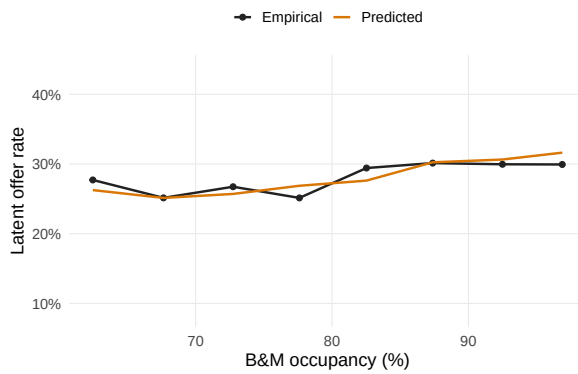
Figure 9: Empirical and Predicted Offer Rates by B&M Occupancy



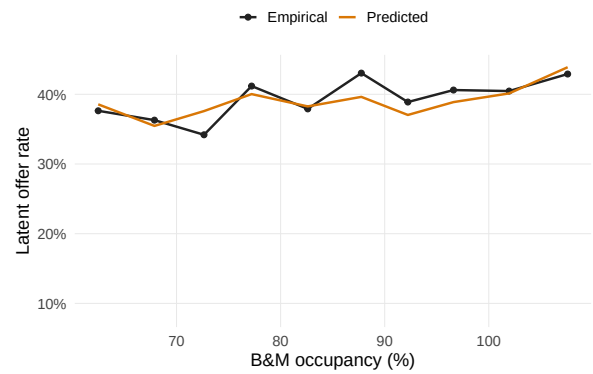
(a) Hospital A



(b) Hospital B

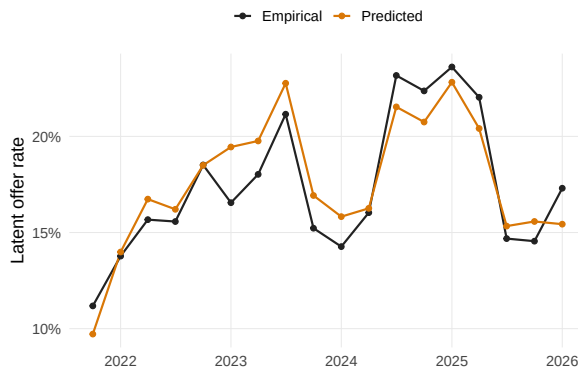


(c) Hospital C

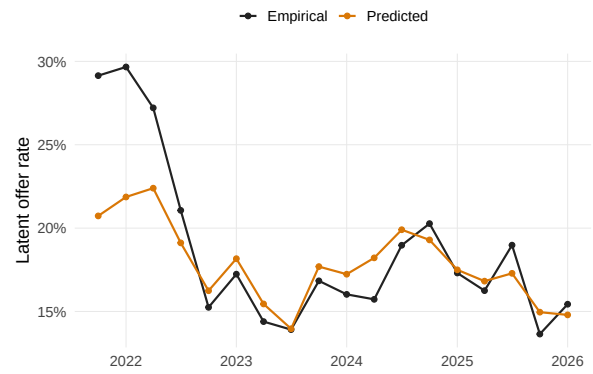


(d) Hospital D

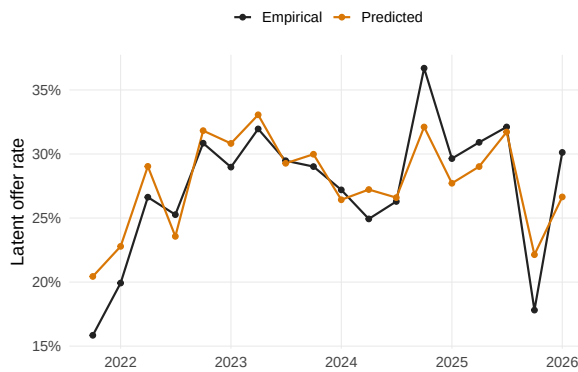
Figure 10: Empirical and Predicted Offer Rates by Quarter



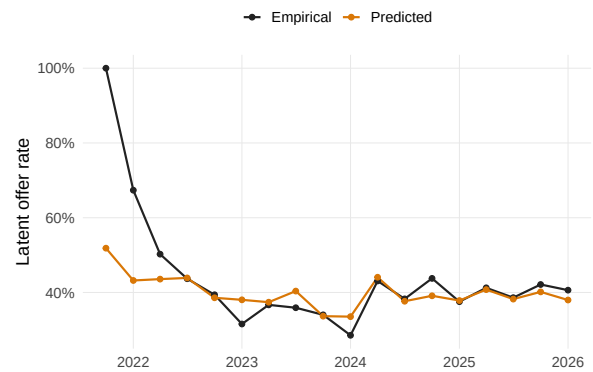
(a) Hospital A



(b) Hospital B

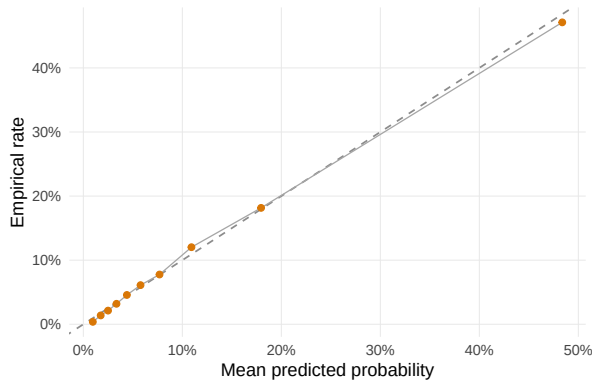


(c) Hospital C

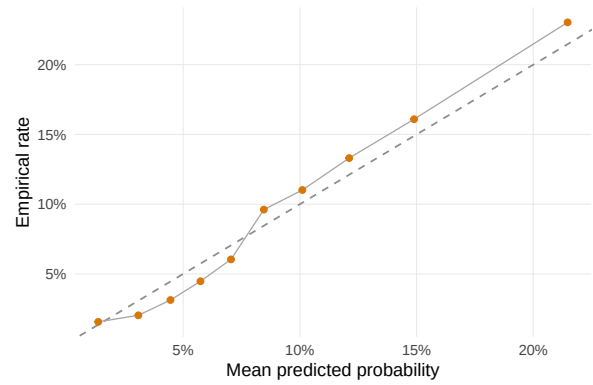


(d) Hospital D

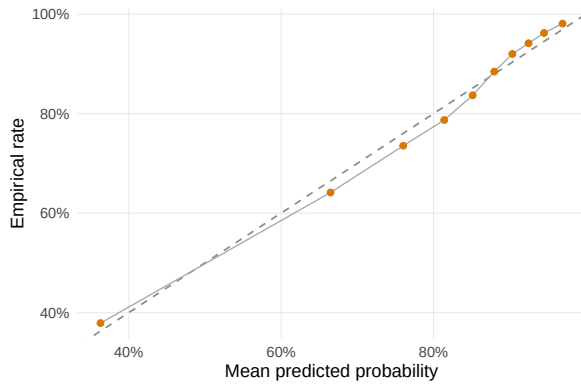
Figure 11: Calibration of Observed Outcomes and True Offers



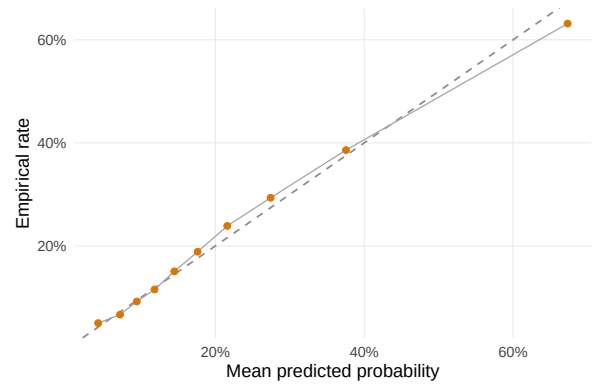
(a) Acceptances



(b) Documented rejections



(c) No observed offers



(d) Latent offers

7 Counterfactual Analysis

The counterfactual analysis examines how provider payment, geographic eligibility, capacity pressure, and information about patient demand affect offers and accepted transfers. I hold the estimated demand and supply parameters fixed, modify the affected patient or hospital inputs, and evaluate the offer policy in the new environment. Except in the information comparison described below, the joint probabilities solve equations (22) and (23).

Outcome accounting. Let $\mathcal{A}$ denote the set of assessment opportunities being evaluated. The mean true-offer rate is $|\mathcal{A}|^{-1} \sum_{i \in \mathcal{A}} p_i$, expected accepted placements are $\sum_{i \in \mathcal{A}} p_i \rho_i$, and acceptance among offers is $\sum_i p_i \rho_i / \sum_i p_i$. The latter weights acceptance by the probability that each patient receives an offer. Let T_i^B denote remaining B&M duration at assessment. Mean potential B&M bed-days released per assessment are

$$\frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} p_i \rho_i \mathbb{E}(T_i^B | \mathcal{J}_i^O). \quad (30)$$

These are undiscounted days of focal B&M occupancy avoided under the model. They are not additional admissions or a prediction that beds remain empty. HaH exposure is calculated analogously using the remaining HaH duration.

Patient option value. Define the node-specific deterministic acceptance utility

$$v_{ig}(p_i) = X_i' \beta_U - \alpha P_i + \sigma_\xi \xi_g + \varphi(1 - p_i).$$

The normalized option value conditional on an offer is

$$Q_i(p_i) = \sum_g w_{ig}^- \log[1 + \exp\{v_{ig}(p_i)\}]. \quad (31)$$

Expected option value created by access is $p_i Q_i(p_i)$. No further acceptance multiplier is needed because the inclusive value already accounts for the option to decline. I report a normalized option value separately from hospital returns and bed-days, where the welfare interpretation is conditional on the demand specification, including the treatment of the selection correction residual. All reference means and standard deviations remain fixed at their estimation values. Counterfactual profit, capacity, duration, and strain inputs are transformed using those same references rather than re-centered after the shock.

A change in supply can affect acceptance among offers through two channels. The offer com-

position changes even when individual preferences remain fixed. In addition, the maintained control-function specification changes individual predicted acceptance when the offer probability changes. I distinguish the full fixed-point response from a comparison holding $\varphi(1 - p_i)$ at its baseline value. This separates selection feedback in the fitted model from reallocation of offers at fixed adjusted utilities.

7.1 Provider Payment

There has been considerable discussion amongst policymakers surrounding the appropriate level of reimbursement for HaH care, with some insurers advocating for and negotiating towards lower than parity reimbursement levels. Therefore, the first experiment scales HaH reimbursement from 90% to 110% of the admission-specific payment received by the hospital, holding B&M reimbursement fixed. Under baseline payment parity, let R_i denote encounter i 's observed total-episode reimbursement under either treatment location. A payment multiplier $m_P \in [0.90, 1.10]$ changes reimbursement for an accepted HaH placement to $m_P R_i$. The counterfactual profit input is therefore

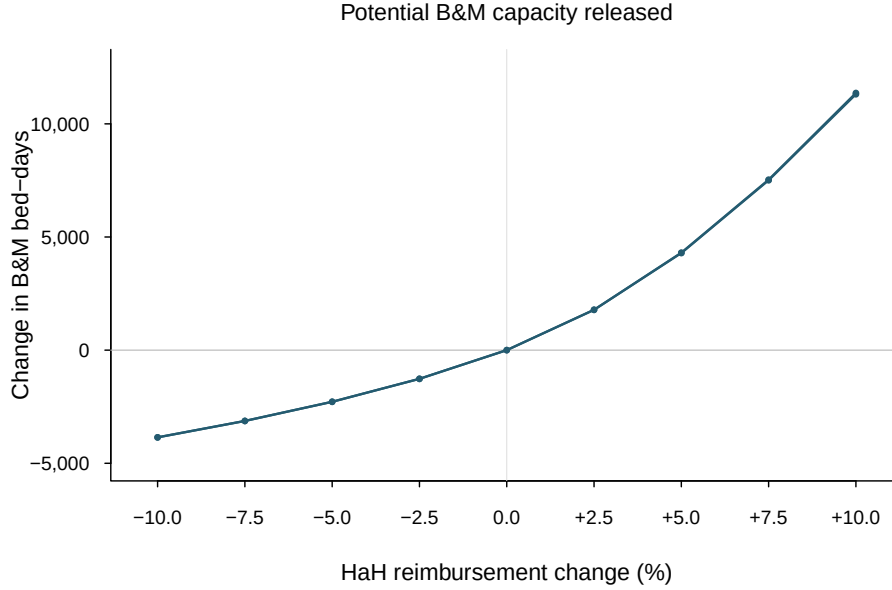
$$\Delta\Pi_i(m_P) = \Delta\Pi_i + \frac{(m_P - 1)R_i}{1,000}, \quad (32)$$

where $m_P = 1$ reproduces the baseline and the estimation centering reference remains fixed. The adjustment applies to encounters eligible for HaH reimbursement at admission. This experiment changes provider returns without directly changing patient prices or other observed demand characteristics.

Figure 12 shows that higher HaH reimbursement increases potential B&M bed-days released. The response is asymmetric: the increase at positive payment changes is larger than the reduction at equally sized negative changes. This is a response of the fitted offer-and-acceptance system, not simply the profit coefficient multiplied by a payment change. Across the 31,944 assessment opportunities for 21,354 patients, baseline potential release is 14,587 bed-days. A 5% reimbursement increase adds 4,289–4,307 days, while a 10% increase adds 11,307–11,364 days. These totals sum conditional assessment-level comparisons with factual histories, preference weights, staffing, and census fixed. The offer and acceptance probabilities and the control function adjust jointly. The narrow bed-day ranges reflect alternative fixed-point solutions rather than sampling uncertainty.

To value the additional throughput that released capacity could support, I apply hospital-specific contribution margins from the observed backfill population. For hospital b , let $\Delta BD_b(m_P)$ denote additional bed-days released, f_b the fraction filled, and c_b contribution dollars per occupied

Figure 12: Provider Reimbursement and Potential B&M Bed-Days Released



day. Potential additional hospital contribution is

$$\Delta V^{\$}(m_P) = \sum_{b \in \mathcal{B}} \Delta BD_b(m_P) f_b c_b. \quad (33)$$

Under persistent excess demand, I assume that 50–100% of released bed-days are filled, allowing for delays, staffing constraints, and bed compatibility. Using historical hospital contribution margins to value this backfill, a 10% reimbursement increase generates \$7.4–\$20.4 million in potential additional contribution across accounting definitions. Additional HaH payments total approximately \$11.4 million, including \$4.14 million for baseline placements, while induced transfers raise direct-variable treatment costs by approximately \$14.9 million. The resulting \$3.5 million reduction in focal contribution leaves a combined hospital cash gain of \$4.0–\$16.9 million. Table 5 separately reports \$9.5–\$9.6 million in incremental profit-equivalent occupancy relief, calculated using equation (29). At a 10% reimbursement increase, the approximately \$9.5 million profit-equivalent occupancy-relief value is substantial when compared with the \$7.4–\$20.4 million accounting profits range, while the \$3.5 million focal contribution loss offsets roughly 17–47% of these potential backfill gains.

TABLE 5: Reimbursement Increases, Released Capacity, and Hospital Contribution

HaH reimbursement increase	5%	10%
Additional B&M bed-days	[4,289, 4,307]	[11,307, 11,364]
<i>Hospital contribution (\$ millions)</i>		
(1) Potential backfill contribution	[2.80, 7.67]	[7.44, 20.38]
(2) Focal contribution: (2.1)+(2.2)+(2.3)	−2.16	[−3.49, −3.46]
(2.1) Induced transfer cost effect	[−5.83, −5.81]	[−14.94, −14.86]
(2.2) Extra reimbursement: baseline placements	2.07	4.14
(2.3) Extra reimbursement: induced placements	[1.58, 1.60]	[7.24, 7.34]
(3) Combined cash contribution: (1)+(2)	[0.64, 5.51]	[3.95, 16.92]
<i>Structural profit equivalents (\$ millions)</i>		
B&M occupancy relief	[3.61, 3.63]	[9.50, 9.55]

Notes: All changes are relative to baseline reimbursement over the full assessment cohort, in February 2026 U.S. dollars (\$). Backfill ranges combine 50–100% fill with revenue including or excluding estimated settlements and direct or all reported variable costs. Fixed overhead is excluded. Focal contribution uses undiscounted remaining direct-variable cost differences and extra reimbursement on both baseline and induced HaH placements. Combined cash endpoints use coherent assessment-level fixed-point choices. Structural values use capacity and duration inputs with the model’s $\delta = 0.99$ discount factor and are not added to cash contribution. Ranges describe scenarios and fixed-point solutions, not confidence intervals or annual forecasts.

7.2 Geographic Eligibility

Baseline geographic eligibility follows the operational ZIP-code list described in Appendix B; it is not identical to a reconstructed 30-minute isochrone around the admitting hospital. Let G_i^0 denote eligibility under that list and let d_i denote home-to-hospital travel time under the counterfactual travel-time measure. To evaluate an expansion without removing currently eligible patients, define

$$\tilde{\chi}_i^H(\tilde{d}) = \mathbb{1}\{\text{condition eligible}\} \mathbb{1}\{\text{payor eligible}\} \mathbb{1}\{G_i^0 = 1 \vee d_i \leq \tilde{d}\}, \quad \tilde{d} \in \{45, 60\}. \quad (34)$$

Newly eligible patients are those outside the operational baseline area who satisfy the expanded rule. Their observed covariates enter the estimated demand and supply functions.

The exercise distinguishes responses conditional on assessment from changes in program volume. Population effects additionally require assumptions about how newly eligible patients enter screening and assessment. The baseline extension applies observed condition- and hospital-specific assessment rates to the added population, with alternative rates used to assess sensitivity. Longer travel can also raise delivery costs and staffing exposure. Outcomes include the composition of assessments, offers, accepted placements, potential bed-days released, and patient option

value by distance and demographic group.

7.3 Hospital Capacity Stress

A capacity-stress environment combines higher B&M occupancy, fewer staffed beds, and a shift in overall patient composition towards infectious respiratory conditions. For hospital b , the altered state is

$$\tilde{N}_i^B = N_i^B + \Delta N_{b(i)}, \quad \tilde{K}_i = m_{K,b(i)} K_i, \quad 0 < m_{K,b(i)} \leq 1. \quad (35)$$

I reconstruct the hospital-specific capacity inputs M_{ib}^B at this state using the original centering references. A case-mix change additionally alters the distribution of assessed patients and their predicted remaining B&M and HaH exposure. Examining occupancy, staffed capacity, and case mix separately before combining them distinguishes the sources of the response.

7.4 Information About Patient Demand

The information experiment changes how the hospital uses knowledge of persistent patient preferences while holding underlying patient types and experienced treatment histories fixed. It compares pooled, baseline, and perfect knowledge of the persistent effect. The hospital need not observe the transitory patient choice shock even under perfect knowledge of persistent type.

To isolate the information channel, hold the adjusted utility term $\varphi(1 - p_i^0)$ at its baseline value, where p_i^0 is the baseline offer probability. Define

$$v_{ig}^0 = X_i' \beta_U + \sigma_\xi \xi_g + \varphi(1 - p_i^0), \quad a_{ig}^0 = \Lambda(v_{ig}^0).$$

Let w_{ig}^r denote the weights used by the hospital under regime r . Its acceptance belief and perceived option value are

$$\rho_i^r = \sum_g w_{ig}^r a_{ig}^0, \quad (36)$$

$$Q_i^r = \sum_g w_{ig}^r \log[1 + \exp(v_{ig}^0)]. \quad (37)$$

The current supply specification uses ρ_i^r , while Q_i^r describes perceived patient benefit through the inclusive value of introducing HaH to the patient's choice set.

The pooled benchmark uses the population quadrature weights w_g for the persistent effect. The baseline uses the predictive weights w_{ig}^- . Under perfect information, the hospital observes the persistent type and its weights are concentrated on that type. The direct prior-offer term and the

history variables in patient utility remain at their baseline values. Thus pooling restricts the use of history for predicting the persistent effect, and it does not erase patients' actual experiences or remove the direct costs and benefits associated with previous interactions.

Crucially, hospital beliefs and the distribution used to evaluate realized outcomes are separate. In the pooled benchmark, the hospital chooses p_i^C using pooled beliefs, while expected accepted placements for the observed assessment are

$$p_i^C \sum_g w_{ig}^- a_{ig}^0.$$

Under perfect information, I solve a type-specific offer probability p_{ig}^P for each possible persistent type. Expected accepted placements are then

$$\sum_g w_{ig}^- p_{ig}^P a_{ig}^0,$$

and expected patient option value is

$$\sum_g w_{ig}^- p_{ig}^P \log[1 + \exp(v_{ig}^0)].$$

The common evaluation weights preserve the underlying population across regimes, such that information changes targeting, not the distribution of patient preferences.

This comparison evaluates how patient-specific information helps hospitals translate HaH offers into realized capacity relief. Changes in offer volume capture whether better information encourages broader use of HaH, while changes in the allocation of offers capture more selective targeting toward patients likely to accept. The resulting changes in accepted placements and released bed-days measure the operational consequences of anticipating patient willingness to receive at-home care. The policy implication is that expanding effective capacity depends not only on treatment resources and reimbursement, but also on how well hospitals use available information to identify patients willing to receive care at home.

8 Conclusion

Hospital-at-Home provides a flexible margin of inpatient capacity whose deployment depends on hospital selection and patient adoption. Hospitals determine which patients receive offers, while patients determine whether those offers translate into treatment at home and released brick-and-mortar capacity. Using clinical, accounting, census, and staffing data from four hospitals, supplemented by a novel patient survey, this paper models these decisions jointly and examines hospital

offer incentives in both static and dynamic frameworks.

The estimates show that patient willingness and hospital targeting jointly shape effective capacity. Preferences for home treatment vary persistently across patients, and first HaH treatment remains positively associated with subsequent acceptance. Survey evidence connects willingness to accept with expectations about care quality and experience. Hospitals target likely adopters while balancing financial returns, clinical needs, and the value of released capacity. The model-implied private shadow value of capacity rises with occupancy, with economically substantial gross capacity benefits at the two largest hospitals. Together, these findings show that the capacity contribution of HaH depends on matching patients willing to receive home care with transfers that are valuable under prevailing hospital conditions.

The reimbursement counterfactual illustrates how capacity relief can motivate expansion even without treatment-cost savings. A 10 percent increase in HaH reimbursement generates approximately 11,300 additional potential B&M bed-days across the observed assessment cohort, holding patient histories and operating states fixed. Predicted treatment costs rise by more than additional reimbursement, but potential backfill produces a combined hospital cash gain of \$4.0–\$16.9 million under the stated accounting and utilization scenarios. The financial incentive to expand HaH therefore depends on the opportunities created by released capacity as well as the returns from treating the transferred patient.

These findings make selection and adoption central to evaluating policies affecting reimbursement, eligibility, and information. Expanding access creates opportunities for capacity relief, while patient acceptance and the subsequent use of released beds determine the gains realized. Assessing efficiency requires distinguishing hospital financial returns from patient benefits, household costs, and the consequences of capacity availability for other patients. More broadly, the paper provides a framework for understanding flexible supply technologies whose deployment requires consumer adoption. Their value depends on the joint distribution of gains from reallocating production, consumers' willingness to adopt, and the shadow value of conventional capacity.

References

- Adams, Danielle, Ashby J. Wolfe, Jessica Warren, Alexandre Laberge, Adam C. Richards, Kurt Herzer, and Lee A. Fleisher**, “Initial Findings From an Acute Hospital Care at Home Waiver Initiative,” *JAMA Health Forum*, 2023, 4 (11).
- Adelino, Manuel, Katharina Lewellen, and Anant Sundaram**, “Investment Decisions of Non-profit Firms: Evidence from Hospitals,” *Journal of Finance*, 2015, 70 (4), 1583–1628.
- Aguirregabiria, Victor, Allan Collard-Wexler, and Stephen P. Ryan**, “Dynamic Games in Empirical Industrial Organization,” 2021, 4, 225–343.
- **and Pedro Mira**, “Dynamic Discrete Choice Structural Models: A Survey,” *Journal of Econometrics*, 2010, 156 (1), 38–67.
- Allcott, Hunt and Nathan Wozny**, “Gasoline Prices, Fuel Economy, and the Energy Paradox,” *The Review of Economics and Statistics*, December 2014, 96 (5), 779–795.
- Applebaum, Eileen and Batt Rosemary**, “The New Hospital at Home Movement: Opportunity or Threat for Patient Care?,” Technical Report, Center for Economic and Policy Research 2023.
- Arcidiacono, Peter and Paul L. Ellickson**, “Practical Methods for Estimation of Dynamic Discrete Choice Models,” *Annual Review of Economics*, September 2011, 3, 363–394.
- **and Robert A. Miller**, “Conditional Choice Probability Estimation of Dynamic Discrete Choice Models with Unobserved Heterogeneity,” *Econometrica*, November 2011, 79 (6), 1823–1868.
- Armony, Mor, Shlomo Israelit, Avishai Mandelbaum, Yariv N. Marmor, Yulia Tseytlin, and Galit B. Yom-Tov**, “On Patient Flow in Hospitals: A Data-Based Queueing-Science Perspective,” *Stochastic Systems*, 2015, 5 (1), 146–194.
- Arsenault-Lapierre, G., M. Henein, D. Gaid, M. Le Berre, G. Gore, and I. Vedel**, “Hospital-at-Home Interventions vs In-Hospital Stay for Patients With Chronic Disease Who Present to the Emergency Department: A Systematic Review and Metaanalysis,” *JAMA Network Open*, 2021, 4 (6).
- Bagust, Adrian, Michael Place, and John W. Posnett**, “Dynamics of Bed Use in Accommodating Emergency Admissions: Stochastic Simulation Model,” *BMJ*, 1999, 319 (7203), 155–158.
- Ben-Akiva, M., M. Bradley, T. Morikawa, J. Benjamin, T. Novak, H. Oppewal, and V. Rao**, “Combining Revealed and Stated Preferences Data,” *Marketing Letters*, 1994, 5 (4), 335–349.

- Brekke, Kurt R., Luigi Siciliani, and Odd Rune Straume**, “Hospital Competition and Quality with Regulated Prices,” *Scandinavian Journal of Economics*, 2011, 113 (2), 444–469.
- Brody, Abraham A., Alicia I. Arbaje, Linda V. DeCherrie, Albert D. Federman, Bruce Leff, and Albert L. Siu**, “Starting Up a Hospital at Home Program: Facilitators and Barriers to Implementation,” *Journal of the American Geriatrics Society*, 2019, 67 (3), 588–595.
- Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik**, “Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs,” *Econometrica*, 2014, 82 (6), 2295–2326.
- Caplan, Gideon A., Nur S. Sulaiman, Dee A. Mangin, Nicoletta Aimonino Ricauda, Andrew D. Wilson, and Louise Barclay**, “A meta-analysis of hospital in the home,” *Medical Journal of Australia*, 2012, 197 (9), 512–519.
- Capps, Cory, David Dranove, and Mark Satterthwaite**, “Competition and Market Power in Option Demand Markets,” *RAND Journal of Economics*, 2003, 34 (4), 737–763.
- Capron, Andrew, Brian McManus, Bailey Beauclair, Escher Howard-Williams, Carlton Moore, and Patrick O’Shea**, “Patient Outcomes With Hospital-at-Home in Routine Care: Evidence from a Spatial Eligibility Boundary,” 2026. Working paper.
- Carey, Kathleen**, “Stochastic Demand for Hospitals and Optimizing “Excess” Bed Capacity,” *Journal of Regulatory Economics*, 1998, 14 (2), 165–188.
- Centers for Medicare & Medicaid Services**, “CMS Announces Comprehensive Strategy to Enhance Hospital Capacity Amid COVID-19 Surge,” Technical Report, United States Department of Health and Human Services 11 2020.
- , “Report on the Study of the Acute Hospital Care at Home Initiative,” Technical Report, United States Department of Health and Human Services 2024.
- , “Acute Hospital Care at Home,” QualityNet 2026. Acute Hospital Care at Home waiver effective through September 30, 2030.
- Chiu, Kevin**, “The impact of certificate of need laws on heart attack mortality: Evidence from county borders,” *Journal of Health Economics*, 2021, 79, 102518.
- Courtemanche, Charles J. and Joseph Garuccio**, “How Do Certificate-of-Need Laws Affect Hospitals? A Review of the Evidence,” NBER Working Paper 34026, National Bureau of Economic Research 2025.

- Dube, Jean-Pierre, Gunter J. Hitsch, and Peter E. Rossi**, “State Dependence and Alternative Explanations for Consumer Inertia,” *RAND Journal of Economics*, 2010, 41 (3), 417–445.
- Ellis, Randall P. and Thomas G. McGuire**, “Provider Behavior Under Prospective Reimbursement: Cost Sharing and Supply,” *Journal of Health Economics*, 1986, 5 (2), 129–151.
- Ferrario, Beatrice and Stefanie Stantcheva**, “How Americans Think About Health Care and Insurance,” September 2025. Social Economics Lab, Harvard University Working Paper.
- Francetic, Igor, Rachel Meacock, and Matt Sutton**, “Free-for-All: Does Crowding Impact Outcomes Because Hospital Emergency Departments Do Not Prioritise Effectively?,” *Journal of Health Economics*, 2024, 95, 102881.
- Freedman, Seth**, “Capacity and Utilization in Health Care: The Effect of Empty Beds on Neonatal Intensive Care Admission,” *American Economic Journal: Economic Policy*, 2016, 8 (2), 154–185.
- Fuster, Andreas and Basit Zafar**, “To Buy or Not to Buy: Consumer Constraints in the Housing Market,” *American Economic Review*, May 2016, 106 (5), 636–40.
- Gandhi, Ashvin**, “Picking Your Patients: Selective Admissions in the Nursing Home Industry,” *SSRN Electronic Journal*, 2023, Working Paper.
- Gao, Pengjie, Chang Lee, and Dermot Murphy**, “Good for Your Fiscal Health? The Effect of the Affordable Care Act on Healthcare Borrowing Costs,” *Journal of Financial Economics*, 2022, 145 (2), 464–488.
- Gaynor, Martin and Gerard F. Anderson**, “Uncertain Demand, the Structure of Hospital Costs, and the Cost of Empty Hospital Beds,” *Journal of Health Economics*, 1995, 14 (3), 291–317.
- Goeree, Michelle Sovinsky**, “Limited Information and Advertising in the U.S. Personal Computer Industry,” *Econometrica*, 2008, 76 (5), 1017–1074.
- Gourieroux, Christian, Alain Monfort, Eric Renault, and Alain Trognon**, “Generalised Residuals,” *Journal of Econometrics*, 1987, 34 (1–2), 5–32.
- Gowrisankaran, Gautam, Kelli R. Marquardt, and Robert Town**, “Quality, Selection, and Congestion: Equilibrium Hospital Outcomes During COVID-19,” July 2026, (35426).
- Haaland, Ingar, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart**, “Understanding Economic Behavior Using Open-Ended Survey Data,” *Journal of Economic Literature*, 2025. Forthcoming.

- Hahn, Jinyong, Petra Todd, and Wilbert van der Klaauw**, “Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design,” *Econometrica*, 2001, 69 (1), 201–209.
- Hansen, Lars Peter**, “Large Sample Properties of Generalized Method of Moments Estimators,” *Econometrica*, 1982, 50 (4), 1029–1054.
- Heckman, James J.**, “Heterogeneity and State Dependence,” in Sherwin Rosen, ed., *Studies in Labor Markets*, University of Chicago Press, 1981, pp. 91–140.
- Hill, J. D., J. R. Hampton, and J. R. Mitchell**, “A randomised trial of home-versus-hospital management for patients with suspected myocardial infarction,” *Lancet*, 1978, 1 (8069), 837–841.
- Ho, Kate and Ariel Pakes**, “Hospital Choices, Hospital Prices, and Financial Incentives to Physicians,” *American Economic Review*, 2014, 104 (12), 3841–3884.
- Hotz, V. Joseph and Robert A. Miller**, “Conditional Choice Probabilities and the Estimation of Dynamic Models,” *Review of Economic Studies*, 1993, 60 (3), 497–529.
- , —, **Seth Sanders, and Jeffrey Smith**, “A Simulation Estimator for Dynamic Models of Discrete Choice,” *The Review of Economic Studies*, 1994, 61 (2), 265–289.
- Imbens, Guido W. and Joshua D. Angrist**, “Identification and Estimation of Local Average Treatment Effects,” *Econometrica*, 1994, 62 (2), 467–475.
- Jack, William**, “Purchasing Health Care Services from Providers with Unknown Altruism,” *Journal of Health Economics*, 2005, 24 (1), 73–93.
- Janke, Alexander T. and Arjun K. Venkatesh**, “Understanding and Addressing the US Hospital Bed Shortage: Build, Baby, Build,” *JAMA Network Open*, 2025, 8 (2), e2460652.
- Jeziorski, Przemyslaw and Elena Krasnokutskaya**, “Dynamic auction environment with sub-contracting,” *The RAND Journal of Economics*, 2016, 4 (47), 751–791.
- Jofre-Bonet, Mireia and Martin Pesendorfer**, “Estimation of a Dynamic Auction Game,” *Econometrica*, 2003, 71 (5), 1443–1489.
- Joskow, Paul L.**, “The Effects of Competition and Regulation on Hospital Bed Supply and the Reservation Quality of the Hospital,” *Bell Journal of Economics*, 1980, 11 (2), 421–447.
- Kalouptsi, Myrto**, “Time to Build and Fluctuations in Bulk Shipping,” *American Economic Review*, 2014, 104 (2), 564–608.

- Keele, Luke J. and Rocio Titiunik**, “Geographic Boundaries as Regression Discontinuities,” *Political Analysis*, 2015, 23 (1), 127–155.
- Kleiner, Samuel A.**, “Hospital treatment and patient outcomes: Evidence from capacity constraints,” *Journal of Public Economics*, 2019, 175, 94–118.
- Lee, David S. and Thomas Lemieux**, “Regression Discontinuity Designs in Economics,” *Journal of Economic Literature*, 2010, 48 (2), 281–355.
- Leff, Bruce, Lynda Burton, Scott Mader, Bruce Naughton, Jeffrey Burl, Sharon K. Inouye, William B. Greenough, Susan Guido, Christopher Langston, Kevin Frick, Donald Steinwachs, and John R. Burton**, “Hospital at Home: Feasibility and Outcomes of a Program To Provide Hospital-Level Care at Home for Acutely Ill Older Patients,” *Annals of Internal Medicine*, 2005, 143 (11).
- , —, **Susan Guido, William B. Greenough, Donald Steinwachs, and John R. Burton**, “Home hospital program: a pilot study,” *Journal of American Geriatric Society*, 1999, 47 (6), 697–702.
- Leong, Man Qing, Cher Wee Lim, and Yi Feng Lai**, “Comparison of Hospital-at-Home Models: A Systematic Review of Reviews,” *BMJ Open*, 2021, 11 (1), e043285.
- Leuchter, Ryan K., Benedicte A. Delarmente, Sitaram Vangala, Yusuke Tsugawa, and Catherine A. Sarkisian**, “Health Care Staffing Shortages and Potential National Hospital Bed Shortage,” *JAMA Network Open*, 2025, 8 (2), e2460645.
- Levine, David M., Kei Ouchi, Bonnie Blanchfield, Agustina Saenz, Kimberly Burke, Mary Paz, Keren Diamond, Charles T. Pu, and Jeffrey L. Schnipper**, “Hospital-Level Care at Home for Acutely Ill Adults,” *Annals of Internal Medicine*, 2020, 172 (2), 77–85.
- National Guideline Centre**, *Emergency and Acute Medical Care in Over 16s: Service Delivery and Organisation*, National Institute for Health and Care Excellence, 2018.
- Newey, Whitney K. and Daniel McFadden**, “Large Sample Estimation and Hypothesis Testing,” in Robert F. Engle and Daniel L. McFadden, eds., *Handbook of Econometrics*, Vol. 4, Elsevier, 1994, chapter 36, pp. 2111–2245.
- Pesendorfer, Martin and Philipp Schmidt-Dengler**, “Asymptotic Least Squares Estimators for Dynamic Games,” *The Review of Economic Studies*, 2008, 75 (3), 901–928.
- Petrin, Amil and Kenneth Train**, “A Control Function Approach to Endogeneity in Consumer Choice Models,” *Journal of Marketing Research*, 2010, 47 (1), 3–13.

- Pindyck, Robert S.**, “Irreversible Investment, Capacity Choice, and the Value of the Firm,” *American Economic Review*, 1988, 78 (5), 969–985.
- Pratt, Adrian C and Richard M Wood**, “Addressing overestimation and insensitivity in the 85% target for average bed occupancy,” *International Journal for Quality in Health Care*, 07 2021, 33 (3), mzab100.
- Ryan, Stephen P.**, “The Costs of Environmental Regulation in a Concentrated Industry,” *Econometrica*, May 2012, 80 (3), 1019–1061.
- Saenger, Pamela, Alex D. Federman, Linda V. DeCherrie, Sara Lubetsky, Elisse Catalan, Bruce Leff, and Albert L. Siu**, “Choosing Inpatient vs Home Treatment: Why Patients Accept or Decline Hospital at Home,” *Journal of the American Geriatrics Society*, 2020, 68 (7), 1579–1583.
- Shepperd, S., H. Doll, R. M. Angus, M. J. Clarke, S. Iliffe, L. Kalra, N. A. Ricauda, V. Tibaldi, and A. D. Wilson**, “Avoiding hospital admission through provision of hospital care at home: a systematic review and meta-analysis of individual patient data,” *Canadian Medical Association Journal*, 2009, 180 (2), 175–182.
- Stantcheva, Stefanie**, “How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible,” *Annual Review of Economics*, 2023, 15 (1), 205–234.
- Train, Kenneth E.**, *Discrete Choice Methods with Simulation*, 2 ed., Cambridge University Press, 2009.
- Vakkalanka, J. Priyanka, Tracy L. Young, Gabriel Bianchi, Fred Ullrich, Knute D. Carter, Nicholas M. Mohr, and Jeydith Gutierrez**, “Outcomes Associated With Hospital at Home vs Traditional Inpatient Stay,” *JAMA Network Open*, 2026, 9 (5), e2610810.
- Wang, X., C. Stewart, and G. Lee**, “Patients’ and caregivers’ perceptions of the quality of hospital-at-home service: A scoping review,” *Journal of Clinical Nursing*, 2024, 33 (3), 817–838.
- Wedig, Gerard J., Mahmud Hassan, and Frank A. Sloan**, “Hospital Investment Decisions and the Cost of Capital,” *Journal of Business*, 1989, 62 (4), 517–537.
- Woodworth, Lindsey**, “Swamped: Emergency Department Crowding and Patient Mortality,” *Journal of Health Economics*, 2020, 70, 102279.
- **and James F. Holmes**, “Just a Minute: The Effect of Emergency Department Wait Time on the Cost of Care,” *Economic Inquiry*, 2020, 58 (2), 698–716.

Appendices

A Supporting Figures and Tables

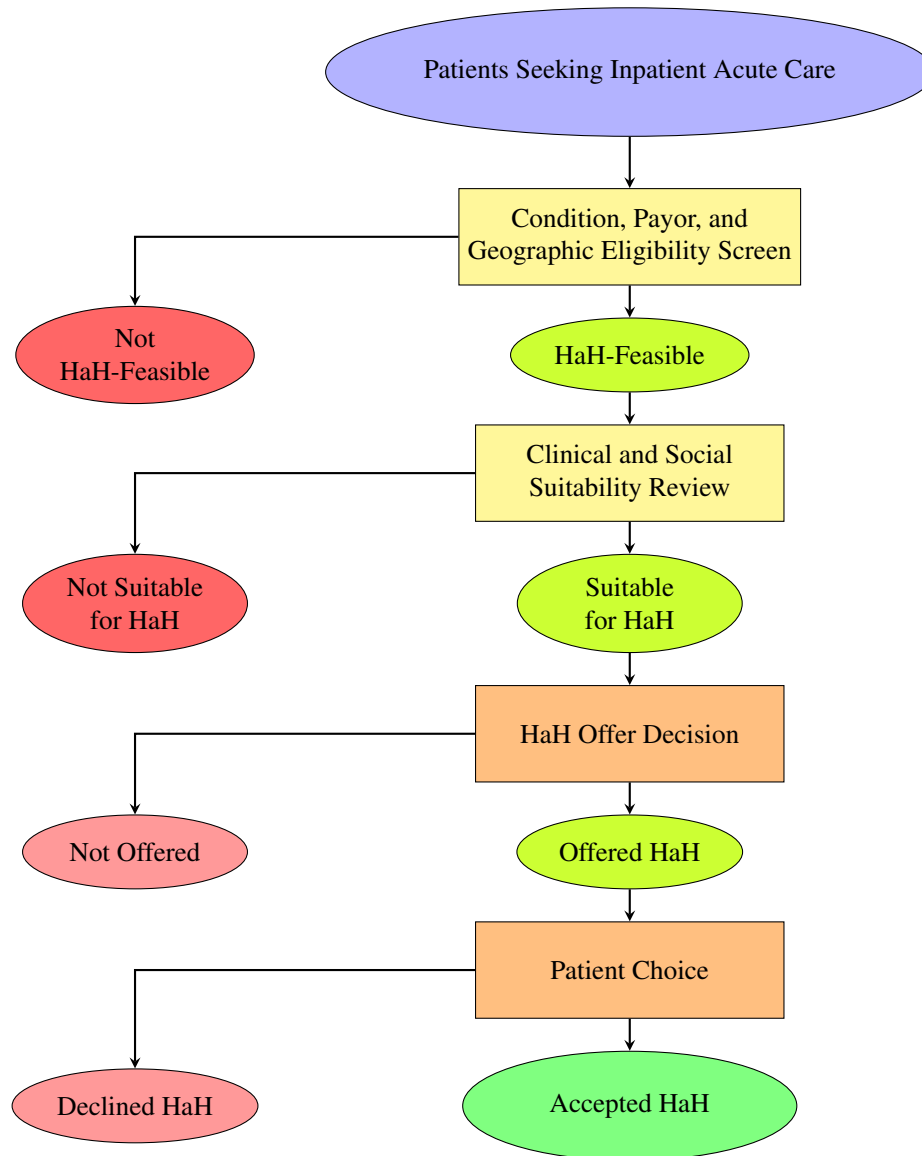
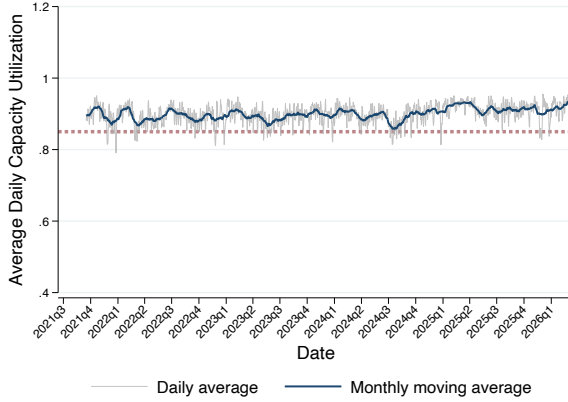


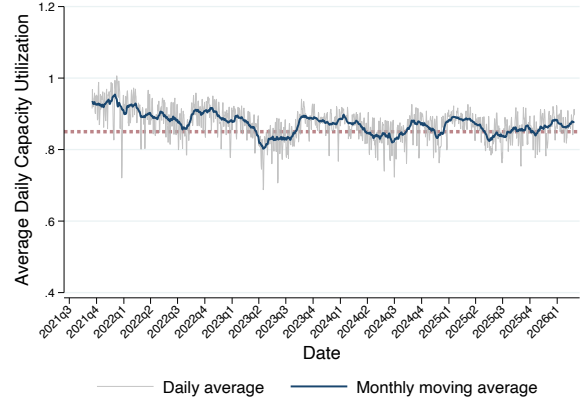
Figure A.1: Hospital-at-Home Patient Screening Process

Notes: Hospital-at-home programs typically allocate patients through four sequential stages. First, hospitals apply condition, payor, and geographic eligibility criteria, although specific rules vary across systems and some programs are payor agnostic. Eligible patients then undergo a clinical and social suitability review, which may assess clinical stability, treatment and monitoring needs, home safety, caregiver support, and access to utilities and communication. Among patients deemed suitable, the hospital decides whether to extend an HaH offer. Offered patients then choose whether to accept home-based care or remain in brick-and-mortar care. Patients who are eligible, suitable, offered HaH, and accept the offer constitute the treated HaH population.

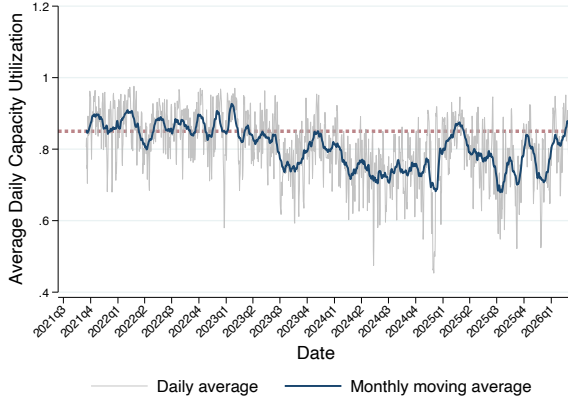
Figure A.2: Brick-and-Mortar Capacity Utilization Over Time



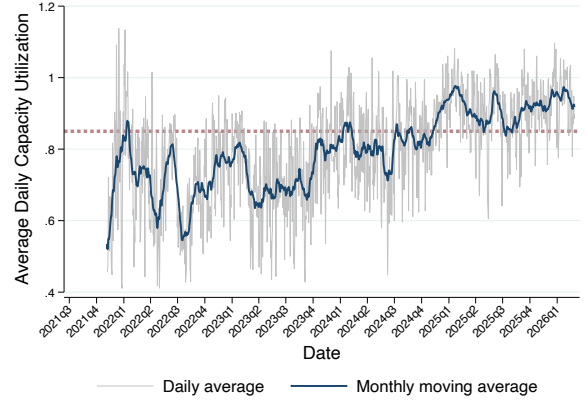
(a) Hospital A



(b) Hospital B



(c) Hospital C



(d) Hospital D

Persistent Preferences and Treatment History. Tables A.1 and A.2 compare alternative descriptions of demand history. Every specification retains persistent patient heterogeneity and posterior updating. Column (1) excludes direct demand-history terms, column (2) adds previously completed treatment, and column (3) instead includes prior acceptance and expected rejection counts. The baseline in column (4) combines these terms. The baseline has the lowest AIC among these specifications. Adding first treatment to the count-history model improves full-sample fit, with no clear change in terminal or first-post-treatment prediction. Expected rejections retain a negative association with acceptance, while additional acceptances show no clear cumulative increase. Persistent heterogeneity remains, although its estimated magnitude falls and becomes less precise. The central supply coefficients remain broadly similar across specifications. These results support the baseline's first-treatment flexibility.

Figure A.3 illustrates how prior choices inform the model's assessment of patient willingness.

TABLE A.1: Demand History and Persistent Preferences

	Normally-distributed heterogeneity			
	(1) No HaH history	(2) First treatment	(3) History counts	(4) Baseline specification
<i>Panel A: Patient acceptance</i>				
Prior acceptance count	–	–	0.089 (0.066)	–0.037 (0.071)
Expected prior rejection count	–	–	–0.270 (0.070)	–0.294 (0.066)
Previously completed treatment	–	0.135 (0.071)	–	0.262 (0.080)
Persistent preference SD	0.938 (0.078)	0.821 (0.096)	0.601 (0.152)	0.466 (0.165)
Control function	–3.512 (0.212)	–3.373 (0.213)	–3.375 (0.214)	–3.272 (0.204)
<i>Panel B: Hospital offers</i>				
Profit (per \$1,000)	0.492 (0.096)	0.480 (0.095)	0.467 (0.094)	0.453 (0.093)
Acceptance belief	2.970 (0.199)	3.030 (0.202)	2.857 (0.189)	2.898 (0.189)
B&M duration exposure	–0.826 (0.104)	–0.801 (0.104)	–0.811 (0.103)	–0.775 (0.102)
Hospital A linear relief	0.317 (0.047)	0.310 (0.046)	0.312 (0.046)	0.301 (0.046)
Hospital B linear relief	0.214 (0.048)	0.210 (0.047)	0.212 (0.047)	0.209 (0.047)
Hospital C linear relief	0.198 (0.046)	0.195 (0.046)	0.195 (0.046)	0.191 (0.046)
Hospital D linear relief	0.062 (0.048)	0.062 (0.047)	0.065 (0.048)	0.065 (0.047)
<i>Fit</i>				
Parameter count	169	170	171	172
Log likelihood	–16,161	–16,159	–16,154	–16,149
AIC	32,659	32,658	32,650	32,642

Notes: All columns use 31,944 likelihood assessments from 21,354 patients and retain 897 history-only assessments. All parameters are jointly re-estimated with common controls, fixed documentation inputs, posterior updating, and supply-history terms. History headings refer only to direct demand terms. Treatment requires a distinct prior HaH-treated encounter discharged before assessment. Counts are cumulative prior acceptances and expected declinations. Standard errors in parentheses use the observed Hessian conditional on generated inputs. Supply inputs retain the baseline units, hospital-specific linear B&M relief, and separate B&M duration exposure.

TABLE A.2: Demand-History Restrictions and Held-Out Acceptance Prediction

<i>Panel A: Nested likelihood-ratio tests</i>			
Expansion (columns)	Restrictions	LR statistic	<i>p</i> -value
First treatment: (2) vs. (1)	1	3.38	0.0662
History counts: (3) vs. (1)	2	13.62	0.00111
Baseline: (4) vs. (3)	1	9.89	0.00166
<i>Panel B: Acceptance log-loss differences</i>			
Expansion minus restriction	Terminal encounters (1,040 recorded offers)	First post-treatment (390 recorded offers)	
First treatment: (2) – (1)	+0.0015 [–0.0003, +0.0033]	+0.0004 [–0.0015, +0.0023]	
History counts: (3) – (1)	–0.0035 [–0.0055, –0.0015]	+0.0103 [+0.0018, +0.0186]	
Baseline: (4) – (3)	+0.0007 [–0.0024, +0.0036]	–0.0053 [–0.0126, +0.0024]	

Notes: Column numbers refer to Table A.1. Panel A tests zero first-treatment coefficients or jointly zero acceptance and rejection count coefficients. The number of restrictions gives the χ^2 degrees of freedom. In Panel B, negative differences favor the expansion. Brackets contain paired patient-bootstrap 95% intervals from 100,000 re-samples, stratified by fold and conditional on fitted models and frozen inputs without retraining. Training excludes each target and its later patient encounters. First-post-treatment prediction uses two patient folds. Acceptance among recorded offers is scored as $\rho_i/[\rho_i + q_i(1 - \rho_i)]$ to account for incomplete declination documentation.

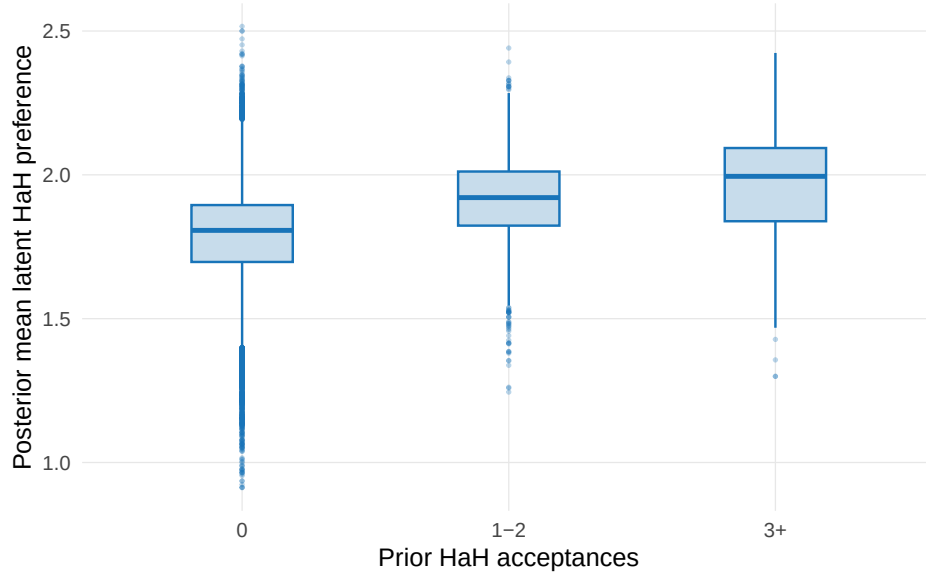
More prior acceptances are associated with higher posterior home preference, but substantial overlap indicates that acceptance counts alone do not fully summarize inferred preferences. This pattern illustrates the information available for patient targeting through posterior updating. Evidence on additional treatment-associated shifts comes from the joint estimates in Table 2.

B Evidence on the Causal Health Effects of HaH

To estimate the causal effect of HaH on health outcomes, I exploit the program’s geographic eligibility rule and implement a fuzzy spatial regression discontinuity design (RDD) (Lee and Lemieux, 2010; Keele and Titiunik, 2015). The complete design and analysis are developed in Capron et al. (2026). I summarize the econometric method, identifying variation, and estimates relevant to the demand and supply models here.

The program served patients residing within a fixed list of ZIP codes constructed to satisfy the 30-minute emergency-response requirement mandated by CMS. Using the operational ZIP-code list, rather than a reconstructed hospital isochrone, allows me to recover the eligibility boundary

Figure A.3: Posterior Home Preference by Prior HaH Acceptance History



actually faced by patients. Panel (a) of Figure B.1 illustrates this boundary and the spatial distribution of the analytic sample and HaH receipt.

Let r_i denote patient i 's signed distance in kilometers from the eligibility boundary, with $r_i < 0$ indicating residence inside the eligible area and $r_i \geq 0$ indicating residence outside. Geographic eligibility is therefore $E_i = \mathbb{1}\{r_i < 0\}$. Within bandwidth h around $r_i = 0$, the outcome equation follows the standard local-polynomial fuzzy-RD specification (Hahn et al., 2001; Calonico et al., 2014):⁸

$$Y_i = \alpha + \tau HaH_i + \gamma' X_i + \sum_{k=1}^K \left[\theta_k^- r_i^k E_i + \theta_k^+ r_i^k (1 - E_i) \right] + \varepsilon_i, \quad (\text{B.1})$$

where $HaH_i = \mathbb{1}\{\text{treated with HaH}\}$ and X_i contains prespecified patient, admission, neighborhood, and hospital characteristics.

Because HaH receipt is endogenously determined, I instrument for HaH_i using geographic eligibility, E_i . Identification requires that potential outcomes and treatment receipt vary smoothly through the boundary absent the eligibility rule, that eligibility affects outcomes only through HaH receipt, and that eligibility weakly increases the probability of treatment. The discontinuity in the probability of receiving an HaH offer is shown in Panel (b) of Figure B.1. This offer discontinuity illustrates the access mechanism; the treatment first stage uses HaH receipt itself. The correspond-

⁸Flexible polynomials of order K are allowed to vary on either side of the cutoff. These polynomials are not required for identification but allow the relationship between distance and the outcome to differ across sides of the boundary.

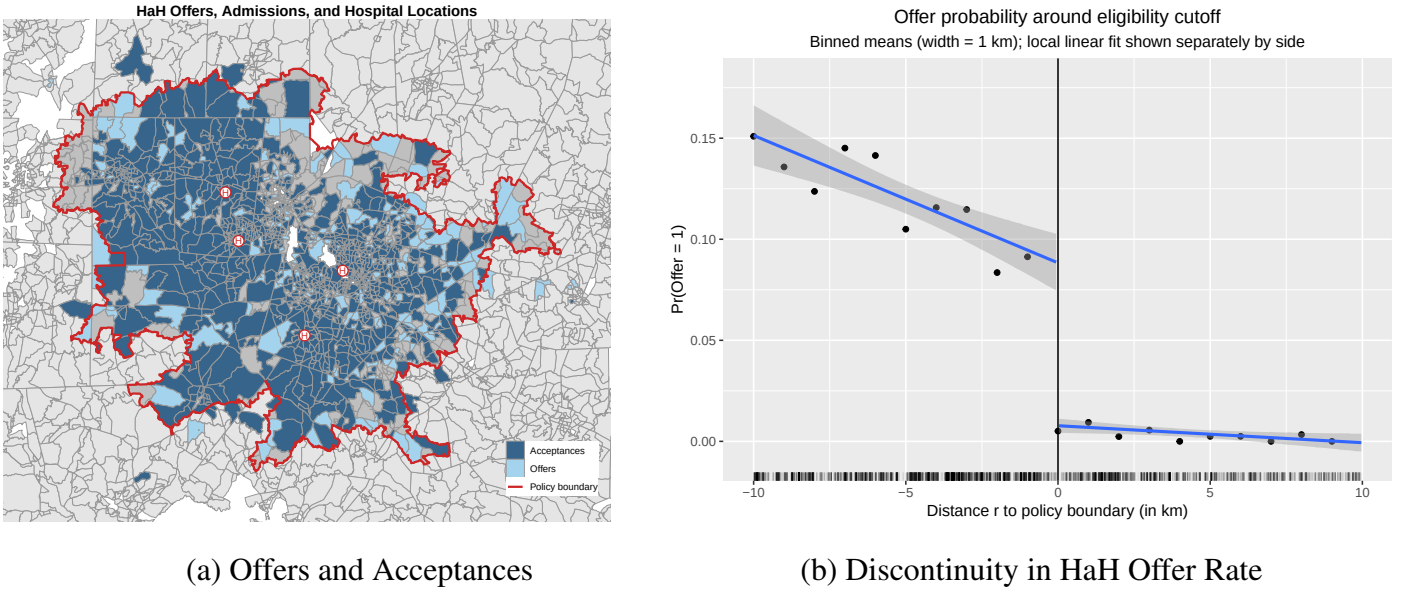
ing first-stage equation is

$$HaH_i = \pi_0 + \pi_1 E_i + \rho' X_i + \sum_{k=1}^K \left[\kappa_k^- r_i^k E_i + \kappa_k^+ r_i^k (1 - E_i) \right] + v_i. \quad (\text{B.2})$$

Letting ϕ_1 denote the reduced-form discontinuity in Y_i , the local average treatment effect for compliers at the boundary is (Imbens and Angrist, 1994)

$$\tau = \frac{\phi_1}{\pi_1} = \frac{\lim_{r \uparrow 0} \mathbb{E}[Y_i | r_i = r] - \lim_{r \downarrow 0} \mathbb{E}[Y_i | r_i = r]}{\lim_{r \uparrow 0} \mathbb{E}[HaH_i | r_i = r] - \lim_{r \downarrow 0} \mathbb{E}[HaH_i | r_i = r]}. \quad (\text{B.3})$$

Figure B.1: Program Treatment Boundary and Discontinuous Offer Probability



The full set of results can be found in Table 1 of Capron et al. (2026). Residence inside the eligibility boundary increased HaH receipt by approximately 4 percentage points at the discontinuity, with first-stage F -statistics ranging from 45 to 159 across studied outcomes. The estimates point to mixed effects across margins: lower mortality and fewer falls during the acute-care episode, but more subsequent inpatient readmissions. Differences in longer-run mortality, length of stay, discharge destination, and several other outcomes were not statistically distinguishable from zero. HaH was also associated with fewer clinical orders, particularly imaging and consultations, without evidence that these services were simply shifted earlier in the inpatient stay prior to transfer home. Taken together, the results suggest broadly similar health outcomes across treatment settings, with evidence of both short-run benefits and downstream utilization costs. These local estimates motivate the baseline abstraction from direct health effects in Section 5, while escalation risk

and acute-care needs remain determinants of HaH use. They do not establish equivalence across treatment settings or identify health effects for patients newly reached under expanded eligibility.

C Model Inputs and Estimation

C.1 Offer Documentation and Empirical Frequencies

Evidence from electronic health records indicates that flowsheet data miss some declined HaH offers, while accepted placements are observed throughout. I first use overlapping reporting fields to construct hospital-regime declination-capture probabilities. These probabilities are fixed inputs to both the joint demand–supply likelihood and a separate binomial model for reconstructing offers. Only the latter produces the bounded counts and rates used in the reduced-form evidence; those reconstructed outcomes are not used as inputs to the joint likelihood.

Capture of declinations across reporting regimes. New documentation procedures that made recording declined offers were rolled out throughout the program’s tenure. Figure C.1 shows that older indicators remain in use after new fields appear. Their continued availability supports comparing capture across regimes. The adjustment assumes that capture by the older fields is transportable across adjacent regimes and that modern documentation is complete.

Let q_{br} be the probability that a declined offer is documented at hospital b in regime r , with $q_i = q_{b(i),r(i)}$. Modern capture is normalized to $q_{b,\text{modern}} = 1$. Within each hospital,

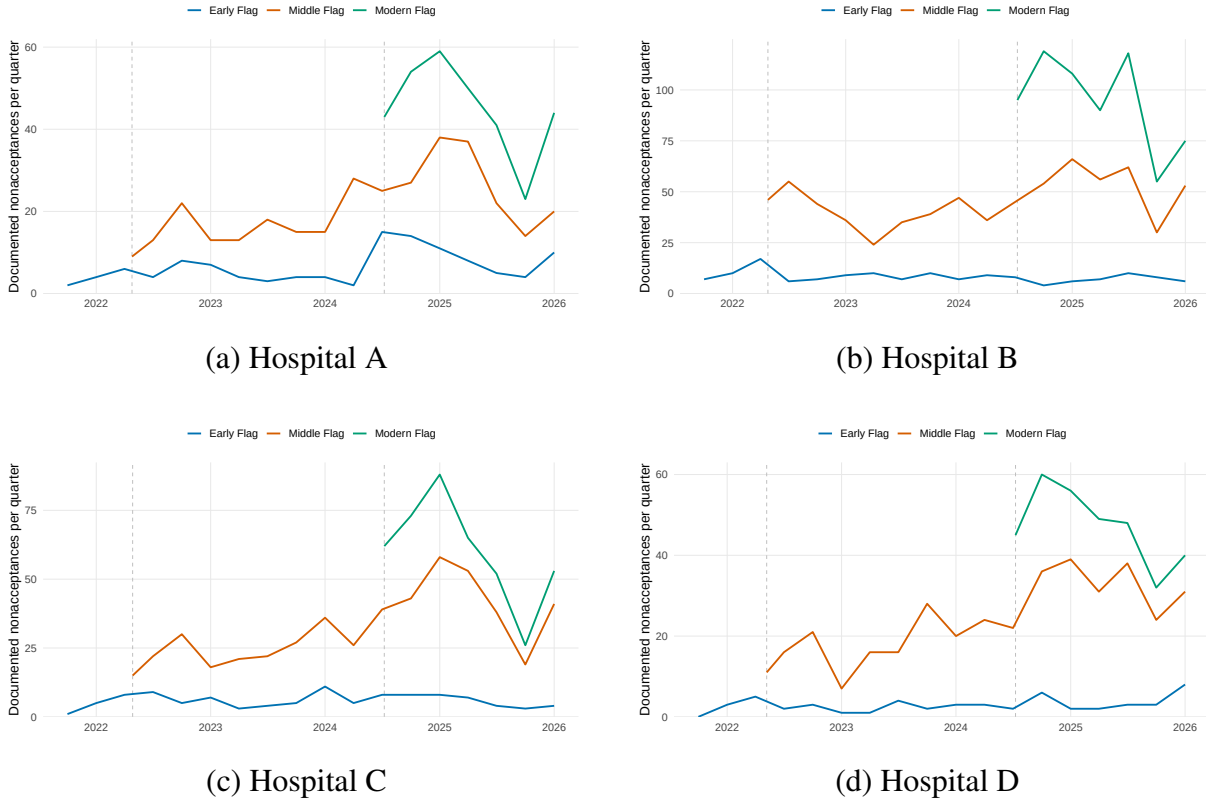
$$q_{b,\text{middle}} = \frac{\text{modern-regime declines with either older flag}}{\text{all modern-regime declines}},$$

$$q_{b,\text{early}} = q_{b,\text{middle}} \times \frac{\text{middle-regime declines with social-stability evidence}}{\text{all observed middle-regime declines}}.$$

Table C.1 reports the declination-capture probabilities used in joint maximum likelihood estimation. Each assessment is assigned $q_i = q_{b(i),r(i)}$ before estimation. These values are then held fixed at every likelihood evaluation and are not estimated with the demand and supply parameters. The joint model fits the observed acceptances, documented declines, and no-observed-offer outcomes using these capture probabilities. The table also reports implied overall offer capture, including acceptances, but only as a descriptive summary that it is not used as an additional likelihood input.

Reconstructing reduced-form offers. Directly scaling documented declines by $1/q_i$ can imply more offers than assessments for certain hospital-regime cells. I therefore fit a separate binomial model that uses the number of non-accepted assessments to limit possible hidden offers and pools

Figure C.1: Offer Documentation Across Reporting Regimes



Notes: Quarterly documented non-acceptances carrying each reporting flag. An assessment may carry multiple flags. Dashed vertical lines mark reporting transitions.

TABLE C.1: Reporting Regimes and Capture Probabilities

Hospital	Declination capture, q_{br}			Implied overall offer capture		
	Early	Middle	Modern	Early	Middle	Modern
A	0.163	0.675	1.000	0.521	0.867	1.000
B	0.108	0.572	1.000	0.421	0.750	1.000
C	0.156	0.665	1.000	0.553	0.829	1.000
D	0.088	0.665	1.000	0.423	0.833	1.000

Notes: Overall capture is $(A + D)/(A + D/q)$ within hospital-regime, where A and D count acceptances and documented declines.

information across months. Let c index hospital–month–regime cells, split at reporting transitions, with M_c total assessments, A_c acceptances, D_c documented declines, and $Z_c = M_c - A_c$ denoting total non-acceptances. If θ_c is the probability of a declined offer among non-accepted assessments,

$$D_c \mid Z_c, \theta_c \sim \text{Binomial}(Z_c, q_c \theta_c), \quad \text{logit}(\theta_c) = \mu_{b(c), r(c)} + u_c, \quad u_c \sim N(0, \tau^2).$$

Hospital-regime means and a shared monthly variance are estimated by marginal maximum likelihood with normal quadrature, using the same fixed capture rates, $q_c = q_{b(c), r(c)}$. This empirical-Bayes fit is performed separately, outside the joint demand–supply estimator. The reconstructed offer count is

$$\tilde{O}_c = A_c + D_c + (Z_c - D_c) \hat{\mathbb{E}} \left[\frac{(1 - q_c) \theta_c}{1 - q_c \theta_c} \mid D_c, Z_c, q_c \right].$$

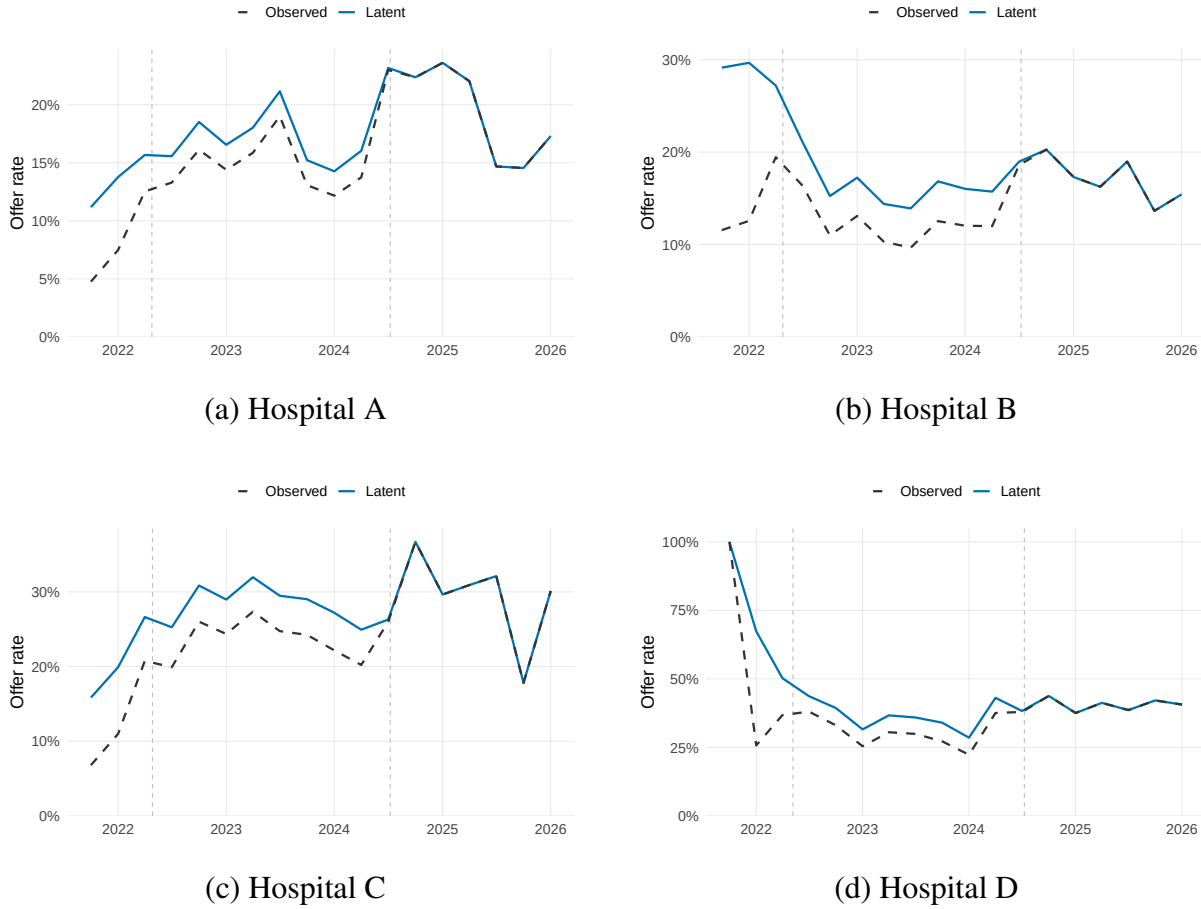
The expectation averages the hidden-offer probability over the cell posterior at the fitted hyperparameters. Thus known offers count as one, while each no-observed-offer assessment receives this posterior probability. Counts lie between observed offers and assessments; modern counts equal observed offers. Group rates divide summed reconstructed counts by summed assessments. These are the reduced-form outcomes, replacing raw inverse-capture totals in the empirical evidence presented throughout. Table C.2 summarizes the reconstruction. Partial pooling stabilizes cell counts with little change in the aggregate total. The resulting latent offers are plotted in Figure C.2.

TABLE C.2: Descriptive Offer Reconstruction

Assessments	31,944
Hospital–month–regime cells	211
Monthly heterogeneity SD, $\hat{\tau}$	0.214
Observed offers	6,169
Raw inverse-capture total (benchmark)	7,071.0
Bounded reconstructed offers	7,070.7
Marginal log likelihood	−558.1
Complete-pooling log likelihood	−571.7
Early-cell held-out log-score difference	−0.052

Notes: Complete pooling sets $\tau = 0$ within hospital-regime. Log likelihoods include binomial constants. The held-out comparison leaves out one early-period cell at a time, with the negative difference indicating no predictive gain over complete pooling in those cells.

Figure C.2: Recovered Latent Offer Rates



Notes: Dashed vertical lines mark reporting transitions.

C.2 Input Transformations

Selected regressors are centered or standardized to express their effects relative to a common baseline and in interpretable units. The reference means and standard deviations used for these transformations are fixed before estimation and retained in counterfactuals, so changes reflect altered inputs rather than a changing benchmark. Census, staffing, and duration enter their physical schedules in raw units; the transformations apply to the resulting regressors. The offer-index equations suppress this transformation in the notation used in the main text.

Standardized covariates. Demand covariates standardized by a reference mean and SD are age, condition weight, home-to-hospital distance, and census block group characteristics (household size, college share, log per-capita income, and share below 150% federal poverty line). Supply inputs use the latest valid NWASS, Morse, and SOFA records strictly before the actual assessment

timing. Care burden averages standardized NWASS and Morse scores and then rescales the resulting index; acute severity uses rescaled SOFA. Missing component scores contribute zero on the standardized scale, with a separate indicator for each of the three measures. No additional clinical scores enter demand, and missing scores do not remove assessments. A standardized coefficient refers to one reference SD, and dividing it by that SD gives the coefficient per original unit.

Centered payoff inputs. The profit regressor is the conditional HaH-minus-B&M dollar return, less its fixed hospital-quarter-condition reference, divided by \$1,000, with equation (C.13) providing the mapping. Common and hospital-specific B&M relief, HaH strain, and escalation risk subtract fixed global means without rescaling. Their coefficients retain the original units.

Acceptance, histories, and indicators. Only the stand-alone acceptance term is centered, using $\rho_i - \bar{\rho}$ with $\bar{\rho} = 0.411$. Payoffs that are conditional on acceptance use uncentered ρ_i . Prior accepted placements, expected rejections, and expected true offers enter as raw cumulative counts. Categorical variables, fixed-effect indicators, flags for missing fields, and the federal shutdown indicator remain untransformed.

References and component values. For clarity here, write $M_i^{B,\text{raw}}$ for the physical capacity vector and $\bar{M}^B$ for its fixed reference. Then

$$\rho_i \lambda'_B (M_i^{B,\text{raw}} - \bar{M}^B) = \rho_i \lambda'_B M_i^{B,\text{raw}} - \rho_i \lambda'_B \bar{M}^B.$$

The constant capacity offset can be allocated to the stand-alone acceptance term, with a corresponding intercept adjustment because that term is centered. Component levels therefore depend on the reference even when the total index is preserved. Section 5 suppresses this distinction, and the capacity-value measure in equation (29) uses the centered input. Differences in that measure across states retain their interpretation under a common reference.

C.3 Exposure Inputs and Numerical Estimation

Remaining treatment exposure. Let T_i^o denote counterfactual remaining duration in days under location $o \in \{B, H\}$, with its distribution predicted from information $\mathcal{I}_i^o$ at assessment. Superscripts B and H refer to B&M and HaH, as in Section 5.3. Let ℓ_i^{LOS} denote elapsed LOS, distinguishing it from the likelihood contribution $\ell_{ig}(Y_i; \Theta)$. The discharge models vary with hospital, condition, location, elapsed LOS, and day of week. Under the baseline zero transfer delay, the HaH path begins at the assessment.

For duration $T \geq 0$, the discounted exposure schedule is

$$A_\delta(T) = \sum_{t=0}^{\lfloor T \rfloor - 1} \delta^t + (T - \lfloor T \rfloor) \delta^{\lfloor T \rfloor}, \quad \delta = 0.99, \quad (\text{C.1})$$

where an empty sum is zero. The last term prorates a fractional final day. The exposure inputs used in the supply model are

$$E_i^o = \mathbb{E} \left[A_\delta(T_i^o) \mid \mathcal{J}_i^o \right], \quad o \in \{B, H\}. \quad (\text{C.2})$$

Thus discounting is applied to each possible remaining duration before averaging over its predictive distribution.

To connect these inputs to discharge predictions, write $S_i^o(u) = \mathbb{P}(T_i^o > u \mid \mathcal{J}_i^o)$ for remaining-stay survival. Then

$$E_i^o = \sum_{t=0}^{\infty} \delta^t \int_t^{t+1} S_i^o(u) du. \quad (\text{C.3})$$

For integer-valued durations this reduces to $\sum_{t \geq 0} \delta^t S_i^o(t)$. If $\mu_i^o(t)$ is the exit probability for this duration during $(t, t+1]$ conditional on remaining in care at t , survival on the daily grid satisfies

$$S_i^o(t) = \prod_{v=0}^{t-1} \{1 - \mu_i^o(v)\}, \quad S_i^o(0) = 1.$$

Remaining B&M and HaH exposure therefore use their respective discharge paths. Section 5.3 multiplies these discounted days by capacity increments evaluated at the assessment's census and staffing state. The undiscounted bed-days in Section 7 instead use $\mathbb{E}[T_i^o \mid \mathcal{J}_i^o] = \int_0^\infty S_i^o(u) du$.

Histories and sequential likelihood. Assessments are processed chronologically within patient. An acceptance increments prior accepted placements by one; a documented rejection increments prior rejections by one. A no-observed-offer outcome increments expected rejections by the posterior $\widehat{O}_i$ in equation (7). Expected true offers increase by $\widehat{O}_i$ for every outcome, with $\widehat{O}_i = 1$ for accepted placements and documented rejections. All increments occur after evaluating the current predictive likelihood, and all parameter-dependent histories are reconstructed at each trial vector $\Theta = (\Theta_D, \Theta_S)$.

Using the likelihood notation of Section 5.5, the objective is

$$\mathcal{L}(\Theta) = \sum_j \sum_{i \in \mathcal{I}_j} \log L_i(\Theta), \quad L_i(\Theta) = \sum_g w_{ig}^- \ell_{ig}(Y_i; \Theta). \quad (\text{C.4})$$

The recursion updates node weights and histories at every retained assessment, including observations with zero direct objective weight. If all such observations precede the included history, this objective equals the integrated patient-level likelihood in equation (25). If they are interspersed, equation (C.4) specifies the selected predictive contributions with the intervening updates retained.

Gauss–Hermite quadrature uses 20 standard-normal nodes ξ_g and normalized prior weights w_g . The posterior mean of the persistent utility component before assessment i is $\sigma_\xi \sum_g w_{ig}^- \xi_g$; the mean of the standardized latent type is $\sum_g w_{ig}^- \xi_g$.

Optimization and inference. I maximize the joint log likelihood using a quasi-Newton algorithm with analytic gradients, solving each offer–acceptance fixed point by bisection. Derivatives account for the fixed point and sequential updates to patient histories and preference weights. Parameters are internally rescaled for numerical stability and returned to their original units for reporting. Standard errors use the observed information matrix and the delta method where needed. They condition on the generated inputs and do not adjust for clustering or uncertainty in the documentation, cost, and discharge models.

Fixed-point regularity. Holding pre-assessment weights, histories, and other inputs fixed, the derivative of the residual in equation (23) is

$$\frac{\partial F_i}{\partial p} = 1 + \varphi \Lambda'(\Gamma_i(\rho_i(p); \Theta_S)) (\gamma_\rho + \Delta W_i) \sum_g w_{ig}^- \rho_{ig}(p) \{1 - \rho_{ig}(p)\}.$$

The numerical solver verifies that the bracket endpoints have opposite residual signs. If this derivative is positive throughout the unit interval, the solution is unique. When it is nonzero at the solution, implicit differentiation gives

$$\frac{\partial p_i}{\partial \Theta} = - \left(\frac{\partial F_i}{\partial p} \right)^{-1} \frac{\partial F_i}{\partial \Theta},$$

where parameter derivatives account for earlier updates to patient histories and preference weights. A derivative close to zero can make the solution sensitive to parameter changes.

C.4 Treatment Cost Model

The profit input in equation (11) uses predicted direct-variable costs remaining at assessment. Its construction combines a common LOS schedule, a HaH transfer increment, and a HaH-day cost difference. These are conditional accounting predictions rather than causal treatment effects.

Episode-level cost specification. For the episode associated with assessment i , let C_i^{ep} denote total direct treatment cost, L_i^{ep} total LOS, and h_i^{ep} HaH days. Let $D_i^H = \mathbb{1}\{h_i^{\text{ep}} > 0\}$ and let $c(i)$ denote the condition group. The specification is

$$C_i^{\text{ep}} = a_{c(i)} + g_{c(i)}(L_i^{\text{ep}}) + D_i^H s_{c(i)} + r_{c(i)} h_i^{\text{ep}} + Z_i^{C'} \gamma_C + \varepsilon_i^C, \quad (\text{C.5})$$

$$g_c(L) = b_{c0}L + \sum_{k \in \{3,7,14,30\}} b_{ck}(L-k)_+, \quad (\text{C.6})$$

where the transfer increment s_c captures the additional episode cost associated with entering HaH, while r_c measures the difference in cost per HaH day relative to the common schedule. Neither coefficient is constrained by its label to be positive. The controls Z_i^C include DRG-family, hospital, fiscal-year, age, sex, ED-entry, and condition-weight variables.

The model is estimated by penalized least squares on episode totals, selecting the ridge penalty by held-out root mean squared error across five calendar blocks. The common LOS schedule comes from the full screened sample used in the joint demand-supply estimation model. The HaH transfer and daily-rate increments come from a the sub-sample of documented-offer encounters. Therefore, I impose the assumption that their conditional predictive relationship extends beyond observed offers in order to form predictions for the full assessment sample.

Allocation to the remaining stay. Using the elapsed LOS and remaining durations defined in Appendix C.3, the common cost schedule for an integer remaining duration T is

$$G_c(\ell^{\text{LOS}}, T) = \sum_{t=0}^{T-1} \delta^t [g_c(\ell^{\text{LOS}} + t + 1) - g_c(\ell^{\text{LOS}} + t)]. \quad (\text{C.7})$$

With zero transfer delay and A_δ from equation (C.1), conditional remaining costs are

$$C_i^B(T) = G_{c(i)}(\ell_i^{\text{LOS}}, T), \quad (\text{C.8})$$

$$C_i^H(T) = G_{c(i)}(\ell_i^{\text{LOS}}, T) + s_{c(i)} + r_{c(i)} A_\delta(T). \quad (\text{C.9})$$

The transfer increment is incurred at transfer, while the daily increment applies only to remaining HaH exposure. Fractional final-day cost increments are retained, using the same exposure convention as E_i^H . The supply model inputs average the two schedules over their respective predicted remaining-duration distributions:

$$\widehat{C}_i^B = \mathbb{E} [G_{c(i)}(\ell_i^{\text{LOS}}, T_i^B) \mid \mathcal{I}_i^O], \quad (\text{C.10})$$

$$\widehat{C}_i^H = \mathbb{E} [G_{c(i)}(\ell_i^{\text{LOS}}, T_i^H) \mid \mathcal{I}_i^O] + s_{c(i)} + r_{c(i)} E_i^H. \quad (\text{C.11})$$

Already-incurred costs and common episode controls cancel from the contrast. The common LOS schedule contributes when predicted duration differs by location, separating the duration channel from the HaH transfer and daily-rate increments.

Profit units and counterfactual use. Let $\Delta\Pi_i^\$$ denote the uncentered dollar profit difference between HaH and B&M conditional on an accepted placement. Under the baseline reimbursement-parity assumption, as supported by negotiated contracts and observed reimbursements, the profit difference becomes

$$\Delta\Pi_i^\$ = \widehat{C}_i^B - \widehat{C}_i^H. \quad (\text{C.12})$$

For the offer equation, the input is

$$\Delta\Pi_i = \frac{\Delta\Pi_i^\$ - \overline{\Delta\Pi}_{b(i),\text{qtr}(i),c(i)}^\$}{1,000}, \quad (\text{C.13})$$

where $\text{qtr}(i)$ denotes the quarter assigned to assessment i for reference-cell construction. Acceptance weights this input once through $\rho_i\Delta W_i$. A positive coefficient favors higher conditional returns, holding other components fixed, rather than implying positive absolute HaH profits. Payment counterfactuals add the dollar reimbursement change before applying the original normalization. Hospital return outcomes use uncentered costs and revenues, keeping hospital accounting returns distinct from social resource savings.

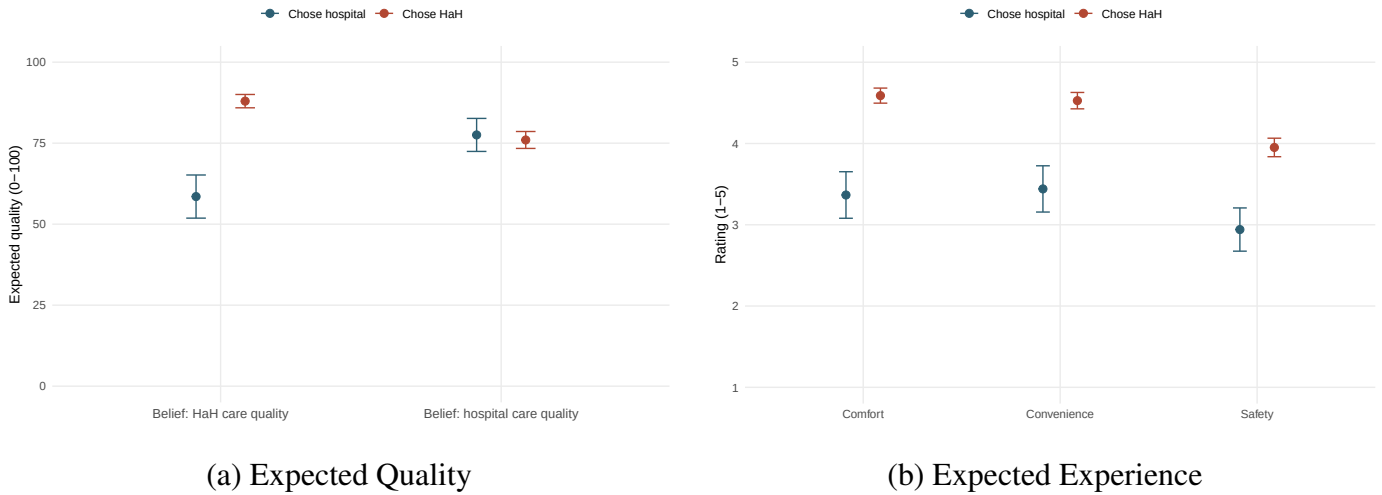
D Survey Augmentation of Patient Choice Model

This appendix augments the joint demand–supply model in Section 5 with a novel survey measuring household beliefs over expected relative care quality and relative comfort, safety, and convenience under HaH, as it compares to B&M care. Survey answers enter the model as part of persistent patient preferences and are not part of the hospital’s information set in estimation. Non-medical spending responses are relatively sparse and therefore enter only as a supplemental specification at the end of this appendix.

TABLE D.1: Survey Respondents and Reported Information

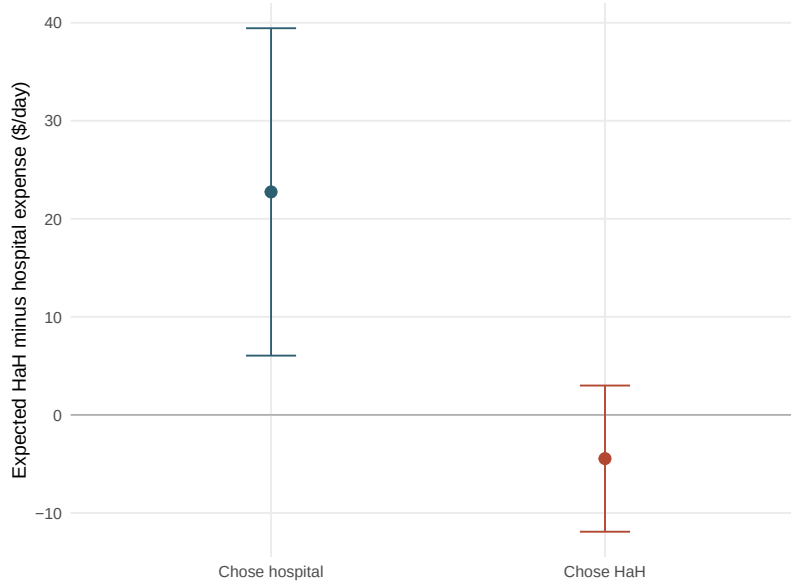
	Count (N)
Encounters linked to a survey	453
Patient respondents	359
Caregiver respondents	94
Complete expected-quality difference	404
Any expected-experience information	453
Complete comfort, safety, and convenience ratings	450
Complete expected non-medical spending responses	297

Notes: Caregiver responses answer questions phrased to address the patient's expected beliefs and therefore are assumed to measure the associated household's care-choice preferences.

Figure D.1: Expected Care Quality and HaH Experience by Reported Choice

Notes: The left panel uses expected care-quality ratings on a 0-to-100 scale. The right panel uses a one-to-five Likert scale for expected comfort, convenience, and safety ratings. Points report respondent means and bars report 95 percent confidence intervals.

Figure D.2: Expected Non-Medical Spending Difference by Reported Choice



Notes: The outcome is expected daily household spending under HaH minus expected daily household spending under hospital care, summed across transportation, meals, child care, pet care, cleaning, and paid caregiver support. Points report means and bars report 95 percent confidence intervals.

D.1 Survey-Augmented Demand Model

Expected quality. Let $\Delta Q_j = Q_j^H - Q_j^B$ denote patient j 's reported expected-quality difference between HaH and B&M, each rated from 0-to-100. I standardize this difference as

$$Q_j = \frac{\Delta Q_j - \bar{\Delta Q}}{s_{\Delta Q}}, \quad \bar{\Delta Q} = 4.61, \quad s_{\Delta Q} = 34.64, \quad (\text{D.1})$$

using the sample mean and standard deviation among the 404 respondents with both ratings. Higher Q_j indicates more favorable expected HaH quality relative to B&M. Conditional on first-assessment covariates X_j^0 , I specify

$$Q_j | X_j^0 \sim N(\mu_j^Q, s_Q^2), \quad s_Q = 0.958,$$

where μ_j^Q is a ridge-regression prediction. The prediction rule and residual standard deviation are held fixed during joint estimation. Observed scores are treated as exact measurements of a persistent quality-expectation trait, while missing scores are integrated out. Hospital beliefs use this conditional distribution, updated with strictly prior assessment outcomes, without observing the later survey response.

Expected experience. Let $F_j \sim N(0, 1)$ denote a persistent expected-experience factor. For component $m \in \{\text{comfort, safety, convenience}\}$, let $Y_{jm}^S \in \{1, \dots, 5\}$ denote the ordered rating. The ordered-probit measurement model is

$$Y_{jm}^S = k \iff \tau_{m,k-1} < \lambda_m^S F_j + e_{jm}^S \leq \tau_{m,k}, \quad e_{jm}^S \sim N(0, 1), \quad (\text{D.2})$$

where $\tau_{m,0} = -\infty$, $\tau_{m,5} = \infty$, and the four finite cutpoints are ordered. Conditional on F_j , component errors are independent, so an observed response contributes

$$h_{jm}(f) = \Phi\left(\tau_{m,Y_{jm}^S} - \lambda_m^S f\right) - \Phi\left(\tau_{m,Y_{jm}^S-1} - \lambda_m^S f\right). \quad (\text{D.3})$$

The loadings and cutpoints are estimated in a first-stage model and then held fixed. The loadings are 2.389 for comfort, 1.228 for safety, and 1.466 for convenience; higher F_j therefore indicates more favorable expected HaH experience. Missing components contribute a factor of one. Quality and experience are patient-specific traits that apply across the patient's assessments. The interpretation treats later answers as measurements of preferences present at the time of choice.

Augmented utility and hospital beliefs. Using the demand notation in Section 5.2, and letting superscript S denote the survey-augmented demand system, the survey adds two terms to persistent patient utility:

$$\omega_j = \sigma_\xi \xi_j + \theta_Q Q_j + \theta_F F_j, \quad \xi_j \sim N(0, 1), \quad (\text{D.4})$$

$$\Delta U_{ij}^S = X_i' \beta_U - \alpha P_i + \varphi(1 - p_i) + \omega_j + \varepsilon_i. \quad (\text{D.5})$$

The baseline non-medical price difference is zero in this extension. Conditional on first-assessment covariates, the prior distribution of (Q_j, F_j, ξ_j) is

$$G_j^0 = N(\mu_j^Q, s_Q^2) \times N(0, 1) \times N(0, 1). \quad (\text{D.6})$$

This distinguishes the residual preference SD, σ_ξ , from the total conditional SD of persistent private utility,

$$s_\omega = \left(\sigma_\xi^2 + \theta_Q^2 s_Q^2 + \theta_F^2\right)^{1/2}. \quad (\text{D.7})$$

Let G_i^- denote the distribution of these traits updated with the patient's strictly prior observed assessment outcomes. The type-specific acceptance probability, hospital acceptance belief, and

offer probability satisfy

$$\rho_i^S(p; \tilde{q}, \tilde{f}, \xi) = \Lambda[X_i' \beta_U - \alpha P_i + \varphi(1 - p) + \theta_Q \tilde{q} + \theta_F \tilde{f} + \sigma_\xi \xi], \quad (\text{D.8})$$

$$\rho_i(p) = \int \rho_i^S(p; \tilde{q}, \tilde{f}, \xi) dG_i^-(\tilde{q}, \tilde{f}, \xi), \quad p_i = \Lambda[\Gamma_i(\rho_i(p_i); \Theta_S)]. \quad (\text{D.9})$$

Here $\tilde{q}$ and $\tilde{f}$ denote draws of the persistent quality and experience traits Q_j and F_j , respectively. Survey answers inform the shared preference parameters through the measurement likelihood, but do not update G_i^- or enter the supply index directly. The historical offer–acceptance filter integrates the Gaussian composite ω_j using 20 Gauss–Hermite nodes. Survey measurement integration is evaluated separately.

Augmented likelihood. Write $\tilde{\Theta} = (\Theta, \theta_Q, \theta_F)$ for the survey-augmented parameter vector. Conditional on a trait vector $\mathbf{z} = (\tilde{q}, \tilde{f}, \xi)$, the contribution of observed assessment outcome $Y_i \in \{a, d, n\}$ is

$$\ell_i^S(Y_i; \mathbf{z}, \tilde{\Theta}) = \begin{cases} p_i \rho_i^S(p_i; \mathbf{z}), & Y_i = a, \\ p_i \{1 - \rho_i^S(p_i; \mathbf{z})\} q_i, & Y_i = d, \\ 1 - p_i + p_i \{1 - \rho_i^S(p_i; \mathbf{z})\} (1 - q_i), & Y_i = n. \end{cases} \quad (\text{D.10})$$

Its predictive probability is $L_i^S = \int \ell_i^S dG_i^-$, and posterior weights update after the current outcome as in Section 5.5. Here q_i retains its baseline meaning as the documentation probability, distinct from the quality draw $\tilde{q}$. The same assessments from the baseline joint demand–supply contribute to the objective. Let $\mathcal{M}_j$ contain the observed experience components and define $H_j(\tilde{f}) = \prod_{m \in \mathcal{M}_j} h_{jm}(\tilde{f})$. For any trait distribution G , the joint density or probability of patient j ’s observed survey information is

$$m_j^S(G) = \begin{cases} g_Q(Q_j^{\text{obs}}; G) \int H_j(\tilde{f}) dG(\tilde{f}, \xi \mid Q_j = Q_j^{\text{obs}}), & Q_j \text{ observed,} \\ \int H_j(\tilde{f}) dG(\tilde{q}, \tilde{f}, \xi), & Q_j \text{ missing,} \end{cases} \quad (\text{D.11})$$

where $g_Q(\cdot; G)$ is the marginal density of Q_j under G . An observed quality score enters as a density evaluated at its reported value. Missing quality is integrated out, and a patient with no observed survey components has $m_j^S = 1$.

Let G_j^Y denote the terminal trait distribution after the patient’s complete observed assessment history, updated without survey answers, and let $\mathcal{R}$ denote the encounters with survey responses.

The normalized survey-augmented log likelihood is

$$\mathcal{L}^S(\tilde{\Theta}) = \sum_j \sum_{i \in \mathcal{I}_j} \log L_i^S(\tilde{\Theta}) + \sum_{j \in \mathcal{R}} \log \left\{ \frac{m_j^S(\mathbf{G}_j^Y)}{m_j^S(\mathbf{G}_j^0)} \right\}. \quad (\text{D.12})$$

Each patient’s survey contributes once, irrespective of their number of assessments. The denominator is the marginal measurement density under the fixed first-stage prior and is constant with respect to the jointly estimated parameters. This normalization reproduces the baseline model’s criterion when $\theta_Q = \theta_F = 0$, without a survey multiplier or population expansion weight. Non-respondents contribute through their outcome histories and the shared parameters. Predictions conditioned on survey answers are used for patient diagnostics separately from the hospital’s historical beliefs.

All demand and supply parameters in the survey-augmented model are jointly estimated. The augmented model assumes that the conditional quality distribution and experience measurement model also apply to non-respondents, and that later reports measure preferences present at the time of choice. Caregiver income, employment, and paid leave remain descriptive survey fields that are explored in the subsection that follows.

D.2 Conditioning on Reported Non-Medical Spending

The survey elicits household spending on transportation, meals, child care, pet care, cleaning, and paid caregiver support under each location. Let $\Delta C_i^{\$}$ denote the signed HaH-minus-B&M difference in dollars per day, and let $D_i^{\$}$ indicate a complete report matched to assessment i . Spending is used in daily units, without multiplication by length of stay, and applies only to the matched assessment.

The 297 matched reports comprise 237 acceptances and 60 documented declinations. This extension retains every assessment and adds a spending slope and reporting indicator to equation (D.5). Denoting this specification by superscript $\{S, \$\}$,

$$\Delta U_{ij}^{S,\$} = \Delta U_{ij}^S + \gamma_P D_i^{\$} \frac{\Delta C_i^{\$}}{100} + \delta_{\$} D_i^{\$}. \quad (\text{D.13})$$

For $D_i^{\$} = 0$, the added terms are defined to be zero without imputing missing spending. The unrestricted slope γ_P measures the utility association with an additional \$100 per day of reported HaH spending relative to B&M. The reporting indicator allows the matched group to have a different utility level. This specification jointly re-estimates 175 parameters using equation (D.12), with acceptance probabilities formed from equation (D.13). Spending and report status enter the offer–acceptance fixed point through the acceptance belief. Neither enters supply directly. Quality and experience answers remain outside the historical filter, and their normalized measurement

contribution is unchanged.

This is a retrospective sensitivity analysis that conditions on report information unavailable in the historical EHR data that informs the offer decision. The reporting indicator absorbs an average group difference without fully correcting for survey selection. Spending answers combine lack exogenous price variation. Thus, γ_p is considered purely associational and is not used to convert utility into willingness to pay or monetary welfare.

D.3 Results

Table D.2 shows that more favorable expectations about HaH quality and experience are associated with greater willingness to accept. A one-SD increase in the reported quality difference raises acceptance utility by 0.133, while a one-SD increase in the experience factor raises it by 0.563. These estimates connect persistent differences in patients' choices to their stated expectations about care. The residual preference term $\sigma_\xi \xi_j$ captures a patient's lasting inclination toward or against HaH after accounting for observed covariates and, in the survey-augmented model, quality and experience expectations. It shifts utility across that patient's assessments, unlike the assessment-specific shock ε_i . In the baseline model, this unobserved component also absorbs differences in expectations that the survey subsequently measures. Its estimated SD, σ_ξ , falls from 0.466 to 0.112 when those measures enter utility. This suggests that the survey accounts for part of the persistent variation previously attributed to unobserved preferences.

The decline in σ_ξ does not imply that patients become more alike. Total conditional persistent-preference variation, s_ω , includes quality and experience as well as the residual, and its estimated SD is 0.588 in the survey-augmented model, rising from 0.466 in the baseline. The model therefore assigns more of this variation to measured expectations. The remaining residual SD is not statistically significant, with a log-scale standard error of approximately 1.54, so the estimates do not establish that other persistent preferences are negligible.

The supply estimates retain the same substantive pattern as the baseline model. Hospitals favor transfers with greater financial returns, while clinical burden and escalation risk discourage offers. The hospital-specific linear capacity terms are positive and statistically significant at Hospitals A–C, while the estimate for Hospital D remains statistically insignificant. HaH strain is small, positive, and statistically significant. Thus, incorporating measured patient expectations leaves the main interpretation of hospital incentives unchanged from the baseline model.

Table D.3 adds reported non-medical spending. Among matched reports, higher HaH spending relative to B&M is associated with lower acceptance utility. The positive reporting-group coefficient indicates that respondents with complete spending reports differ substantially from other assessments, even after the included controls. Quality and experience remain positively associated

TABLE D.2: Baseline and Survey-Augmented Joint Estimates

	Baseline	Survey-augmented	
	Coefficient	Coefficient	SE
<i>Patient acceptance</i>			
Expected-quality difference, θ_Q	—	0.133***	(0.036)
Expected-experience factor, θ_F	—	0.563***	(0.093)
Control function, ϕ	-3.272***	-3.293***	(0.187)
Home-hospital distance	0.012	-0.000	(0.014)
Age	0.101***	0.085***	(0.018)
Male	0.021	0.036	(0.024)
Household size	0.004	0.006	(0.014)
College share	0.032	0.028	(0.019)
Log per-capita income	-0.031	-0.026	(0.023)
Share below 150% FPL	-0.043**	-0.024	(0.017)
Prior expected rejections	-0.294***	-0.255***	(0.060)
Prior accepted placements	-0.037	-0.081	(0.066)
Condition acuity weight	-0.097***	-0.097***	(0.028)
Residual preference SD, σ_ξ	0.466	0.112	
Total persistent-preference SD, s_ω	0.466	0.588	
<i>Hospital offers</i>			
Profit, β_Π	0.453***	0.464***	(0.091)
HaH strain, λ_H	0.003**	0.003**	(0.001)
Escalation risk, β_H	-1.292*	-1.365*	(0.800)
Care burden	-0.182***	-0.181***	(0.013)
Acute severity	-0.080***	-0.079***	(0.012)
B&M stay-duration exposure	-0.775***	-0.771***	(0.100)
Hospital A occupancy release	0.301***	0.299***	(0.045)
Hospital B occupancy release	0.209***	0.205***	(0.046)
Hospital C occupancy release	0.191***	0.190***	(0.045)
Hospital D occupancy release	0.065	0.066	(0.047)
Stand-alone acceptance belief, γ_ρ	2.898***	2.988***	(0.190)
Likelihood assessments	31,944	31,944	
Unique patients	21,354	21,354	
Linked survey patients	—	453	
Outcome log likelihood	-16,149	-16,145	
Normalized joint log likelihood	-16,149	-16,123	
Joint-outcome Brier score	0.269	0.269	

Notes: Both models retain the baseline controls, fixed effects, documentation probabilities, and input transformations. Profit is in centered \$1,000 units. Capacity terms use hospital-specific linear relief slopes and separate B&M duration exposure. Survey-augmented standard errors appear in parentheses and use the observed Hessian conditional on linkage and first-stage measurement inputs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

with acceptance. Changes in the control-function and acceptance-belief coefficients also show that conditioning on these reports affects the model’s separation of patient choice and hospital targeting.

The joint log-likelihood gain of ~ 253 reflects both the spending slope and the reporting indicator. Prediction improves sharply among matched reports but worsens slightly elsewhere. Because every spending report follows a recorded offer, and the conditional model uses information unavailable to the hospital at the time, this gain does not establish better historical prediction and critically does not identify a population price response.

TABLE D.3: Survey-Augmented Estimates with Reported Spending

	Survey-augmented		With reported spending	
	Coefficient	SE	Coefficient	SE
Spending per \$100/day, γ_p	–		–0.319**	(0.142)
Complete-report indicator, δ_s	–		1.578***	(0.130)
Expected-quality difference, θ_Q	0.133***	(0.036)	0.128***	(0.025)
Expected-experience factor, θ_F	0.563***	(0.093)	0.618***	(0.081)
Control function, φ	–3.293***	(0.187)	–2.308***	(0.154)
Profit, β_Π	0.464***	(0.091)	0.434***	(0.085)
Likelihood assessments	31,944		31,944	
Matched spending reports used	–		297	
Outcome log likelihood	–16,145		–15,900	
Normalized joint log likelihood	–16,123		–15,870	
Joint Brier: all assessments	0.269		0.264	
Joint Brier: matched-report assessments	0.960		0.368	
Joint Brier: other assessments	0.263		0.263	

Notes: Both columns augment the baseline model with quality and experience. The second also conditions on reported non-medical spending and report status in the offer–acceptance fixed point, jointly re-estimating all demand and supply parameters. Spending is measured per day. Standard errors in parentheses use the observed Hessian conditional on linkage and first-stage measurement inputs. Fit comparisons are in-sample and use different conditioning information. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Dynamic Offer Model

This appendix develops the dynamic counterpart to Section 5.3. The hospital values the same accounting, capacity, clinical, and offer-stage consequences, but now accounts explicitly for future assessment opportunities and operating states. The dynamic specification replaces the static model's valuation of remaining capacity exposure at the current state with interval payoffs and continuation values. It retains Θ_S and the payoff functions defined in Section 5; the supply coefficients are estimated again under the dynamic specification.

E.1 The Hospital's Dynamic Offer Problem

Assessment timing and state. The risk set and patient information are those of the joint model. Let τ_i denote assessment time in hours. With the daily discount factor $\delta = 0.99$, the continuation discount and interval flow factor are

$$d_i = \delta^{(\tau_{i+1} - \tau_i)/24}, \quad a_i = \int_0^{(\tau_{i+1} - \tau_i)/24} \delta^s ds = \frac{1 - d_i}{-\log \delta}. \quad (\text{E.1})$$

Exactly tied assessments share an event time; their census changes are grouped before evaluating nonlinear capacity flows. Below, binary choice values refer to the focal assessment conditional on the other decisions at that event.

Write z_i for the full information state: the B&M census and staffed capacity at each facility, HaH census and staffing, current patient characteristics, calendar variables, and the prior-history and preference weights used in Section 5. Admitted patients occupying ED beds remain part of B&M census. Including patient histories distinguishes physical capacity changes from information acquired through an offer. The next-event law F_A governs the time to the next assessment, its patient characteristics, and the background operating state. It is estimated from consecutive assessment events, without constructing a separate daily admission model. Future assessment timing, patient mix, and staffing are uncertain; the hospital does not observe their future realizations.

Current payoffs. The direct offer terms $Z_i' \kappa$, remaining accounting return $\Delta \Pi_i$, and escalation risk H_i retain their definitions in equations (11)–(12). Accounting returns are booked once at the placement decision because $\Delta \Pi_i$ already values the expected remaining episode. Escalation remains an ex ante risk term; the baseline does not separately simulate return transfers. No explicit backfill-profit payoff is added.

For capacity flows, let $m^B(z)$ collect the sum of the common B&M scarcity functions across facilities and the individual hospital tail functions, using equation (17). Let $m^H(z)$ be the HaH

strain function in equation (14). These are the existing functions evaluated at the evolving state. Conditional on an accepted transfer, define the capacity differences over the first interval as

$$M_i^{B,0} = \mathbb{E} \left[\int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s \{m^B(z_i^B(s)) - m^B(z_i^H(s))\} ds \mid z_i, A_i = 1 \right], \quad (\text{E.2})$$

$$M_i^{H,0} = \mathbb{E} \left[\int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s \{m^H(z_i^H(s)) - m^H(z_i^B(s))\} ds \mid z_i, A_i = 1 \right]. \quad (\text{E.3})$$

Here B and H denote the focal patient's alternative treatment paths. If the state is constant over the interval, its discounted flow is multiplied by a_i . Survival and capacity reuse determine how long the two paths differ. The current offer-minus-no-offer payoff is

$$g_i(\Theta_S) = Z_i' \kappa + \rho_i \left[\beta_\Pi \Delta \Pi_i + \beta_H H_i + \beta_E E_i + \lambda_B' M_i^{B,0} + \lambda_H M_i^{H,0} \right]. \quad (\text{E.4})$$

The acceptance probability enters once. The interval capacity terms replace, rather than supplement, the full remaining-exposure terms in the static index. Their subsequent flows enter continuation value.⁹

Dynamic problem. Let $\bar{\Gamma}_i(o, z_i; \Theta_S)$ be the expected current payoff from action o . Its capacity-flow component is the interval integral of $-\lambda_B' m^B(z) + \lambda_H m^H(z)$, so that $\bar{\Gamma}_i(1, z_i; \Theta_S) - \bar{\Gamma}_i(0, z_i; \Theta_S) = g_i(\Theta_S)$. These flow levels are retained in the value function; normalizing them away at every state would remove consequences of earlier transfers. The choice-specific value and Bellman equation are

$$v_i(o, z_i; \Theta_S) = \bar{\Gamma}_i(o, z_i; \Theta_S) + \mathbb{E}[d_i V_{i+1}(z_{i+1}; \Theta_S) \mid z_i, O_i = o], \quad (\text{E.5})$$

$$V_i(z_i; \Theta_S) = \mathbb{E}_{\varepsilon_i} \left[\max_{o \in \{0,1\}} \{v_i(o, z_i; \Theta_S) + \varepsilon_i(o)\} \right]. \quad (\text{E.6})$$

The first expectation integrates over future event timing, acceptance, survival, capacity reuse, staffing, and patient information. The discount factor remains inside this expectation because the next event time is uncertain. The second expectation integrates over current private offer shocks. Interval flows are not added again to the continuation term.

⁹Input units follow Appendix C.2: accounting returns are in \$1,000 units and capacity and risk measures retain their units. Among acceptance-probability terms, only the standalone term is centered; placement multipliers use raw ρ_i . Reference offsets in centered placement features enter once in the assessment payoff, not again in every subsequent interval of that placement. The physical state-flow functions themselves are evaluated before this assessment-level centering.

Let $z_{i+1}^{a,i}$, $z_{i+1}^{d,i}$, and $z_{i+1}^{0,i}$ denote next states following acceptance, rejection, and no offer. Then

$$\begin{aligned} \Delta v_i(\Theta_S) = & g_i(\Theta_S) + \rho_i \mathbb{E} \left[d_i \{ V_{i+1}(z_{i+1}^{a,i}; \Theta_S) - V_{i+1}(z_{i+1}^{0,i}; \Theta_S) \} \mid z_i, A_i = 1 \right] \\ & + (1 - \rho_i) \mathbb{E} \left[d_i \{ V_{i+1}(z_{i+1}^{d,i}; \Theta_S) - V_{i+1}(z_{i+1}^{0,i}; \Theta_S) \} \mid z_i, A_i = 0, O_i = 1 \right]. \end{aligned} \quad (\text{E.7})$$

The counterfactual no-offer state is evaluated using the same underlying draws in each conditional comparison. Rejection and no offer have the same physical census path, but can imply different subsequent histories and beliefs. If these information effects are held fixed, the final line is zero and the continuation difference reduces to the acceptance-weighted HaH-minus-B&M comparison. Future value functions already incorporate acceptance by future patients; they do not multiply the focal ρ_i again.

Counterfactual state paths. An accepted transfer initially removes one patient from B&M census and adds one to HaH census. Let $s_i^B(h)$ and $s_i^H(h)$ be the remaining-stay survival probabilities at elapsed days h . The expected physical footprints are

$$w_i^B(h) = s_i^B(h) e_{b(i)}, \quad w_i^H(h) = s_i^H(h) e_h + s_i^B(h) e_{b(i)}, \quad (\text{E.8})$$

where the unit vectors add census to the originating hospital or HaH. At later assessment j , the physical states are constructed as

$$\mathcal{S}_j^{B,i} = \mathcal{S}_j^{-i} + w_i^B(h_{ij}), \quad \mathcal{S}_j^{H,i} = \mathcal{S}_j^{-i} + w_i^H(h_{ij}), \quad h_{ij} = \frac{\tau_j - \tau_i}{24}. \quad (\text{E.9})$$

The common background $\mathcal{S}_j^{-i}$ removes the focal patient's realized footprint and any capacity use attributed to its release before either path is overlaid. This avoids adding the same replacement patient twice. Reuse affects the duration of the census difference, not a separate profit payoff. The B&M coefficient continues to value capacity on a composite basis.¹⁰

Both paths use common draws from $\hat{F}_A$, with the fitted offer policy reevaluated at their respective states. Patient histories and preference weights update after simulated outcomes using the joint model's filtering rule. The maintained background arrival law does not respond to the focal offer beyond the specified capacity-reuse footprint. Changes in admission behavior or assessment intensity therefore require an extension of that law.

¹⁰Evaluating nonlinear capacity functions at these expected footprints is an approximation. It differs from averaging the functions over discrete survival and reuse realizations. The same convention must be used in current flows and continuation returns.

E.2 Conditional Choice Probabilities and Two-Step GMM

Conditional choice probabilities. I use a CCP estimator following Hotz and Miller (1993); Hotz et al. (1994); the equation-matching formulation is related to Pesendorfer and Schmidt-Dengler (2008). These methods represent continuation values using conditional choice probabilities and state transitions, avoiding repeated solution of the dynamic optimization problem during estimation (Aguirregabiria and Mira, 2010; Arcidiacono and Ellickson, 2011; Aguirregabiria et al., 2021). The joint model supplies the fitted ex ante true-offer policy $\hat{p}(1 | z) = \hat{p}(z)$ and acceptance belief $\hat{p}(z)$. The former is a logit-based fitted probability, not a documented offer indicator or the posterior classification in equation (7). The documentation probabilities q_i remain empirical measurement inputs.

Conditional on z_i , assume independent standard Type-I extreme-value action shocks, with unit scale, and the maintained acceptance and transition laws. The Hotz–Miller inversion gives

$$\Delta \hat{c}v_i = \log \left(\frac{\hat{p}_i}{1 - \hat{p}_i} \right). \quad (\text{E.10})$$

The first-stage policy must capture the state dependence relevant to the dynamic decision. Using the joint model’s fitted policy does not equate its static payoff coefficients with the dynamic coefficients.

Fixed-policy returns. I hold the first-stage policy, demand parameters, and transition estimates fixed during supply estimation. At each simulated state, the fitted offer–acceptance fixed point is evaluated using these first-stage parameters. A trial Θ_S changes payoff weights, not the policy or acceptance map used to construct returns. This is a two-step estimator, distinct from joint maximization of the static likelihood.

For fixed policy $\hat{p}$, the policy-value recursion includes the expected selected offer shock:

$$\begin{aligned} V^{\hat{p}}(z; \Theta_S) &= \sum_o \hat{p}(o | z) \left\{ \bar{\Gamma}(o, z; \Theta_S) + \mathbb{E}[dV^{\hat{p}}(z'; \Theta_S) | z, o] \right\} + \eta(z), \\ \eta(z) &= \gamma_E - \sum_o \hat{p}(o | z) \log \hat{p}(o | z), \end{aligned} \quad (\text{E.11})$$

where γ_E is Euler’s constant under the standard extreme-value location normalization.

I simulate returns over 90 days, using common transition draws across branches and an approximate stationary tail. A future event j is discounted by $\delta^{(\tau_j - \tau_i)/24}$. Each return component is projected onto a flexible basis of the current state and patient characteristics. Because payoffs are

linear in Θ_S under the fixed policy,

$$\Delta v_i(\Theta_S) \approx \Delta \hat{v}_i^0 + \hat{\mathcal{D}}_i' \Theta_S, \quad r_i(\Theta_S) = \Delta \hat{c} v_i - \Delta \hat{v}_i^0 - \hat{\mathcal{D}}_i' \Theta_S. \quad (\text{E.12})$$

The vector $\hat{\mathcal{D}}_i$ collects current and continuation exposures to every coefficient in equation (E.4); $\Delta \hat{v}_i^0$ includes the shock-value contribution. The tail approximation must preserve this affine decomposition.

Two-step GMM. Let $\mathcal{M}_i$ contain projected policy-value derivatives and any additional excluded state instruments used for overidentification. Define

$$g_n(\Theta_S) = \frac{1}{n} \sum_{i=1}^n \mathcal{M}_i r_i(\Theta_S), \quad \tilde{\Theta}_S \in \arg \min_{\Theta_S} g_n(\Theta_S)' g_n(\Theta_S). \quad (\text{E.13})$$

At $\tilde{\Theta}_S$, I aggregate assessment-level moments by calendar date and estimate their long-run covariance $\hat{\Omega}$ using a Bartlett-kernel HAC estimator. The second step is

$$\hat{\Theta}_S \in \arg \min_{\Theta_S} g_n(\Theta_S)' \hat{\Omega}^{-1} g_n(\Theta_S). \quad (\text{E.14})$$

Objective coefficients are estimated without sign restrictions.

E.3 Identification and Inference

Identifying variation. The offer-shock normalization fixes the scale of Θ_S . Conditional on the generated demand and transition objects, remaining accounting returns identify β_Π through variation distinct from capacity consequences. The hospital-specific B&M terms use variation in census, capacity, remaining exposure, and their continuation paths. HaH census, staffing, and remaining HaH exposure distinguish λ_H from B&M relief; residual clinical variation identifies β_H and the clinical offer shifters. The coefficient on stand-alone acceptance is identified separately from acceptance-weighted placement consequences only when their payoff exposures are sufficiently distinct. As in Section 5.3, it can reflect mission-related targeting, but cannot be directly interpreted as a weight on patient surplus. The common offer term remains a net baseline friction under the stated centering. History and fixed-effect coefficients capture policy heterogeneity without separate welfare labels.

Local identification requires

$$\text{rank} \left\{ \mathbb{E}[\mathcal{M}_i \hat{\mathcal{D}}_i'] \right\} = \dim(\Theta_S). \quad (\text{E.15})$$

This condition also requires valid moments, adequate offer-policy support, and a state and transition law that capture the hospital's relevant information. CCP inversion alone does not establish these assumptions. Diagnostics include derivative correlations, the minimum singular value and condition number of the moment Jacobian, and profile objectives. Particular attention is given to accounting versus capacity exposure, and acceptance targeting versus placement benefits.

Inference. For $\mathbf{m}_i(\Theta_S) = \mathcal{M}_i r_i(\Theta_S)$, define $\mathbf{m}_t(\Theta_S) = \sum_{i:\text{date}(\tau_i)=t} \mathbf{m}_i(\Theta_S)$, retaining dates without assessments with zero moments. The HAC matrix allows dependence within dates and serial dependence across nearby dates. Standard errors use a calendar-date block multiplier bootstrap, assigning a common weight to assessments in each block while preserving chronology. Each replication refits the joint demand–offer model, estimated accounting, survival, reuse, escalation, and transition inputs, and the return projections, then reconstructs the HAC matrix and repeats both GMM steps. Empirical documentation probabilities remain fixed in this conditional inference; propagating their sampling uncertainty requires reconstructing them as well. The covariance of the replication estimates provides standard errors, including uncertainty in the reestimated generated inputs.

E.4 Dynamic Results and Comparison with the Static Model

Private capacity value. For comparison with the static model, the dynamic private capacity value is the offer-value difference between the full calculation and a no-release calculation that retains focal HaH treatment and HaH strain but removes the B&M capacity change and its continuation consequences. Focal acceptance, history updates, and payoff reference offsets are held common. Its dollar-equivalent value per accepted transfer is

$$\widehat{CV}_i^{\text{Dyn}} = 1,000 \frac{\Delta \widehat{v}_i^{\text{release}} - \Delta \widehat{v}_i^{\text{no release}}}{\widehat{\rho}_i \widehat{\beta}_\Pi}. \quad (\text{E.16})$$

TABLE E.1: Dynamic Hospital Offer Estimates

	Estimate	Standard error
Accounting profits (per \$1,000)	0.489***	(0.007)
HaH program strain	0.0009***	(0.0003)
HaH escalation risk	−2.553***	(0.095)
Care-burden	−0.1827***	(0.0004)
Acute-severity	−0.0787***	(0.0004)
B&M stay-duration exposure	−0.622***	(0.009)
Hospital A released-capacity	0.230***	(0.004)
Hospital B released-capacity	0.151***	(0.005)
Hospital C released-capacity	0.160***	(0.005)
Hospital D released-capacity	0.046***	(0.004)
Strictly prior expected offers	0.295***	(0.001)
Acceptance belief	2.572***	(0.009)
Baseline net offer term	−0.848***	(0.003)
Assessments	31,944	
Unique patients	21,354	
Moments used in GMM	31	
<i>Conditional fit statistics</i>		
Offer-CCP approximation RMSE	0.0103	
Mean continuation contribution	0.1028	

Notes: Two-step conditional-choice-probability (CCP) GMM estimates of the fixed-policy dynamic supply approximation. The baseline offer CCPs, demand parameters, acceptance map, posterior histories, and 46 department effects are held fixed. The dynamic stage re-estimates 26 supply coefficients using 31 moments, including condition and hospital effects, flags for missing clinical fields, and the shutdown indicator. An additional 897 history-only assessments initialize the patient filter. Current payoff inputs retain the baseline units, centers, and acceptance weighting; accounting profits are in \$1,000 units, and care burden and acute severity are standardized. The daily discount factor is 0.99. Standard errors in parentheses condition on generated CCPs, demand, transition and reward models, survival inputs, and fixed department effects, omitting first-stage estimation uncertainty. They are not directly comparable to the main-text joint-estimation standard errors. RMSE compares dynamic offer probabilities with the baseline fitted CCPs on the estimation sample. Mean continuation is in offer-index units. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Acceptance and Capacity in Hospital Offer Decisions

This appendix clarifies how patient acceptance and hospital congestion interact in the offer model. It first characterizes the direct capacity incentive and then distinguishes that incentive from the response of the jointly determined offer and acceptance probabilities.

Offer values. Retain the main-text notation and the fixed input transformations used in estimation. Write the component of the offer index independent of acceptance as

$$\Gamma_i^0 = \kappa_O + \kappa_{\text{hist}} \text{Hist}_i^O + \gamma_S S_i + \zeta_i' \theta - \gamma_\rho \bar{\rho},$$

where the final term accounts for centering the standalone acceptance belief. The offer index can then be written as

$$\Gamma_i(\rho_i) = \Gamma_i^0 + \rho_i [\gamma_\rho + \Delta W_i], \quad p_i = \Lambda(\Gamma_i(\rho_i)), \quad (\text{F.1})$$

with

$$\Delta W_i = \beta_\Pi \Delta \Pi_i + \beta_H H_i + \beta_E E_i^B + \lambda_B' M_i^B + \lambda_H M_i^H.$$

This regrouping retains both standalone targeting through γ_ρ and the acceptance-contingent consequences in ΔW_i . All placement inputs retain their estimation centers; only the fixed offset from centering the standalone belief is absorbed into Γ_i^0 .

Congestion and acceptance. Let $u_i^B = N_i^B / K_i$ denote occupancy at the admitting hospital $b(i)$. Consider changes in u_i^B holding staffed capacity K_i , expected remaining duration E_i^B , patient characteristics, histories, and other payoff inputs fixed. The raw hospital-specific capacity component is

$$M_{ib}^{B, \text{raw}} = \mathbb{1}\{b(i) = b\} E_i^B \left(2u_i^B - \frac{1}{K_i} \right).$$

Because the centering references remain fixed, the change in placement value between two occupancy levels satisfies

$$\Delta W_i(u_1) - \Delta W_i(u_0) = 2\lambda_{B, b(i)} E_i^B (u_1 - u_0). \quad (\text{F.2})$$

The comparison therefore isolates occupancy from the separate remaining-stay effect $\beta_E E_i^B$.

Proposition F.1 (Acceptance and congestion in the offer index). *Under the preceding comparison, suppose $\lambda_{B, b(i)} \geq 0$ and $E_i^B > 0$.*

(i) *Holding acceptance fixed, the offer index and offer probability are nondecreasing in occu-*

pancy. Occupancy and acceptance are complements in the offer index:

$$\left. \frac{\partial \Gamma_i}{\partial u_i^B} \right|_{\rho_i} = 2\rho_i \lambda_{B,b(i)} E_i^B, \quad \frac{\partial^2 \Gamma_i}{\partial u_i^B \partial \rho_i} = 2\lambda_{B,b(i)} E_i^B \geq 0. \quad (\text{F.3})$$

(ii) If $\Gamma_i^0 < 0$ and $\gamma_\rho + \Delta W_i(u_i^B) > 0$, deterministic offer value is nonnegative if and only if

$$\rho_i \geq \rho_i^*(u_i^B) \equiv \frac{-\Gamma_i^0}{\gamma_\rho + \Delta W_i(u_i^B)}. \quad (\text{F.4})$$

This threshold is nonincreasing in occupancy. If it exceeds one, no feasible acceptance probability yields nonnegative deterministic offer value.

Proof. Differentiate (F.1) using (F.2) to obtain part (i); monotonicity of Λ gives the probability result. Under the sign conditions in part (ii), rearranging $\Gamma_i \geq 0$ gives (F.4). Its numerator is positive and fixed, while its denominator is nondecreasing in occupancy. $\square$

The proposition describes how the same potential bed release becomes more valuable as congestion rises, with a larger direct effect when acceptance is more likely. Greater congestion can consequently make an offer worthwhile at a lower acceptance probability, holding other patient and payoff inputs fixed. The threshold characterizes the deterministic index, equivalently $p_i \geq 1/2$; actual offers also depend on the logistic offer shock v_i . Moreover, complementarity applies to offer *log-odds*: the occupancy gradient in probability additionally depends on $p_i(1 - p_i)$ and need not increase with acceptance.

Response of the joint model. In estimation and counterfactuals, acceptance also changes with the offer probability. Holding demand covariates and strictly prior preference weights fixed, the acceptance map satisfies

$$\rho_i'(p) = -\varphi \sum_g w_{ig}^- \rho_{ig}(p) [1 - \rho_{ig}(p)], \quad (\text{F.5})$$

where φ is the coefficient on $1 - p$ in patient utility. Under the maintained exclusion of occupancy from that utility, occupancy enters acceptance through p alone. Define the fixed-point residual as

$$F_i(p, u_i^B) = p - \Lambda(\Gamma_i^0 + \rho_i(p) [\gamma_\rho + \Delta W_i(u_i^B)]).$$

At a solution with $\partial F_i / \partial p \neq 0$, implicit differentiation gives the local response

$$\frac{dp_i}{du_i^B} = \frac{2p_i(1 - p_i)\rho_i \lambda_{B,b(i)} E_i^B}{1 - p_i(1 - p_i) [\gamma_\rho + \Delta W_i] \rho_i'(p_i)}. \quad (\text{F.6})$$

The denominator is $\partial F_i / \partial p$. When it is positive, the full local offer response has the same sign as the direct capacity incentive. For $\varphi < 0$ and $\gamma_p + \Delta W_i > 0$, the feedback amplifies the direct probability response whenever this denominator remains positive. Holding the selection adjustment at its baseline value sets $\rho'_i(p) = 0$ for this comparison and removes that feedback.

The direct-index results apply to the static specification's valuation of remaining capacity at the current operating state. The dynamic counterpart also values changes in future states and offer decisions. Its total congestion response consequently depends on the estimated transitions and continuation values, in addition to the acceptance weighting emphasized here.